Table 10.5 Material for Bicycle Forks

Material	Strength σ_f (MPa)	Density ρ (kg/m^3)	Shape Factor ϕ_B^f	Index* $\sigma_f^{2/3}/\rho$ (MPa)$^{2/3}$/(Mg/m^3)	Index M_2* $(\phi_B^f \sigma_f)^{2/3}/\rho$ (MPa)$^{2/3}$/(Mg/m^3)
Spruce (Norwegian)	75	450	1.5	39	51
Bamboo	70	700	2.2	24	41
Steel (Reynolds 531)	880	7,850	7.5	12	46
Alu (6061-T6)	250	2,700	5.9	15	49
Titanium 6%Al-4%V	950	4,420	5.9	22	72
Magnesium AZ 61	165	1,810	4.25	17	45
CFRP	375	1,550	4.25	33	87

* *The values of the indices are based on mean values of the material properties.*

Related reading: Bicycle design

Oliver, T. (1992). *Touring bikes, a practical guide*. Crowood Press, ISBN 1-85223-339-7.
A good source of information about bike materials and construction, with tables of data for the steels used in tube sets.

Sharp, A. (1979). *Bicycles and tricycles, an elementary treatise on their design and construction*. MIT Press, ISBN 0-262-69066-7.
A far-from-elementary treatise, despite its title, first published in 1977. It is the place to look if you need the mechanics of bicycles.

Watson R., & Gray, M. (1978). *The Penguin book of the bicycle*. Penguin, ISBN 0-1400-4297-0.
Watson and Gray describe the history and use of bicycles. Not much on design, mechanics, or materials.

Whitt, F.R., & Wilson, D.G. (1982). *Bicycling science* (2nd ed.). The MIT Press, ISBN 0-262-73060-X.
A book by two MIT professorial bike enthusiasts, more easily digestible than Sharp, and with a good chapter on materials.

Wilson, D.G. (1986). A short history of human powered vehicles. *American Scientist, 74*, p. 350.
Typical *Scientific American* article: *good content, balance, and presentation. A good starting point.*

Bike magazines such as Mountain Bike, Which Bike? *and* Cycling and Mountain Biking *carry extensive tables of available bikes and their characteristics—type, maker, cost, weight, gear set, etc.*

Related case studies

6.5 "Cost: Structural materials for buildings"
10.2 "Spars for human-powered planes"
10.4 "Floor joists: Wood, bamboo, or steel?"

10.4 FLOOR JOISTS: WOOD, BAMBOO, OR STEEL?

Floors are supported on joists: beams that span the space between the walls (Figures 6.7 and 10.4). Let us suppose that a joist is required to support a specified bending load (the "floor loading") without sagging excessively or

failing; it must be cheap. Traditionally, joists in the United States and Europe are made of wood with a rectangular section of aspect ratio 2:1, giving an elastic shape factor (refer to Table 9.3) of $\phi_B^e = 2$. In Asian countries bamboo, with a "natural" shape factor of about $\phi_B^e = 3.2$, is a replacement for wood in smaller buildings. But as wood becomes scarcer and buildings need to be larger, steel replaces wood and bamboo as the primary structural material. Standard steel I-section joists have shape factors in the range $5 < \phi_B^e < 25$ (special I-sections can have much larger values). Are steel I-joists a better choice than wooden ones? Table 10.6 contains a summary of the design requirements.

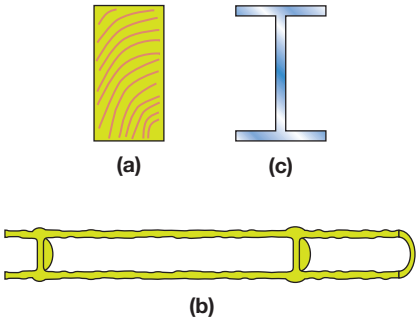


FIGURE 10.4
The cross-sections of a wooden beam ($\phi_B^e = 2$), a steel I-beam ($\phi_B^e = 10$), and bamboo ("natural" shape factors of $\phi_B^e = 3.2$). The values of ϕ_B^e are calculated from the ratios of dimensions of each beam, using the formulae of Table 9.3.

The translation and the selection Consider stiffness first. The cheapest beam, for a given stiffness, is that with the largest value of the index (from Table 10.1 with ρ replaced by $C_m\rho$ to minimize cost):

$$M_1 = \frac{(\phi_B^e E)^{1/2}}{C_m \rho} \tag{10.3}$$

Data for the modulus E, the density ρ, the material cost C_m, and the shape factor ϕ_B^e are listed in Table 10.7, together with the values of the index M_1 with and without shape. The steel beam with $\phi_B^e = 25$ has a slightly larger value of M_1 than wood, meaning that it is a little cheaper for the same stiffness.

But what about strength? The best choice for a light beam of specified strength is one that maximizes the material index:

$$M_2 = \frac{(\phi_B^f \sigma_f)^{2/3}}{C_m \rho} \tag{10.4}$$

The quantities of failure strength σ_f shape factor ϕ_B^f, and index M_2 are also given in the table. Wood performs better than even the most efficient steel I-beam.

Postscript So the conclusion: As far as performance per unit material cost is concerned, there is not much to choose between the standard wood and

Table 10.6 Design Requirements for Floor Joists

Function	Floor joist
Constraints	Length specified Stiffness specified Strength specified
Objective	Minimum material cost
Free variables	Choice of material Section shape

Table 10.7(a) Materials for Floor Joists

Material	Density ρ (kg/m^3)	Cost C_m US$/kg	Flexural Modulus E (GPa)	Flexural Strength σ_f (MPa)
Wood (pine)	490	1.0	9.5	41
Bamboo	700	1.9	17	42
Steel (1020)	7850	0.65	205	355

Table 10.7(b) Indices for Floor Joists

Material	Shape Factor ϕ_B^e	Shape Factor ϕ_B^f	Index $E^{1/2}/C_m\rho$	Index* M_1 $(\phi_B^e E)^{1/2}/C_m\rho$	Index $\sigma_f^{2/3}/C_m\rho$	Index* M_2 $(\phi_B^f \sigma_f)^{2/3}/C_m\rho$
Wood (pine)	2	1.4	6.3	8.9	24	30
Bamboo	3.2	2	3.1	5.5	9	14
Steel (1020)	15	4	2.8	11	9.8	25

* *The values of the indices are based on means of the material properties. The units of the indices for elastic deflection are* (GPa)$^{1/2}$/kg/m^3; those for failure are (MPa)$^{2/3}$/kg/m^3.

the standard steel sections used for joists. As a general statement, this is no surprise—if one were much better than the other, the other would no longer exist. But—looking a little deeper—wood dominates in certain market sectors, bamboo in others, and, in others still, steel. Why?

Wood and bamboo are indigenous to some countries and grow locally; steel has to come further, with associated transport costs. Assembling wood structures is easier than assembling those of steel; wood is more forgiving of mismatches of dimensions, it can be trimmed on site, you can hammer nails into it anywhere. It is a user-friendly material.

But wood is a variable material, and, like us, is vulnerable to the ravages of time, prey to fungi, insects, and small mammals with sharp teeth. The problems they create in a small building—a family home, say—are easily overcome, but in a large commercial building—an office block, for instance—they create greater risks and are harder to fix. Here, steel wins.

Related reading

Cowan, H.J., & Smith, P.R. (1988). *The science and technology of building materials*. Van Nostrand, ISBN 0-442-21799-4.
A broad survey of materials for the structure, cladding, insulation, and interior surfacing of buildings—an excellent introduction to the subject.
Farrelly, D. (1984). *The book of bamboo*. Sierra Club Books, ISBN 0-87156-825-X.
An introduction to bamboo and its many varieties.
Janssen, J.J. (1995). *Building with bamboo: A handbook*. Practical Action, ISBN 1-85339-203-0.
Bamboo remains a building material of great importance, as well as the material of flooring, matting, and basketmaking. The techniques of building with bamboo have a long history, documented here.

Related case studies

6.5 "Cost: Structural materials for buildings"
10.2 "Spars for human-powered planes"
10.3 "Forks for a racing bicycle"

10.5 TABLE LEGS YET AGAIN: THIN OR LIGHT?

Luigi Tavolino's table (Section 6.4 and Figure 6.5) is a great success. He decides to develop a range of less expensive tubular-legged furniture. Some are to have thin legs, others have to be light. He needs a more general way of exploring material and shape.

The translation and the selection Tubes can be made with almost any radius r and wall thickness t, though (as we know from Chapter 9) it is impractical to make them with r/t too large because they buckle locally, and that is bad. Luigi chooses GFRP as the material for his legs, and for this material the maximum available shape factor is $\phi_B^e = 10$ corresponding to a value of $r/t = 10.5$ (refer to Table 9.3). The cross-sectional area A and the second moment of area I of a thin-walled tube are

$$A = 2\pi r t$$

and

$$I = \pi r^3 t$$

from which

$$r = \left(\frac{2I}{A}\right)^{1/2} \tag{10.5}$$

This allows contours of constant r to be plotted on the A—I quadrant of the four quadrant diagram shown in Figure 10.5 (they form a family of lines of slope 1), which can now be used to select either for minimizing mass or for minimizing tube radius.

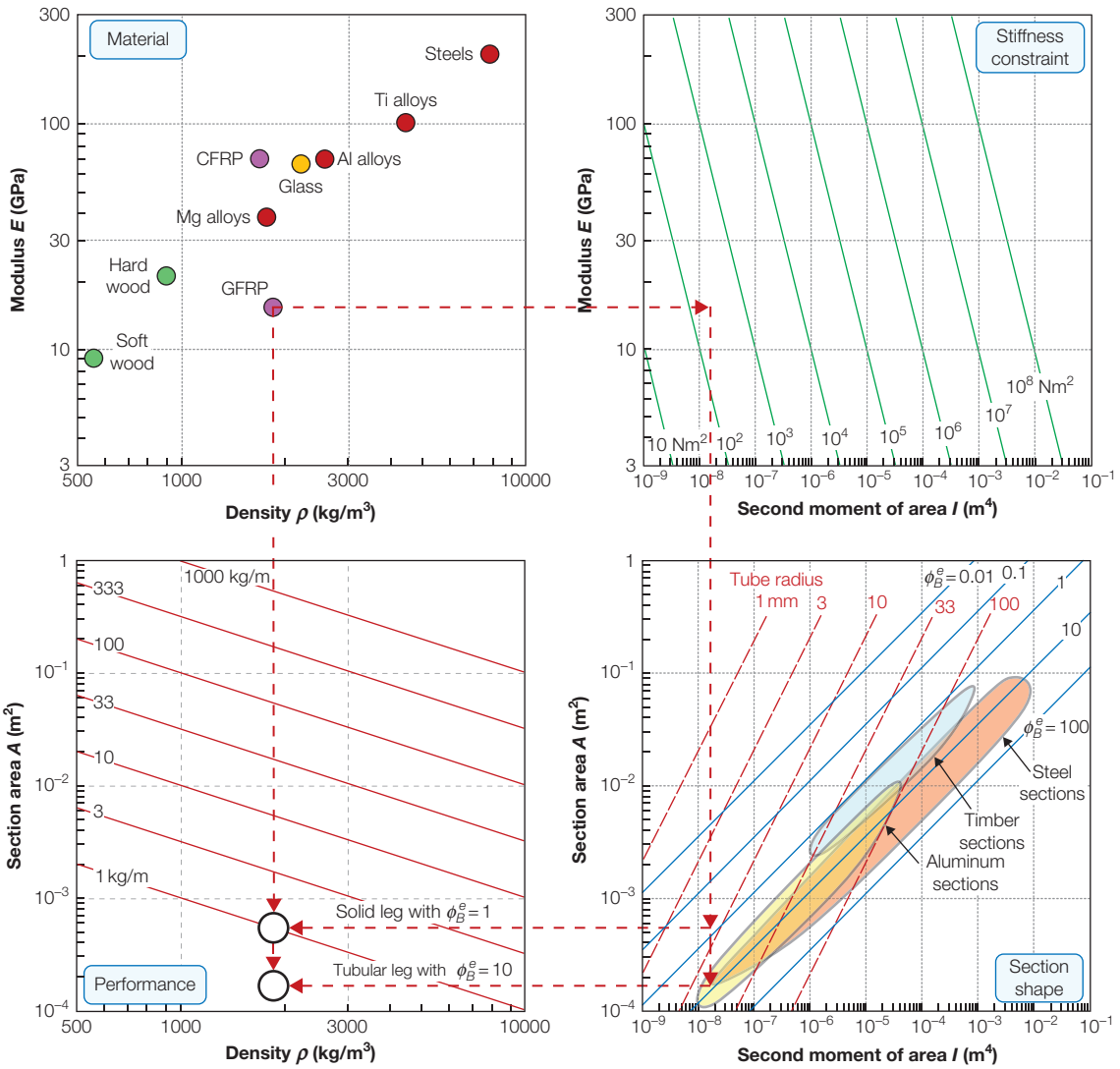


FIGURE 10.5

The chart assembly for exploring stiffness-limited design with tubes. The *I*-*A* quadrant now has contours of tube radius *r*. The thinnest GFRP leg is the one with $\phi_B^e = 1$. The lightest one is that with $\phi_B^e = 10$, the maximum value for GFRP.

Luigi might use it in the following way. He designs a table with cylindrical, unbraced legs, each with length $L = 1$ m. For safety, each leg must support 50 kg without buckling, requiring that

$$\frac{\pi^2}{4}\frac{E\,I}{\ell^3} \geq 500\ \text{N}$$

from which

$$E\,I \geq 200\ \mathrm{N \cdot m^2}$$

Luigi now plots a rectangular selection path on the four-quadrant diagram in the way shown in Figure 10.5. The lightest leg is that with the largest allowable value of ϕ_B^e (10 in the case of GFRP), but this gives a fat leg, one nearly 20 mm in radius, as read from the radius contours. The thinnest leg is a solid one, and that has $\phi_B^e = 1$. It is much thinner—only 5 mm in radius—but it is also almost three times heavier.

Postscript So Luigi will have to compromise: fat and light or thin and heavy—or somewhere in between. Or he could use another material. The strength of the four-quadrant method is that he can explore other materials with ease. Repeating the construction for aluminum, putting an upper limit on ϕ_B^e of, say, 15, or for CFRP with the same upper limit as GFRP is only a moment's work.

Related case studies

6.4 "Materials for table legs"
8.5 "Conflicting objectives: table legs again"

10.6 INCREASING THE STIFFNESS OF STEEL SHEET

How can you make steel sheet stiffer? There are many reasons you might wish to do so. The most obvious: to enable stiffness-limited sheet structures to be lighter than they are now; to allow panels to carry larger in-plane compressive loads without buckling; and to raise the natural vibration frequencies of sheet structures. Bending stiffness is proportional to $E.I$ (E is Young's modulus, I is the second moment of area of the sheet, equal to $t^3/12$ per unit width where t is the sheet thickness). There is nothing much you can do to change the modulus of steel, which is always close to 210 GPa. The answer is to add a bit of shape, increasing I. So consider the design brief of Table 10.8.

The translation and the selection The age-old way to make sheet steel stiffer is to corrugate it, giving it a roughly sinusoidal profile. The corrugations increase the second moment of area of the sheet about an axis normal to the corrugations themselves. The resistance to bending in one direction is thereby increased, but in the cross-direction it is not changed at all.

Corrugations are the clue, but—to be useful—they must stiffen the sheet in all directions, not just one. A hexagonal grid of dimples (Figure 10.6)

Table 10.8 Design Requirements for Stiffened Steel Sheet	
Function	Steel sheet for stiffness-limited structures
Constraints	Profile limited to a maximum deviation ± 5 times the sheet thickness from flatness Cheap to manufacture
Objective	Maximize bending stiffness of sheet
Free variables	Section profile

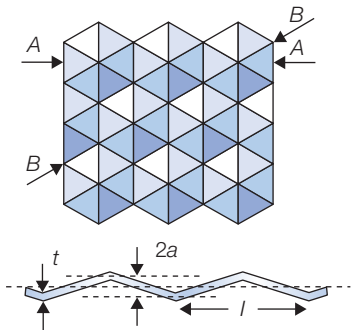


FIGURE 10.6
A sheet with a profile of adjacent hexagonal dimples that increases its bending stiffness and strength. Shape factors for the section *A–A* are calculated in the text. Those along other trajectories are lower but still significantly greater than 1.

achieves this. There is now no direction of bending that is not dimpled. The dimples need not be hexagons; any pattern arranged in such a way that you cannot draw a straight line across it without intersecting dimples will do. Hexagons are probably about the best.

Consider an idealized cross-section as in the lower part of Figure 10.6, which shows the section *A–A* enlarged. As before, we define the shape factor as the ratio of the stiffness of the dimpled sheet to that of the flat sheet from which it originated. The second moment of area of the flat sheet per unit width is

$$I_o = \frac{t^3}{12} \tag{10.6}$$

That of the dimpled sheet with amplitude a is

$$I \approx \frac{1}{12}(2a+t)^2\, t \tag{10.7}$$

giving a shape factor, defined as before as the ratio of the stiffness of the sheet before and after corrugating, of

$$\phi_B^e = \frac{I}{I_o} \approx \frac{(2a+t)^2}{t^2} \tag{10.8}$$

Note that the shape factor has the value unity when the amplitude is zero, but increases as the amplitude increases. The equivalent shape factor for failure in bending is

$$\phi_B^f = \frac{Z}{Z_o} \approx \frac{(2a+t)}{t} \tag{10.9}$$

These equations predict large gains in stiffness and strength. The reality is a little less rosy. This is because, while all cross-sections of the sheet are dimpled, only those that cut through the peaks of the dimples have an

amplitude equal to the peak height (all others have less) and, even among these, only some have adjacent dimples; the section *B–B*, for example, does not. Despite this, and limits set by the onset of local buckling, the gain is real.

Postscript Dimpling can be applied to most rolled-sheet products. It is done by making the final roll pass through mating rolls with meshing dimples, adding little to the cost. It is most commonly applied to sheet steel. Here it finds applications in the automobile industry for bumper armatures, seat frames, and side impact bars, allowing weight saving without compromising mechanical performance. Stiffening sheet also raises its natural vibration frequencies, making them harder to excite, thus helping to suppress vibration in panels.

But a final word of warning: Stiffening the sheet may change its failure mechanism. Flat sheet yields when bent; dimpled sheet, if thin, can fail by a local buckling mode. It is this that ultimately limits the useful extent of dimpling.

Related reading: stiffening sheet by dimpling

Fletcher, M. (1998). Cold-rolled dimples gauge strength. *Eureka* (May).

A brief account of the shaping of steel panels to improve bending stiffness and strength.

Related case study

10.7 "Shapes that flex: Leaf and strand structures"

10.7 SHAPES THAT FLEX: LEAF AND STRAND STRUCTURES

Flexible cables, leaf and helical springs, and flexural hinges require low, not high, structural efficiency. Here the requirement is for low flexural or torsional stiffness about one or two axes while retaining high stiffness and strength in other directions. The simplest example is the torsion bar (Figure 10.7(a)), with a shape factor $\phi_T^e \approx 1$. The single-leaf spring (Figure 10.7(b)) allows much lower values of ϕ_B^e ($\phi_B^e = h/w$; the dimensions are defined in the figure).

Multistrand cables and multileaf assemblies do much better. Consider the change in efficiency when a solid square beam of section $A = b^2$ is subdivided into n cylindrical strands, each of radius r (Figures 10.7(c)(d)), such that

$$n\pi r^2 = b^2 \tag{10.10}$$

The axial stiffness of the original bar is proportional to EA, and its bending stiffness is S_o to EI_o, where E is the modulus of the material and I_o is

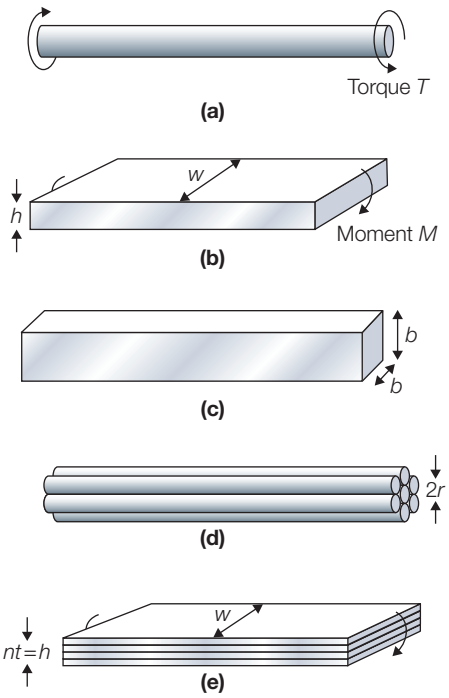


FIGURE 10.7
A simple torsion bar (a); a flat panel of width w used in the definition of ϕ_B^e for the single-leaf structure (b); the standard shape used in the definition of ϕ_B^e for the cable (c); a cable with the same section area as the standard shape of (c) but made up of n cylindrical strands (d); the multileaf structure (e) with the same width and cross-sectional area as the panel in (b).

the second moments of area. When made into a cable of n parallel cylindrical strands, the axial stiffness remains unchanged, but the bending stiffness S falls to a value proportional to nEI, where I is the second moment of a single strand. Thus

$$S \propto n E \frac{\pi r^4}{4} = \frac{1}{4\pi} E \frac{b^4}{n} \tag{10.11}$$

The structural efficiency of the cable is therefore

$$\phi_B^e = \frac{S}{S_o} = \frac{nEI}{EI_o} = \frac{3}{n\pi} \tag{10.12}$$

The number of strands, n, can be very large, allowing the flexural stiffness to be adjusted over a large range while leaving the axial stiffness unchanged.

Multileaf assemblies allow even more dramatic anisotropy. If the thick leaf shown at (b) in the figure is divided into the stack of thin leaves of the same width shown at (e), the bending stiffness changes from

$$S_o = E w h^3/12$$

to

$$S = n E w t^3/12$$

where $t = h/n$. The shape efficiency of the stack is

$$\phi_B^e = \frac{S}{S_o} = \frac{n t^3}{h^3} = \frac{1}{n^2} \tag{10.13}$$

Thus a stack of 10 layers is 100 times less stiff than a simple leaf of the same total thickness, while the in-plane stiffness is unchanged.

Postscript When a segmented structure is bent the segments slide against each other. The analysis we have just done assumed that there was no friction opposing the sliding. A large friction prevents the sliding, and, until sliding starts, the bending stiffness is the same as that of an unsegmented bar or plate. Leaf springs are lubricated or are interleaved with Teflon shims to make sliding easy.

Related case studies

10.6 "Increasing the stiffness of steel sheet"
10.8 "Ultra-efficient springs"

10.8 ULTRA-EFFICIENT SPRINGS

Springs, we deduced in Section 6.7, store energy. They are best made of a material with a high value of σ_f^2/E, or, if mass is more important than volume, of $\sigma_f^2/\rho E$. Springs can be made more efficient still by shaping their section. Just how much more is revealed below.

We take as a measure of performance the energy stored per unit volume of solid of which the spring is made; we wish to maximize this energy. Energy per unit weight and per unit cost are maximized by similar procedures (Table 10.9).

The translation and the selection Consider a single-leaf spring first (Figure 10.8(a)). A leaf spring is an elastically bent beam. The energy stored in a bent beam, loaded by a force F, is

$$U = \frac{1}{2}\frac{F^2}{S_B} \tag{10.14}$$

where S_B, the bending stiffness of the spring, is given by Equation (9.17) or, after replacing I by ϕ_B^e, by Equation (9.18), which, repeated, is

$$S_B = \frac{C_1}{12\ L^3}\ E\,\phi_B^e A^2 \tag{10.15}$$

The force F in Equation (10.14) is limited by the onset of yield; its maximum value is

$$F_f = C_2\, Z\,\frac{\sigma_f}{L} = \frac{C_2}{6\ L}\ \sigma_f\ \phi_B^f\ A^{3/2} \tag{10.16}$$

Table 10.9 Design Requirements for Ultra-efficient Springs

Function	Material-efficient spring
Constraint	Must remain elastic under design loads
Objective	Maximum stored energy per unit volume (or mass or cost)
Free variables	Choice of material Section shape

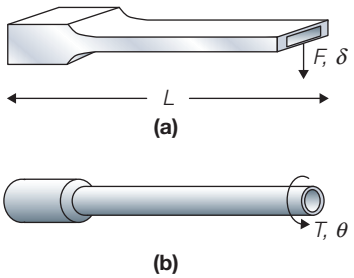


FIGURE 10.8
Hollow springs use material more efficiently than solid springs (a). Best in bending is the hollow rectangular or elliptical section; best in torsion is the tube (b).

(The constants C_1 and C_2 are tabulated in Appendix B, Tables B.3 and B.4.) Assembling these gives the maximum energy the spring can store:

$$\frac{U_{\max}}{V} = \frac{{C_2}^2}{6\,C_1}\left(\frac{(\phi_B^f\,\sigma_f)^2}{\phi_B^e\,E}\right) \qquad (10.17)$$

where $V = A\,L$ is the volume of solid in the spring. The best material and shape for the spring—the one that uses the least material—is that with the greatest value of the quantity

$$M_1 = \frac{(\phi_B^f\,\sigma_f)^2}{\phi_B^e\,E} \qquad (10.18)$$

For a fixed section shape, the ratio involving the two ϕs is a constant: The best choice of material is that with the greatest value of σ_f^2/E —the same result as before. When shape is a variable, the most efficient shapes are those with large $(\phi_B^f)^2/\phi_B^e$. Values for these ratios are tabulated for common section shapes in Table 10.10; hollow-box and elliptical sections are up to three times more efficient than solid shapes.

Torsion bars and helical springs are loaded in torsion (Figure 10.3(b)). A similar calculation gives

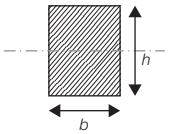
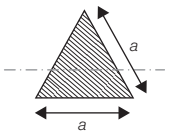
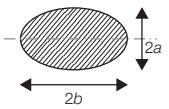
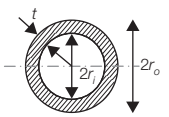
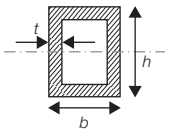
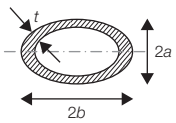
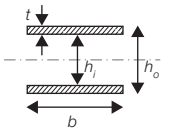
$$\frac{U_{\max}}{V} = \frac{1}{6.5}\,\frac{(\phi_T^f\,\sigma_f)^2}{\phi_T^e\,G} \qquad (10.19)$$

The most efficient material and shape for a torsional spring is that with the largest value of

$$M_2 = \frac{(\phi_T^f\,\sigma_f)^2}{\phi_T^e\,E} \qquad (10.20)$$

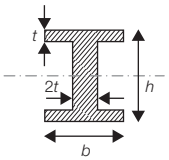
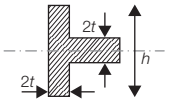
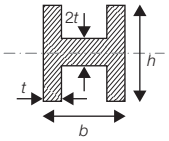
(where G has been replaced by $3E/8$). The criteria are the same: When shape is not a variable, the best torsion bar materials are those with high values of σ_f^2/E. Table 10.10 shows that the best shapes are hollow tubes, with a ratio of $(\phi_T^f)^2/\phi_T^e$ that is twice that of a solid cylinder; all other shapes are less efficient. Springs that store the maximum energy per unit weight (instead of unit volume) are selected with indices given by replacing E by $E\rho$ in Equations (10.18) and (10.20). For maximum energy per unit cost, replace E by $E\,C_m\,\rho$ where C_m is the material cost per kg.

Table 10.10 Shape Factors for the Efficiency of Springs

Section Shape	$(\phi_B^f)^2/\phi_B^e$	$(\phi_T^f)^2/\phi_T^e$
	1	$1.08\,\frac{b^2}{h^2}\,\frac{1}{\left(1+0.6\,\frac{b}{h}\right)^2\left(1-0.58\,\frac{b}{h}\right)}$
	0.38	0.83
	0.75	1.6
	0.75	$0.8\left(1+\frac{a^2}{b^2}\right)$ $(a < b)$
	1.5	3.2
	$\frac{(1+3b/h)}{(1+b/h)}$ $(h,b >> t)$	$3.32\,\frac{1}{(1-t/h)^4}$ .....$(h,b >> t)$
	$\frac{3}{4}\,\frac{(1+3b/a)}{(1+b/a)}$ $(a,b >> t)$	$3.2\,\frac{(1+a^2/b^2)}{(1+a/b)}\left(\frac{b}{a}\right)^{3/2}$ $(a,b >> t)$
	3	–

Continued

Table 10.10 Shape Factors for the Efficiency of Springs *continued*

Section Shape	$(\phi_B^f)^2/\phi_B^e$	$(\phi_T^f)^2/\phi_T^e$
	$\frac{(1+3b/h)}{(1+b/h)}$ $(h,b >> t)$	$1.07\,\frac{(1+4h/b)}{(1+h/b)}$ $(h,b >> t)$
	$1.13\,\frac{(1+4bt^2/h^3)}{(1+b/h)}$ $(h >> t)$	$0.54\,\frac{(1+8b/h)}{(1+b/h)}$ $(h,b >> t)$
	$1.13\,\frac{(1+4bt^2/h^3)}{(1+b/h)}$ $(h >> t)$	$1.07\,\frac{(1+4h/b)}{(1+h/b)}$ $(h,b >> t)$

Postscript Hollow springs are common in vibrating and oscillating devices and for instruments in which inertial forces must be minimized. The hollow elliptical section is widely used for springs loaded in bending; the hollow tube, for those loaded in torsion. More about this problem can be found in the classic paper by Boiten (1963).

Related reading: Design of efficient springs

Boiten, R.G. (1963). Mechanics of instrumentation. *Proc. I. Mech. E., 177*, 269.
A definitive analysis of the mechanical design of precision instruments.

Related case studies

6.7 "Materials for springs"
10.7 "Shapes that flex: Leaf and strand structures"

10.9 SUMMARY AND CONCLUSIONS

In designing components that are loaded such that they bend, twist, or buckle, the designer has two groups of variables with which to optimize performance: *material* and *section shape*. The best choice of material depends on the shapes in which it is available or to which it could potentially be formed. The procedure of Chapter 9 gives a method for optimizing the coupled choice of material and shape.

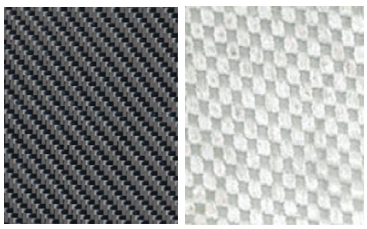
The procedure is illustrated in this chapter. Often the designer has available certain stock materials in certain shapes. If so, the one with the greatest value of the appropriate material index (of which a number were listed in Table 10.1) maximizes performance. Sometimes sections can be specially designed; then material properties and design loads determine a maximum practical value for the shape factor above which local buckling leads to failure; again, the procedure gives an optimal choice of material and shape.

Further gains in efficiency are possible by combining microscopic with macroscopic shape—something we return to later.

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CHAPTER 11

Designing Hybrid Materials



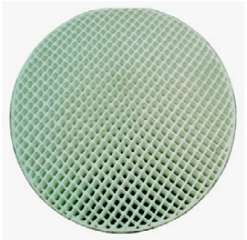
Composite structure



Sandwich structure



Segmented structure



Cellular structure

Hybrid materials.

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11.1 INTRODUCTION AND SYNOPSIS

Why do horse breeders cross a horse with a donkey, delivering a mule? Why do farmers prefer hybrid corn to the natural strain? Mules, after all, are best known for their stubbornness, and—like hybrid corn—they cannot reproduce, so you have to start again for each generation. So—why? Because, although they have some attributes that are not as good as those of their forbears, they have others—hardiness, strength, resistance to disease—that are better. The botanical phrase "hybrid vigor" sums it up.

So let us explore the idea of hybrid materials—combinations of two or more materials assembled in such a way as to have attributes not offered by either one alone (Figure 11.1, central circle). Like the mule, we may find that some attributes are less good (the cost, for example), but if the ones we want are better, something is achieved. Particulate and fibrous composites are examples of one type of hybrid, but there are many others: sandwich structures, lattice structures, segmented structures, and more. Here we explore ways of designing hybrid materials, emphasizing the choice of components, their configuration, their relative volume fraction, and their scale (Table 11.1). The new variables expand design space, allowing the creation of new "materials" with specific property profiles.

And that highlights one of the challenges. How are we to compare a hybrid like a sandwich with monolithic materials like, say, polycarbonate or titanium? To do this we must think of the sandwich not only as a hybrid

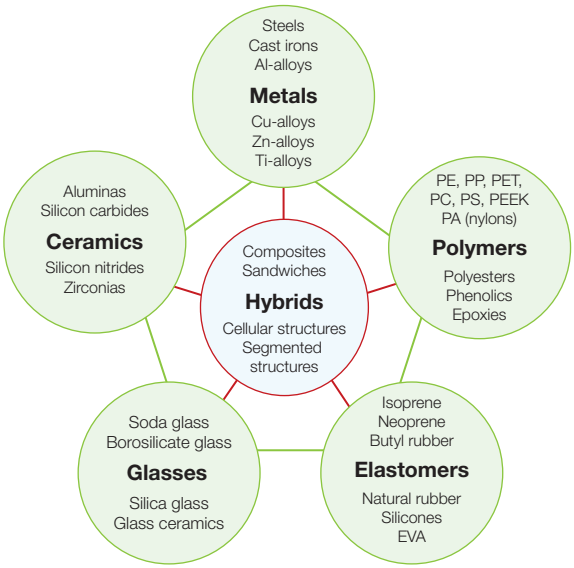


FIGURE 11.1
Hybrid materials combine the properties of two (or more) monolithic materials or of one material and space. They include fibrous and particulate composites, foams and lattices, sandwiches, and almost all natural materials.

Table 11.1 Ingredients of Hybrid Design

Components	The choice of materials to be combined
Configuration	The shape and connectivity of the components
Relative volumes	The volume fraction of each component
Scale	The length scale of the structural unit

with faces on one material bonded to a core of another, but as a "material" in its own right, with its own set of effective properties; it is these that allow the comparison.

The approach adopted here is one of breadth rather than precision. The aim is to assemble methods to allow the properties of alternative hybrids to be scanned and compared, seeking those that best meet a given set of design requirements. Once materials and configuration have been chosen, standard methods such as optimization routines and finite element analyses can be used to refine them. But what the standard methods are *not* good at is the quick scan of alternative combinations. That is where the approximate methods developed below, in which material and configuration become the variables, pay off.

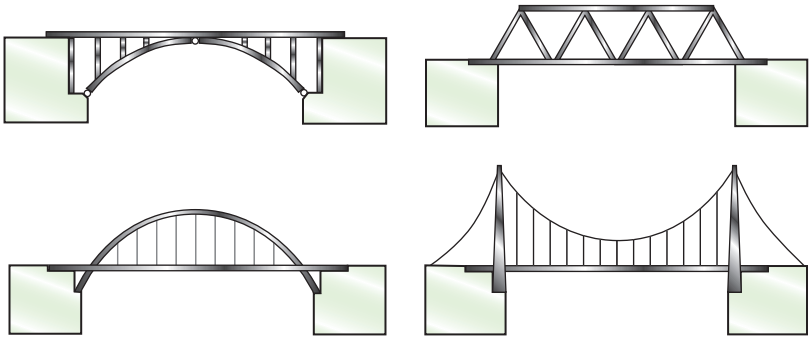


FIGURE 11.2
Four configurations for a bridge. The design variables describing the performance of each differ. Optimization of performance becomes possible only when a configuration has been chosen.

The word "configuration" requires elaboration. Figure 11.2 shows four different configurations of bridge. In the first all members are loaded in compression. In the second, members carry both tension and compression, depending on how the bridge is loaded. In the third and fourth, the suspension cables are loaded purely in tension. Any one of these can be optimized, but no amount of optimization will cause one to evolve into another because this involves a discrete jump in configuration, each characterized by its own set of variables.[1]

Hybrid design has the same feature: The classes of hybrid are distinguished by their configuration. Here we focus on four classes, each with a number of discrete members. The images on the first page of this chapter suggest what they look like. To avoid a mouthful of words every time we refer to one, we use the shorthand: *composite*, *sandwich*, *cellular*, and *segmented structures*. *Composites* combine two solid components, one (the reinforcement) as fibers or particles, contained in the other (the matrix). Their properties are some average of those of the components, and, on a scale large compared to that of the reinforcement, they behave as if they were homogeneous materials. *Sandwiches* have outer faces of one material supported by a core of another, usually a low-density material—a configuration that gives a flexural stiffness per unit weight that is greater than that offered by either component alone.

[1] Numerical tools are emerging that allow a degree of *topological optimization*, that is, the development of a configuration. They work like this. Start with an envelope—a set of boundaries—and fill it with a homogenous "material" with a relative density initially set at 0.5, with properties that depend linearly on relative density. Impose constraints, meaning the mechanical, thermal, and other loads the structure must support, give it a criterion of excellence, and let it condense out into regions of relative density 1 and regions where it is 0 retaining only changes that increase the measure of excellence. The method is computationally intensive but has had some success in suggesting configurations that use a material efficiently (see "Further reading" for more information).

Cellular structures are combinations of material and space (which can, of course, be filled with another material). We distinguish two types: In the first, the low connectivity of the struts allows them to bend when the lattice is loaded; in the other, the higher connectivity suppresses bending, forcing the struts to stretch. *Segmented structures* are materials subdivided in one, two, or three dimensions; the subdivisions both lower the stiffness and, by dividing the material into discrete units, impart damage tolerance.

The approach we adopt is to use bounding methods to estimate the properties of each configuration. With these, the properties of a given pair of materials in a given configuration can be calculated. These can then be plotted on material selection charts, which become tools for selecting both configuration and material.

11.2 HOLES IN MATERIAL-PROPERTY SPACE

All the charts of Chapter 4 have one thing in common: Parts of them are populated with materials and parts are not—there are *holes* (Figure 11.3). Some parts are inaccessible for fundamental reasons that relate to the size of atoms and the nature of the forces that bind them together. But other parts are empty even though, in principle, they could be filled.

Is anything to be gained by developing materials (or material combinations) that lie in these holes? The material indices show where this is profitable. A grid of lines of one index—E/ρ—is plotted in Figure 11.3. If the filled areas can be expanded in the direction of the arrow (i.e., to greater values of E/ρ) it will enable lighter, stiffer structures to be made. The arrow lies normal to the index lines. It defines a *vector for material development*.

One approach to filling holes—the long-established one—is that of developing new metal alloys, new polymer chemistries, and new compositions of glass and ceramic so as to extend the populated areas of the property charts; Figure 1.2 illustrated how this has evolved over time. But developing new materials can be an expensive and uncertain process, and the gains tend to be incremental rather than step-like. An alternative is to combine two or more existing materials so as to allow a superposition of their properties—in short, to create hybrids. The spectacular success of carbon and glass-fiber–reinforced composites at one extreme, and of foamed materials at another (hybrids of material and space) in filling previously empty areas of the property charts is encouragement enough to explore ways in which such hybrids can be designed.

What might we hope to achieve? Figure 11.4 shows two materials, M_1 and M_2, plotted on a chart with properties P_1 and P_2 as axes. The figure shows four scenarios, each typical of a certain class of hybrid. Depending on the

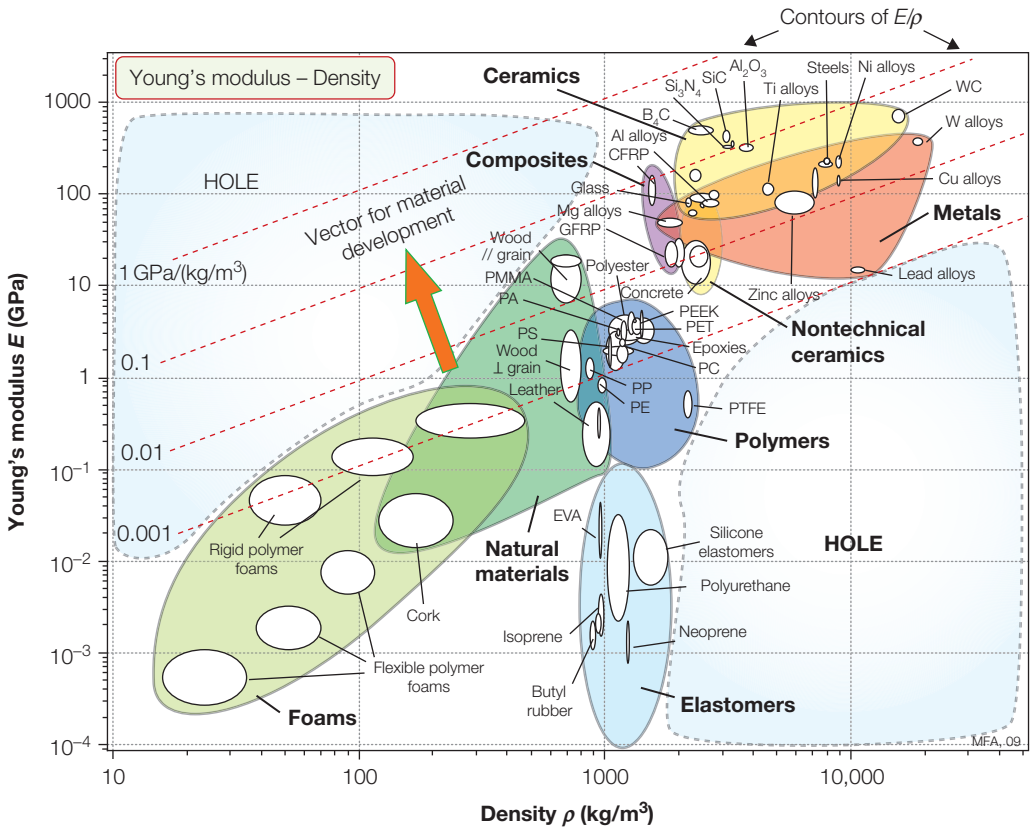


FIGURE 11.3

Holes in modulus-density space, with contours of specific modulus E/ρ. Material development that extended the occupied territory in the direction of the arrow (the "vector for development") allows components with greater stiffness to weight than any current material allows.

configuration of the materials and the way they are combined, we may find any one of the following.

- *"The best of both" scenario.* The ideal is often the creation of a hybrid with the best properties of both components. There are examples, most commonly when a bulk property of one material is combined with the surface properties of another. Zinc-coated steel has the strength and toughness of steel with the corrosion resistance of zinc. Glazed pottery exploits the formability and low cost of clay with the impermeability and durability of glass.
- *"The rule of mixtures" scenario.* When bulk properties are combined in a hybrid, as in structural composites, the best that can be obtained is often the arithmetic average of the properties of the components, weighted by

their volume fractions. Thus unidirectional fiber composites have an axial modulus (the one parallel to the fibers) that lies close to the rule of mixtures.

- *"The weaker link dominates" scenario*. Sometimes we have to live with a lesser compromise, typified by the stiffness of particulate composites, in which the hybrid properties fall below those of a rule of mixtures, lying closer to the harmonic than the arithmetic mean of the properties. Although the gains are less spectacular, they are still useful.
- *"The least of both" or weakest-link scenario*. Sometimes it is not the greatest but the least of the properties that we seek. Fire sprinkler systems, for example, use a wax-metal hybrid designed to fail, releasing the spray when the melting point of the lower-melting material (the wax) is exceeded.

These set certain fixed points, but the list is not exhaustive. Other combinations are possible. These will emerge in the following.

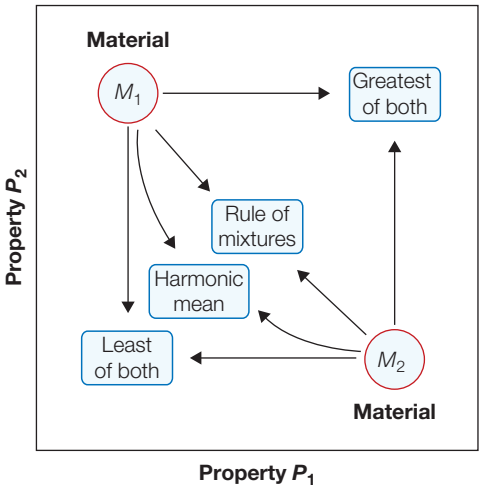


FIGURE 11.4
The possibilities of hybridization. The properties of the hybrid reflect those of its component materials, combined in one of several possible ways.

When is a hybrid a "material"? There is a certain duality in how hybrids are considered and discussed. Some, like filled polymers, composites, or wood are treated as materials in their own right, each characterized by its own set of material properties. Others, such as galvanized steel, are seen as one material (steel) to which a coating of a second (zinc) has been applied, even though it could be regarded as a new material with the strength of steel but the surface properties of zinc ("stinc," perhaps?). Sandwich panels illustrate the duality, sometimes viewed as two sheets of face material separated by a core material, and sometimes—to allow comparison with bulk materials—as "materials" with their own density, flexural stiffness, and strength. To call any one of these a "material" and to characterize it as such is a useful shorthand, allowing designers to use existing methods when designing with them. But if we are to design the hybrid itself, we must deconstruct it and think of it as a combination of materials (or of material and space) in a chosen configuration.

11.3 THE METHOD: "A + B + CONFIGURATION + SCALE"

First, a working definition: A hybrid material is a combination of two or more materials in a predetermined configuration, relative volume, and scale, optimally serving a specific engineering purpose," which we paraphrase as "A + B + configuration + scale." Here we allow for the widest possible choice of A and

B, including the possibility that one of them is a gas or simply space. These new design variables expand design space, allowing an optimization of properties that is not possible if the choice is limited to single, monolithic materials.

The basic idea, illustrated in Figure 11.5, is this. Monolithic materials offer a certain portfolio of properties on which much engineering design is based. Design requirements isolate a sector of material-property space. If this sector contains materials, the requirements can be met by choosing one of them. But if the design requirements are sufficiently demanding, no single material may be found that can meet them all: The requirements lie in a hole in property space. Then the way forward is to identify and separate the conflicting requirements, seeking optimal material solutions for each, and then combine them in ways that retain the desirable attributes of both. The best choice is the one that ranks most highly when measured by the performance metrics that motivate the design: minimizing mass or cost or maximizing some aspect of performance (criterion of excellence). The alternative combinations are examined and assessed, using the criterion of excellence to rank them. The output is a specification of a hybrid in terms of its component materials and their configurations.

The four classes of hybrid structures—composites, sandwiches, cellular, and segmented—are explored in the next four sections, applying the method.

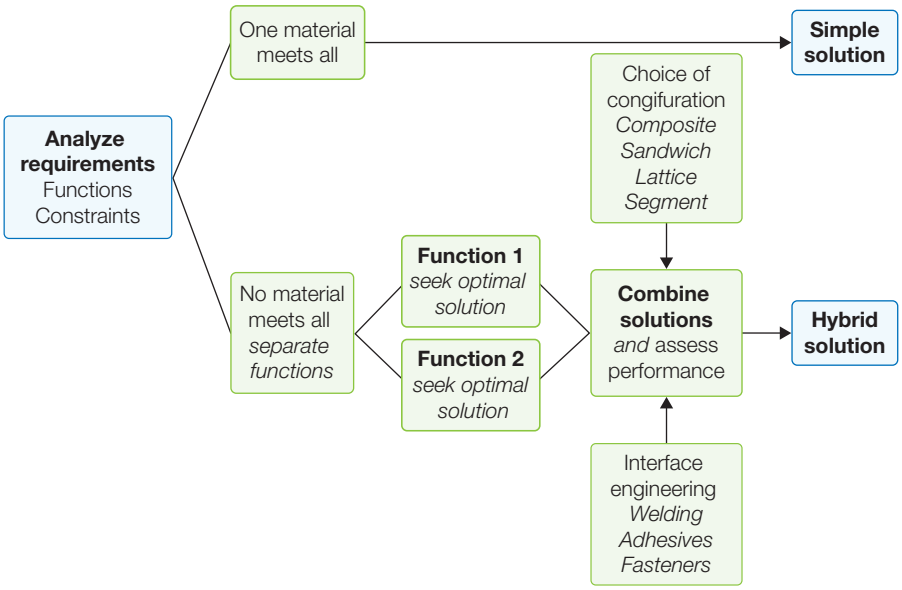


FIGURE 11.5
The steps in designing a hybrid to meet given design requirements.

11.4 COMPOSITES

Aircraft engineers, automobile makers, and designers of sports equipment all have one thing in common: They all want materials that are stiff, strong, tough, and light. The single-material choices that best achieve this are the *light alloys*: alloys that are based on magnesium, aluminum, and titanium. Much research aims at improving their properties. But they are not all that light—polymers have much lower densities. Nor are they all that stiff—ceramics are much stiffer and, particularly in the form of small particles or thin fibers, much stronger. These facts are exploited in the family of structural hybrids that we usually refer to as *particulate and fibrous composites*.

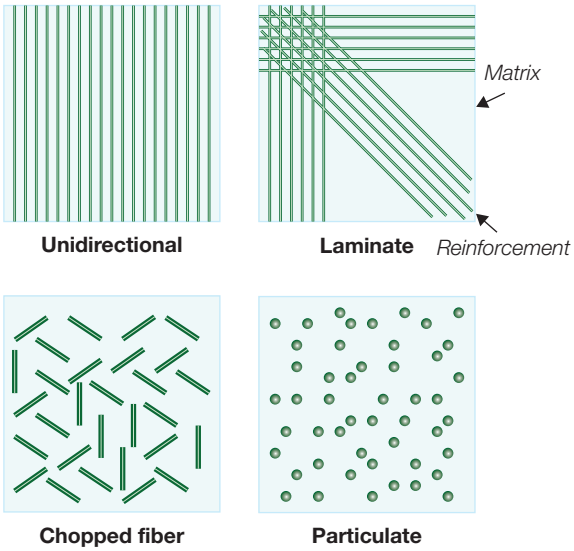


FIGURE 11.6
Schematic of hybrids of the composite type: unidirectional fibrous, laminated fiber, chopped fiber, and particulate composites. Bounds and limits, described in the text, bracket the properties of all of these.

Any two materials can, in principle, be combined to make a composite, and they can be mixed in many geometries (Figure 11.6). In this section we restrict the discussion to *fully dense, strongly bonded composites* such that there is no tendency for the components to separate at their interfaces when the composite is loaded, and to those in which the scale of the reinforcement is large compared to that of the atom or molecule size and the dislocation spacing, allowing the use of continuum methods.

On a macroscopic scale—one that is large compared to that of the components—a composite behaves like a homogeneous solid with its own set of mechanical, thermal, and electrical properties. Calculating these precisely can be done, but is difficult. It is much easier to bracket them by *bounds* or *limits*: upper and lower values between which the properties lie. The term "bound" will be used to describe a rigorous boundary, one that the value of the property *cannot*—subject to certain assumptions—exceed or fall below. It is not always possible to derive bounds; the best that can be done is to derive "limits" outside which it is *unlikely* that the value of the property lies. The important point is that the bounds orlimits bracket the properties of *all* the configurations of matrix and reinforcement shown in Figure 11.6; by using them we escape from the need to model individual geometries.

Criteria of excellence We need criteria of excellence to assess any given hybrid's merit. These are provided by the material indices of Chapter 5. If a possible hybrid has a value of any one of these that exceeds those of existing materials, it achieves our goal.

Density When a volume fraction f of a reinforcement r (density ρ_r) is mixed with a volume fraction $(1-f)$ of a matrix m (density ρ_m) to form a composite with no residual porosity, the composite density $\widetilde{\rho}$ is given exactly by a rule of mixtures (an arithmetic mean weighted by volume fraction):

$$\widetilde{\rho} = f\,\rho_r + (1-f)\rho_m. \tag{11.1}$$

The geometry or shape of the reinforcement does not matter except in determining the maximum packing fraction of reinforcement and thus the upper limit for f.

Modulus The modulus of a composite is bracketed by the well-known Voigt and Reuss bounds. The upper bound, $\widetilde{E}_u$, is obtained by postulating that, on loading, the two components suffer the same strain; the stress is then the volume average of the local stresses, and the composite modulus follows a rule of mixtures:

$$\widetilde{E}_u = f\,E_r + (1-f)E_m. \tag{11.2a}$$

Here E_r is the Young's modulus of the reinforcement and E_m that of the matrix. The lower bound, $\widetilde{E}_L$, is found by postulating instead that the two components carry the same stress; the strain is the volume average of the local strains, and the composite modulus is

$$\widetilde{E}_L = \frac{E_m\,E_r}{f\,E_m + (1-f)E_r} \tag{11.2b}$$

More precise bounds are possible, but the simple ones are adequate to illustrate the method.

To see how the bounds are used, consider the following example.

Composite design for stiffness at minimum mass

A beam is at present made of an aluminum alloy. Beryllium is both lighter and less dense than aluminum; the ceramic alumina (Al_2O_3) is also stiffer but denser. Can hybrids of aluminum with either of these offer improved performance, measured by the criterion of excellence $E^{1/2}/\rho$ derived in Chapter 5?

The three materials are plotted on a segment of the $E-\rho$ property space in Figure 11.7. Composites made by mixing them have densities given exactly by Equation (11.1) and moduli that are bracketed by the bounds of Equations (11.2a) and (11.2b). Both of these moduli depend on the volume fraction of reinforcement and, through this, on the density. Upper and lower bounds for the modulus-density relationship can thus be plotted on the $E-\rho$ chart using volume fraction f as a parameter, as shown in Figure 11.7. Any composite made by combining aluminum

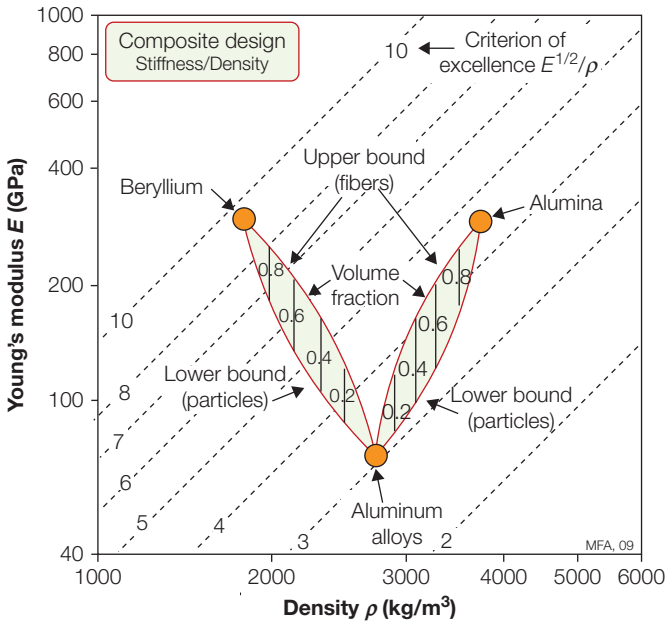


FIGURE 11.7
Part of the $E - \rho$ property chart, showing aluminum alloys, beryllium, and alumina (Al_2O_3). Bounds for the moduli of hybrids made by mixing them are shown. The diagonal contours plot the criterion of excellence $E^{1/2}/\rho$.

with alumina will have a modulus contained in the envelope for Al–Al_2O_3; the same for Al-Be. Fibrous reinforcement gives a modulus in a direction parallel to the fibers near the upper bound; particulate reinforcement or transversely loaded fibers give moduli near the lower one.

The criterion of excellence, $E^{1/2}/\rho$, is plotted as a grid in the Figure 11.7. The bound-envelope for Al–Be composites extends almost normal to the grid, while that for Al–Al_2O_3 lies at a shallow angle to it. Beryllium fibers improve performance (as measured by $E^{1/2}/\rho$) roughly four times as much as alumina fibers do, for the same volume fraction. The difference for particulate reinforcement is even more dramatic. The lower bound for Al-Be lies normal to the contours: 30% of particulate beryllium increases $E^{1/2}/\rho$ by a factor of 1.5. The lower bound for Al–Al_2O_3 is, initially, parallel to the $E^{1/2}/\rho$ grid: 30% of particulate Al_2O_3 gives almost no gain. The underlying reason is that both beryllium and Al_2O_3 increase the modulus, but only beryllium decreases the density; the criterion of excellence is more sensitive to density than to modulus.

The commercial alloy AlBeMet (62% Be, 38% Al) exploits this idea. The two metals are mutually insoluble, creating a 2-phase composite of Al and Be with $E^{1/2}/\rho$ = 6.5 compared with 3.1 for Al alone.

Strength Of all the bounds and limits described in this chapter, those for strength are the least satisfactory. The nonlinearity of the problem, the

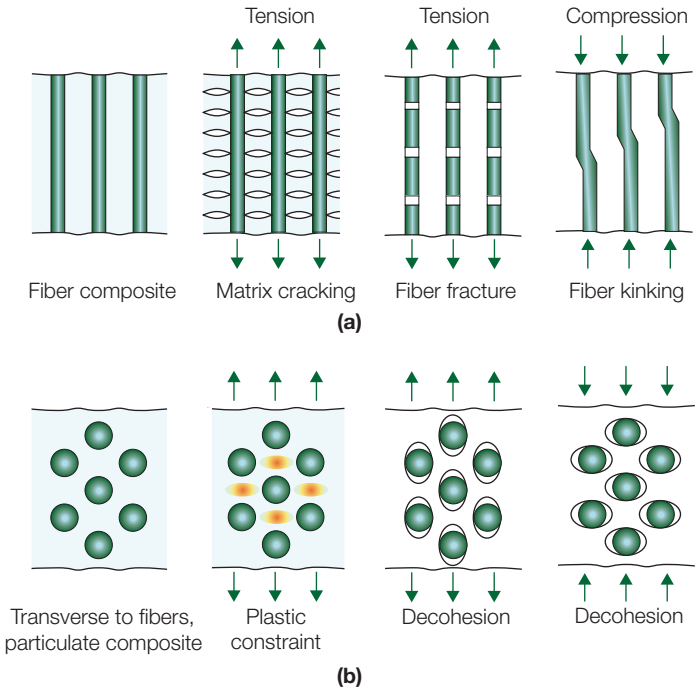


FIGURE 11.8
Failure modes in composites.

multitude of failure mechanisms[2] and the sensitivity of strength and toughness against impurities and processing defects makes accurate modeling difficult. The literature contains many calculations for special cases: reinforcement by unidirectional fibers or by a dilute dispersion of spheres. We wish to avoid models that require detailed knowledge of how a particular architecture behaves, and we seek less restrictive limits.

As the load on a continuous fiber composite is increased, it is redistributed between the components until one suffers general yield or fracture (Figure 11.8(a)). Beyond this point the composite has suffered permanent deformation or damage but can still carry load; final failure requires yielding or fracture of both. The composite is strongest if both reach their failure state simultaneously. Thus the upper bound for a continuous fiber ply like that labeled "Unidirectional" in Figure 11.6 loaded parallel to the fibers (the axial strength in tension, subscript a) is a rule of mixtures:

$$(\tilde{\sigma}_f)_{u,a} = f(\sigma_f)_r + (1-f)(\sigma_f)_m \qquad (11.3a)$$

[2] The treatment here is the simplest that allows the method to be demonstrated. The full complexity of the modeling of composite failure is documented in the texts in "Further reading."

where $(\sigma_f)_m$ is the strength of the matrix and $(\sigma_f)_r$ is that of the reinforcement. If one fails before the other, the load is carried by the survivor. Thus a lower bound for strength in tension is given by

$$(\tilde{\sigma}_f)_{L,a} = \text{Greater of } (f(\sigma_f)_r, (1-f)(\sigma_f)_m) \tag{11.3b}$$

Creating anisotropy

The elastic and plastic properties of bulk monolithic solids are frequently anisotropic, but weakly so—the properties do not depend strongly on direction. Hybridization gives a way of creating controlled anisotropy, and it can be large. We have already seen an example in Figure 11.7, which shows the upper and lower bounds for the moduli of composites. The longitudinal properties of unidirectional long-fiber composites lie near the upper bound; the transverse properties, near the lower one. The vertical width of the band between them measures the anisotropy. A unidirectional continuous-fiber composite has a maximum anisotropy ratio R_a given by the ratio of the bounds—in this example

$$R_a = \frac{\tilde{E}_u}{\tilde{E}_L} = f^2 + (1-f)^2 + f(1-f)\left(\frac{E_m}{E_r} + \frac{E_r}{E_m}\right)$$

In Figure 11.7 the maximum R_a is only 1.5. A more dramatic example involving thermal properties is given in Chapter 12.

Determining the transverse strength (Figure 11.8(b)) is more difficult. It depends on interface bond strength, fiber distribution, stress concentrations, and voids. In general the transverse strength is less than that of the unreinforced matrix, and the strain to failure, too, is less. In a continuous ductile matrix containing strongly bonded, nondeforming, particles or fibers, the flow in the matrix is constrained. The constraint increases the stress required for flow in the matrix, giving an upper-limiting tensile strength of

$$\text{Lesser of} \begin{cases} (\tilde{\sigma}_f)_{u,t} \approx (\sigma_f)_m \left(\dfrac{1}{1-f^{1/2}}\right) \\ (\tilde{\sigma}_f)_{u,t} \approx (\sigma_f)_r \end{cases} \tag{11.4a}$$

More usually the transverse strength is lower than that of the matrix alone because of stress concentration and debonding at the fiber-matrix interface. Hull gives the approximate lower limit for tensile strength as

$$(\tilde{\sigma}_f)_{L,t} \approx (\sigma_f)_m (1-f^{1/2}) \tag{11.4b}$$

The two pairs of limits allow the potential of a given choice of reinforcement and matrix to be explored. Figure 11.9 shows the limits for axial and transverse strength of an epoxy-glass composite ply.

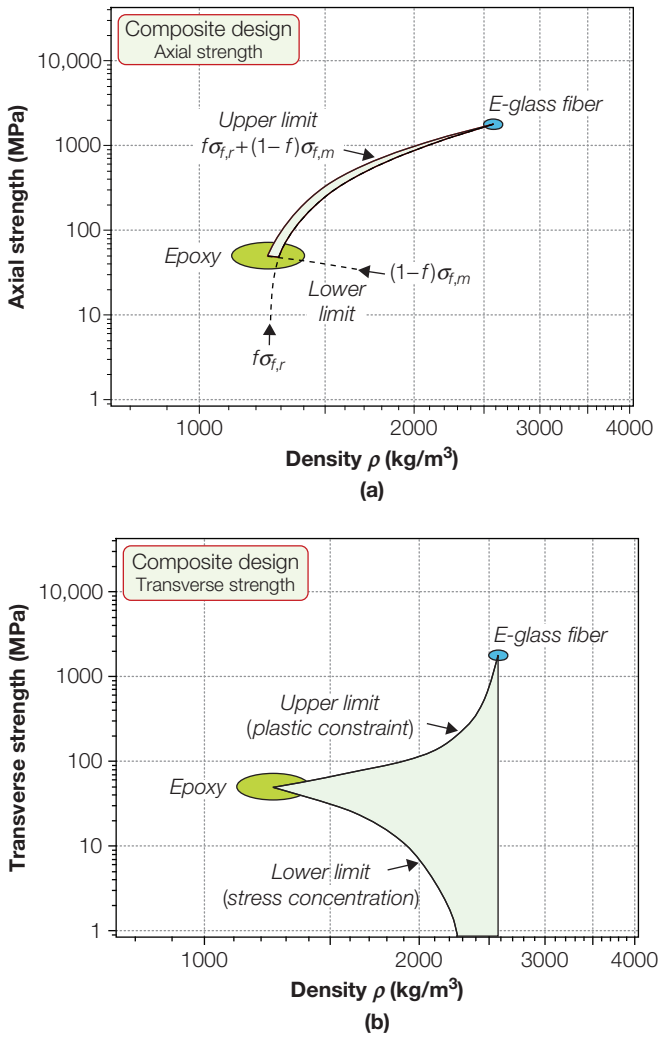


FIGURE 11.9
The limits for axial (a) and transverse (b) strength of a composite ply.

Continuous fiber composites can fail in compression by fiber kinking (Figure 11.8(a), extreme right). The kinking is resisted by the shear strength of the matrix, approximately $(\sigma_f)_m/2$. This leads an axial compressive stress for buckling of fibers of

$$(\widetilde{\sigma}_c)_{u,a} = \frac{1}{\vartheta}\frac{(\sigma_f)_m}{2} \approx 14(\sigma_f)_m \qquad (11.5)$$

Here ϑ is the initial misalignment of the fibers from the axis of compression, in radians. Experiments show that a typical value, in carefully aligned

composites, is $\vartheta \approx 0.035$, giving the final value shown on the right of the equation. We identify the upper bound with the lesser of this and Equation (11.3a). When, instead, misalignment is severe, meaning $\vartheta \approx 1$, the strength falls greatly. We identify the lower bound for compressive failure with that of the matrix, $(\sigma_f)_m$.

Specific heat The specific heats of solids at constant pressure, C_p, are almost the same as those at constant volume, C_v. If they were identical, the heat capacity per unit volume of a composite would, like the density, be given exactly by a rule-of-mixtures:

$$\widetilde{\rho}\,\widetilde{C}_p = f\,\rho_r(C_p)_r + (1-f)\rho_m(Cp)m \tag{11.6}$$

where $(C_p)_r$ is the specific heat of the reinforcement and $(C_p)_m$ is that of the matrix (the densities enter because the units of C_p are J/kg.K). A slight difference appears because thermal expansion generates a misfit between the components when the composite is heated; the misfit creates local pressures on the components and thus changes the specific heat. The effect is very small and need not concern us further.

Thermal expansion coefficient The thermal expansion of a composite can, in some directions, be greater than that of either component; in others, less. This is because an elastic constant—Poisson's ratio—couples the principal elastic strains; if the matrix is prevented from expanding in one direction (by embedded fibers, for instance), then it expands more in the transverse directions. For simplicity we shall use the approximate lower bound:

$$\widetilde{\alpha}_L = \frac{E_r\alpha_r f + E_m\alpha_m(1-f)}{E_r f + E_m(1-f)} \tag{11.7}$$

(it reduces to the rule of mixtures when the moduli are the same) and the upper bound:

$$\widetilde{\alpha}_u = f\alpha_r(1+\nu_r) + (1-f)\alpha_m(1+\nu_m) - \alpha_L[f\nu_r + (1-f)\nu_m] \tag{11.8}$$

where α_r and α_m are the two expansion coefficients and ν_r and ν_m are the Poisson's ratios.

Thermal conductivity Thermal conductivity determines heat flow at a steady rate. A composite of two materials, bonded to give good thermal contact, has a thermal conductivity λ that lies between those of the individual components, λ_m and λ_r. Not surprisingly, a composite containing parallel continuous fibers has a conductivity, parallel to the fibers, given by a rule of mixtures:

$$\widetilde{\lambda}_u = f\lambda_r + (1-f)\lambda_m \tag{11.9}$$

This is an upper bound: In any other direction the conductivity is lower. The transverse conductivity of a parallel-fiber composite (again assuming good

bonding and thermal contact) lies near the lower bound first derived by Maxwell:

$$\tilde{\lambda}_L = \lambda_m \left(\frac{\lambda_r + 2\lambda_m - 2f(\lambda_m - \lambda_r)}{\lambda_r + 2\lambda_m + f(\lambda_m - \lambda_r)} \right) \tag{11.10}$$

Particulate composites, too, have a conductivity near this lower bound. Poor interface conductivity can make λ drop below it. Debonding or an interfacial layer between reinforcement and matrix can cause this; so, too, can a large difference of modulus between reinforcement and matrix (because this reflects phonons, creating an interface impedance) or a structural scale that is shorter than the phonon wavelengths.

Thermal diffusivity The thermal diffusivity

$$a = \frac{\lambda}{\rho C_p}$$

determines heat flow when conditions are transient, that is, when the temperature field changes with time. It is formed from three of the earlier properties: λ, ρ, and C_p. The second and third of these are given exactly by Equations (11.1) and (11.6), allowing the diffusivity to be expressed as

$$\tilde{a} = \frac{\tilde{\lambda}}{f\rho_r (C_p)_r + (1-f)\rho_m (C_p)_m} \tag{11.11}$$

Its upper and lower bounds are found by substituting those for $\tilde{\lambda}$ (Equations (11.9) and (11.10)) into this equation.

Composite design for controlled thermal response

Thermo-mechanical design involves the specific heat, C_p, the thermal expansion, α, the conductivity, λ, and the diffusivity, a. These composite properties are bounded by Equations (11.6) through (11.11). They are involved in a number of indices. One is the criterion for minimizing thermal distortion derived in Section 6.16: maximizing the index λ/α.

Figure 11.10 shows a small part of the $\alpha - \lambda$ materials selection chart, with a grid showing the criterion of excellence, λ/α, superimposed on it. Three materials are shown: aluminum, boron nitride (BN), and silicon carbide (SiC). The thermal properties of Al–BN and Al–SiC composites are bracketed by envelopes calculated from the bounding equations. (Both α and λ have upper and lower bounds so there are four possible combinations for each material pair. Those shown in the figure are the outermost pair of the four.) The plot reveals immediately that SiC reinforcement in aluminum increases performance (as measured by λ/α); reinforcement with BN decreases it.

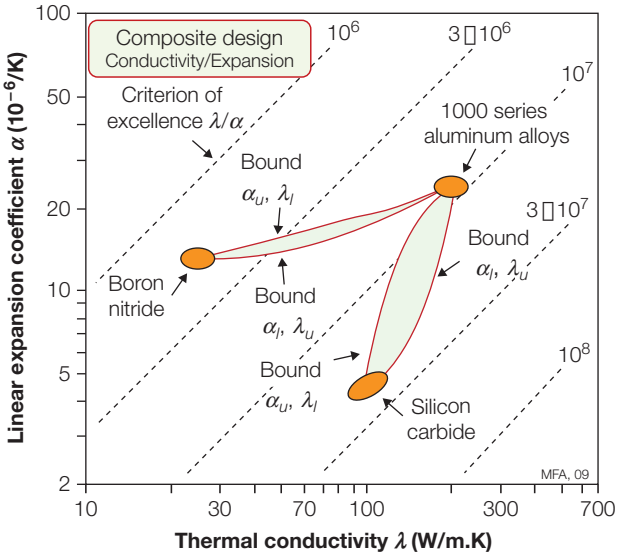


FIGURE 11.10
A part of expansion-coefficient/conductivity space showing aluminum alloys, boron nitride, and silicon carbide. The properties of Al-BN and Al-SiC composites are bracketed by the bounds of Equations (11.7)–(11.10). The Al-SiC composites enhance performance; the Al-BN composites reduce it.

Dielectric constant The dielectric constant $\widetilde{\varepsilon}_d$ is given by a rule of mixtures:

$$\widetilde{\varepsilon}_d = f\varepsilon_{d,r} + (1-f)\varepsilon_{d,m} \tag{11.12}$$

where $\varepsilon_{d,r}$ is the dielectric constant of the reinforcement and $\varepsilon_{d,m}$ that of the matrix.

Electrical conductivity and percolation When the electrical conductivities κ of the components of a composite are of comparable magnitude, bounds for the electrical conductivity are given by those for thermal conductivity with λ replaced by κ. When, instead, they differ by many orders of magnitude (a metallic powder dispersed in an insulating polymer, for instance), questions of *percolation* arise. Percolation is discussed in Section 12.3.

Filling property space with composites We end this section with two figures illustrating how the development of composites has filled out the occupancy of material-property space. The first, Figure 11.11, is a modulus-density ($E-\rho$) section. The areas filled by unreinforced metals and polymers appear as pale red and pale blue envelopes; members are labeled in gray (they are taken from the original $E-\rho$ chart in Figure 4.3). Polymer matrix composites occupy the bolder purple zone; metal matrix composites, the

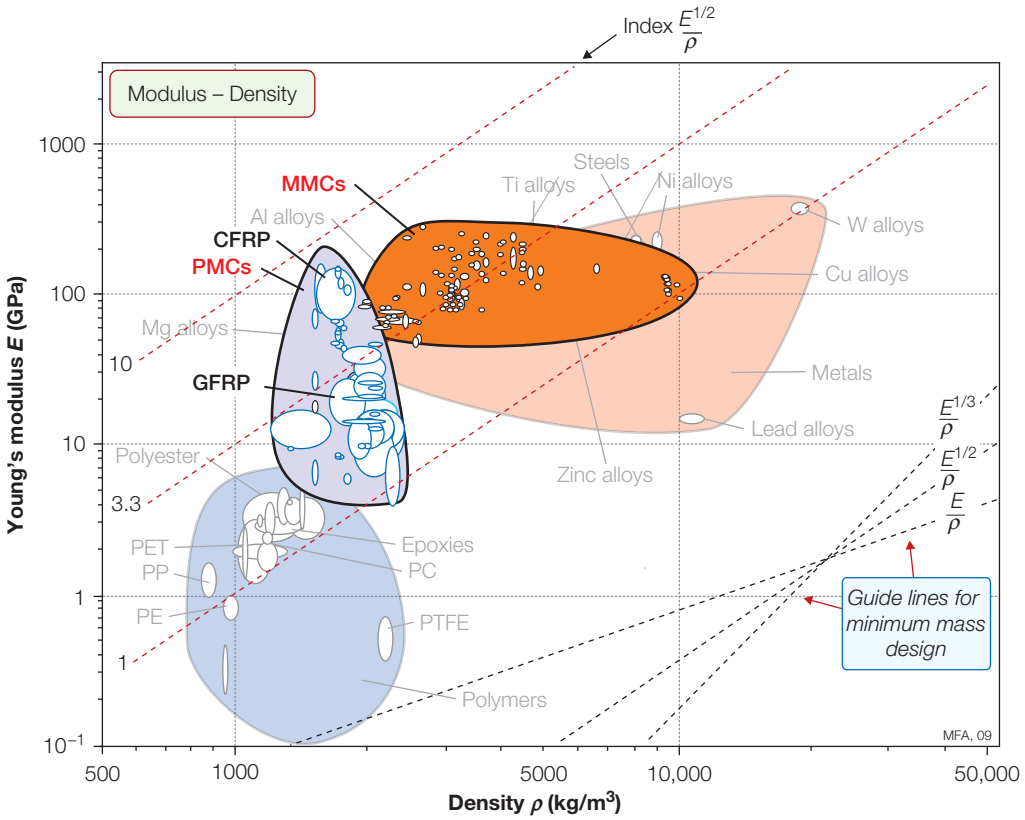


FIGURE 11.11

Polymer (PMC) and metal (MMC) matrix composites expand the occupied area of modulus-density space. (Each of the small bubbles in the bold envelopes labeled PMCs and MMCs describe a composite. Data from the CES Edu '09 database.)

darker red one. Both extend into areas that were previously empty. Using any one of the indices for light, stiff structures (E/ρ, $E^{1/2}/\rho$ and $E^{1/3}/\rho$) as a criterion of excellence, we find that composites offer performance that was not previously attainable.

Figure 11.12 paints a similar picture for the strength-density ($\sigma_y - \rho$) section. The color coding is the same as that in Figure 11.11. Composites again expand the populated area in a direction that, using the indices for light strong structures (σ_y/ρ, $\sigma_f^{2/3}/\rho$, and $\sigma_f^{1/2}/\rho$) as a criterion, offers enhanced performance.

11.5 SANDWICH STRUCTURES

A sandwich panel epitomizes the concept of a hybrid. It combines two materials in a specified geometry and scale, configured such that one forms the faces and the other the core to give a structure of high bending stiffness

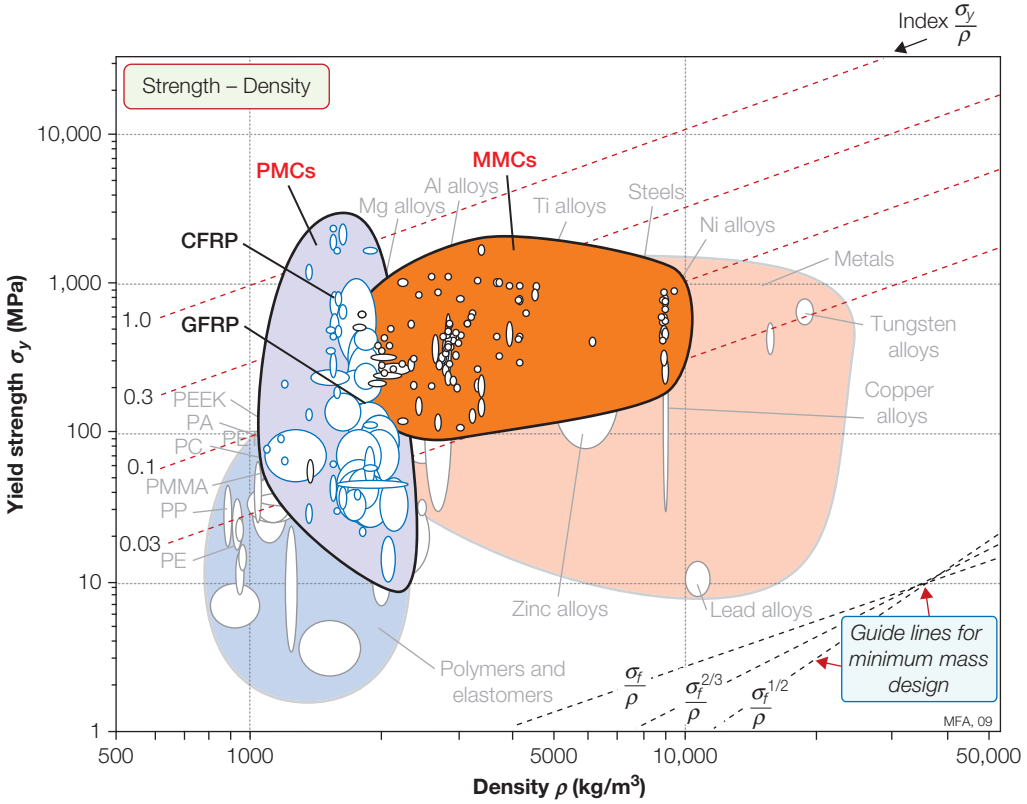


FIGURE 11.12
Polymer (PMC) and metal (MMC) matrix composites also expand the occupied area of strength-density space. (Each of the small bubbles in the bold envelopes labeled PMCs and MMCs describe a composite. Data from the CES Edu '09 database.)

and strength at low weight Figure 11.13). The separation of the faces by the core increases the moment of inertia of the section, I, and its section modulus, Z, producing a structure that resists bending and buckling loads well. Sandwiches are used where weight saving is critical: in aircraft, trains, motor vehicles, portable structures, and sports equipment. Nature, too, makes use of sandwich designs: Sections through the human skull, the wing of a bird, and the stalk and leaves of many plants show a low-density foam-like core separating solid faces.[3]

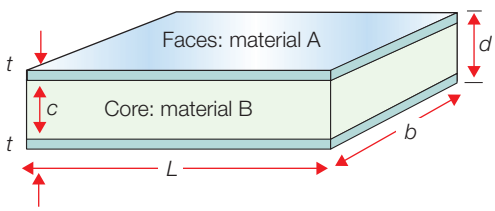


FIGURE 11.13
The sandwich. The face thickness is t, the core thickness is c, and the panel thickness is d.

[3] Allen (1969) and Zenkert (1995) give good introductions to the design of sandwich panels for engineering applications. Gibson et al. (2010) do the same for their use in nature.

Table 11.2 The Symbols

Symbol	Meaning and Usual Units
t, c, d	Face thickness, core thickness, and overall panel thickness (m)
L, b	Panel length and width (m)
m_a	Mass per unit area of panel (kg/m^2)
$f = 2t/d$	Relative volumes occupied by faces
$(1 - f) = c/d$	Relative volume occupied by core
I	Second moment of area (m^4)
ρ_f, ρ_c	Densities of face and core material (kg/m^3)
$\widetilde{\rho}$	Equivalent density of panel (kg/m^3)
E_f	Young's modulus of faces (GN/m^2)
E_c, G_c	Young's modulus and shear modulus of core (GN/m^2)
$\widetilde{E}_{in\text{-}plane}$, $\widetilde{E}_{flex}$	Equivalent in-plane and flexural modulus of panel (GN/m^2)
σ_f	Yield strength of faces (MN/m^2)
σ_c, τ_c	Yield strength and shear-yield strength of core (MN/m^2)
$\widetilde{\sigma}_{in\text{-}plane}$	Equivalent in-plane strength of panel (MN/m^2)
$\widetilde{\sigma}_{flex1}, \widetilde{\sigma}_{flex2}, \widetilde{\sigma}_{flex3}$	Equivalent flexural strength of panel, depending on mechanism of failure (MN/m^2)

The faces, each of thickness t, carry most of the load, so they must be stiff and strong; because they form the exterior surfaces of the panel they must also tolerate the environment in which it operates. The core, of thickness c, occupies most of the volume; it must be light and stiff and strong enough to carry the shear stresses necessary to make the whole panel behave as a load-bearing unit, (if the core is much thicker than the faces, these stresses are small).

A sandwich as a "material" So far we have spoken of the sandwich as a *structure*: faces of material A supported on a core of material B, each with its own density, modulus, and strength. But we can also think of it as a *material* with its own set of properties, and this is useful because it allows comparison with more conventional materials. To do this we calculate *equivalent material properties* for the sandwich, and we identify them, like composites, by a tilde (e.g., $\widetilde{\rho}$, $\widetilde{E}$). The quantities $\widetilde{\rho}$ and $\widetilde{E}$ can be plotted on the modulus-density chart, allowing a direct comparison with all the other materials on the chart. All the constructions using material indices apply unchanged. The symbols that appear in this section are defined in Table 11.2.

Finding equivalent properties of structured materials by experiment

Consider a sandwich structure with solid skins separated by a cellular core. The panel has an equivalent density equal to its mass divided by its volume,

m_a/d, where m_a is its mass per unit area and $d = 2t + c$ is its overall thickness. It has a flexural stiffness EI, measured by loading the panel in bending and recording the deflection. We define an equivalent homogeneous material with $\widetilde{\rho} = \rho$ and $\widetilde{E}\widetilde{I} = EI$, where $\widetilde{I} = bd^3/12$ is the second moment of area for a *homogeneous* panel with the same dimensions as the real one. The equivalent density and modulus are then

$$\widetilde{\rho} = \frac{m_a}{d} \tag{11.13}$$

$$\widetilde{E} = \frac{12\,EI}{b\,d^3} \tag{11.14}$$

Loading the panel to failure allows the failure moment, M_f, to be measured experimentally. It is then possible to define an equivalent flexural strength via $\widetilde{Z}\widetilde{\sigma}_{flex} = M_f$, where $\widetilde{Z} = bd^2/4$ is the (fully plastic) section modulus of the panel. The equivalent flexural strength is then

$$\widetilde{\sigma}_{flex} = \frac{4\,M_f}{b\,d^2} \tag{11.15}$$

A brief example will illustrate the method.

Tests on a carbon-aramid sandwich panel used as flooring in Boeing aircraft gave the results in the following table.

Face material	0.25 mm carbon/phenolic
Core material	3.2 mm cell, 147 kg/m^3, aramid honeycomb
Panel weight per unit area, m_a	2.69 kg/m^2
Panel length, L	510 mm
Panel width, b	51 mm
Panel thickness, d	10.0 mm
Flexural stiffness, EI	122 Nm^2
Failure moment, M_f	196 Nm

The equivalent density from Equation (11.13) is

$$\widetilde{\rho} = \frac{m_a}{d} = 269\,\text{kg/m}^3$$

The equivalent modulus $\widetilde{E}$ from Equation (11.14) is

$$\widetilde{E} = \frac{12\,EI}{b\,d^3} = 28.8\,\text{GPa}$$

The equivalent flexural strength $\widetilde{\sigma}_{flex}$ from Equation (11.15) is

$$\widetilde{\sigma}_{flex} = \frac{4\,M_f}{b\,d^2} = 154\,\text{MPa}$$

Equivalent properties of sandwich structures by analysis

In this section we develop equations for the stiffness and strength of sandwich panels and express them as properties of an equivalent homogeneous material. The symbols are defined earlier in Table 11.2.

Equivalent density The equivalent density of the sandwich (its mass divided by its volume) is

$$\tilde{\rho} = f\rho_f + (1-f)\rho_c \tag{11.16}$$

Here f is the volume fraction occupied by the faces: $f = 2t/d$.

Mechanical properties Sandwich panels are designed to be stiff and strong during bending. In thinking of the panel as a "material," we must therefore distinguish the in-plane modulus and strength from those in bending. The effective in-plane modulus $\tilde{E}_{in\text{-}plane}$ and strength $\tilde{\sigma}_{in\text{-}plane}$ are given, to an adequate approximation, by the rule of mixtures.

Equivalent flexural modulus Flexural properties are quite different. The flexural compliance (the reciprocal of the stiffness) has two contributions: one from the bending of the panel as a whole, the other from the shear of the core (Figure 11.14). They add. The bending stiffness is

$$EI = \frac{b}{12}(d^3 - c^3)E_f + \frac{bc^3}{12}E_c$$

The shear stiffness is

$$AG = \frac{bd^2}{c}G_c$$

Here the dimensions, d, c, t, and L are identified in Figure 11.13, E_f is Young's modulus of the face sheets, G_c is the shear modulus of the core, and A is the area of its cross-section.

Summing deflections gives

$$\delta = \frac{12PL^3}{B_1 b\{(d^3 - c^3)E_f + c^3E_c\}} + \frac{PLc}{B_2 d^2 b G_c} \tag{11.17}$$

The load configuration determines the values of the constants B_1 and B_2, as

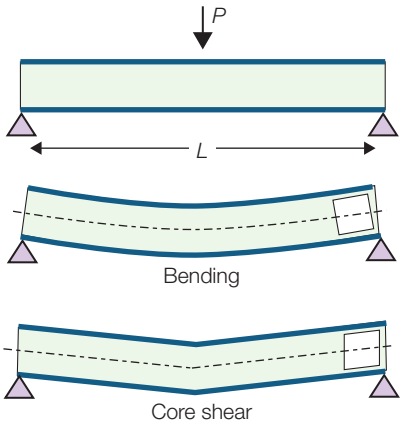
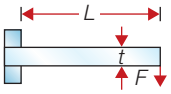
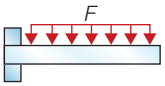
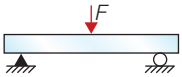
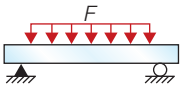
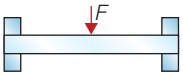
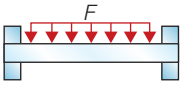


FIGURE 11.14
Sandwich panel flexural stiffness. There are contributions from bending and core shear.

Table 11.3 Constants to Describe Modes of Loading

Mode of Loading	Description	B_1	B_2	B_3	B_4
	Cantilever, end load	3	1	1	1
	Cantilever, uniformly distributed load	8	2	2	1
	Three-point bend, central load	48	4	4	2
	Three-point bend, uniformly distributed load	384/5	8	8	2
	Ends built in, central load	192	4	8	2
	Ends built in, uniformly distributed load	384	8	12	2

summarized in Table 11.3. Comparison with $\delta = \frac{12PL^3}{\widetilde{E}d^3b}$ for the "equivalent" material gives

$$\frac{1}{\widetilde{E}_{flex}} = \frac{1}{E_f\left\{\left(1-(1-f)^3\right)+\frac{E_c}{E_f}(1-f)^3\right\}} + \frac{B_1}{B_2}\left(\frac{d}{L}\right)^2\frac{(1-f)}{G_c} \qquad (11.18)$$

Note that, except for the bending-to-shear balancing term $(d/L)^2$, the equivalent property is scale-independent (as a material property should be); the only variable is the relative thickness of faces and core, f. The bending stiffness (EI) is recovered by forming the $\widetilde{E}\widetilde{I}$ where $\widetilde{I}$ is the second moment of a homogeneous panel ($\widetilde{I} = bd^3/12$).

Equivalent flexural strength Sandwich panels can fail in many different ways (Figure 11.15). The failure mechanisms compete, meaning that the one at the lowest load dominates. We calculate an *equivalent flexural strength* for each mode, then seek the lowest.

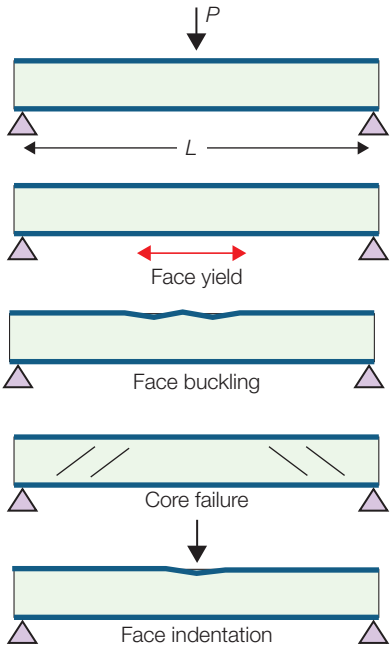


FIGURE 11.15
Failure modes of sandwich panels in flexure.

Face yield The fully plastic moment of the sandwich is

$$M_f = \frac{b}{4}\{(d^2 - c^2)\,\sigma_f + c^2\sigma_c\}$$

Using the fact that $c/d = (1-f)$, Equation (11.15) gives the following equivalent failure strength when face yielding is the dominant failure mode:

$$\tilde{\sigma}_{flex\,1} = \left(1-(1-f)^2\right)\,\sigma_f + (1-f)^2\sigma_c \qquad (11.19)$$

which, again, is independent of scale.

Face buckling In flexure, one face of the sandwich is in compression (Figure 11.16). If it buckles, the sandwich fails. The face stress at which this happens[4] is

$$\sigma_b = 0.57\,(E_f E_c^2)^{1/3} \qquad (11.20)$$

Buckling is a problem only when faces are thin and the core offers little support. The failure moment M_f is then well approximated by

$$M_f = 2\,\sigma_b\,t\,b\,c = 1.14\,(E_f E_c^2)^{1/3}\,t\,b\,c$$

Which, via the previous Equation (9.2), gives

$$\tilde{\sigma}_{flex\,2} = 1.14\,f(E_f E_c^2)^{1/3} \qquad (11.21)$$

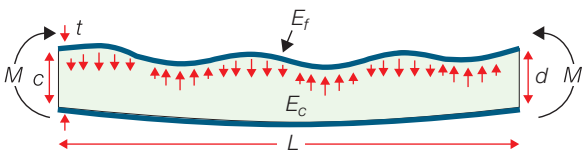


FIGURE 11.16
Face buckling.

Core shear Failure by core shear (Figure 11.17) occurs at the load

$$P_f = B_4 bc\,(\tau_c + \frac{t^2}{cL}\sigma_f)$$

[4] Derivations of this and the other equations cited here can be found in Ashby et al. (2000) and Gibson and Ashby (1997) in "Further reading" at the end of this chapter.

Here the first term results from shear in the core, the second from the formation of plastic hinges in the faces. Equating to

$$P_f = \frac{B_3 b d^2}{4L}\tilde{\sigma}_3$$

gives the equivalent strength when failure is by shear:

$$\tilde{\sigma}_{flex\,3} = \frac{B_4}{B_3}\left\{4\frac{L}{d}(1-f)\,\tau_c + f^2\sigma_f\right\} \qquad (11.22)$$

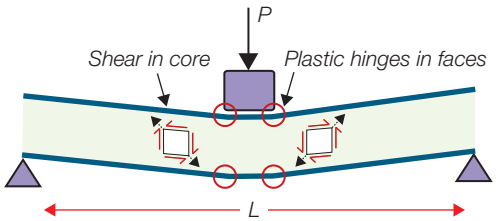


FIGURE 11.17
Core shear.

(The load configuration determines the constant values B_3 and B_4, as summarized earlier in Table 11.3.) When the core material is roughly isotropic (as foams are), τ_c can be replaced by $\sigma_c/2$. When it is not (an example is a honeycomb core), τ_c must be retained.

Indentation The indentation pressure $P_{ind} = P/a$ is

$$\frac{F}{ab} = p_{ind} = \frac{2t}{a}(\sigma_\gamma^f \sigma_\gamma^c)^{1/2} + \sigma_\gamma^c \qquad (11.23)$$

from which we find the minimum face thickness to avoid indentation (refer to Ashby et al., 2000, in Section 11.9).

The efficiency of sandwich structures Sandwiches are compared with monolithic materials as illustrated in Figures 11.18 and 11.19. In the first of these shows the equivalent density $\tilde{\rho}$ and equivalent flexural modulus, $\tilde{E}$ (Equations (11.16) and (11.18)) for sandwiches, using the data in Table 11.4. Here CFRP face sheets are combined with a high-performance foam core in different ratios, stepping through values of $2t/d$ for a chosen ratio of d/L to give the trajectory shown. Its doubly curved shape arises because of the interplay of bending and shear modes of deformation. Contours show values of the index for a light, stiff panel:

$$M_3 = \frac{E^{1/3}}{\rho}$$

The optimum panel, from a stiffness per unit weight perspective, is the one at which a contour is tangent to the trajectory. The figure shows that this occurs at $f \approx 0.04$, giving a panel that is 2.8 times lighter than a solid CFRP panel of the same stiffness (or $(2.8)^3 = 22$ times stiffer for the same mass).

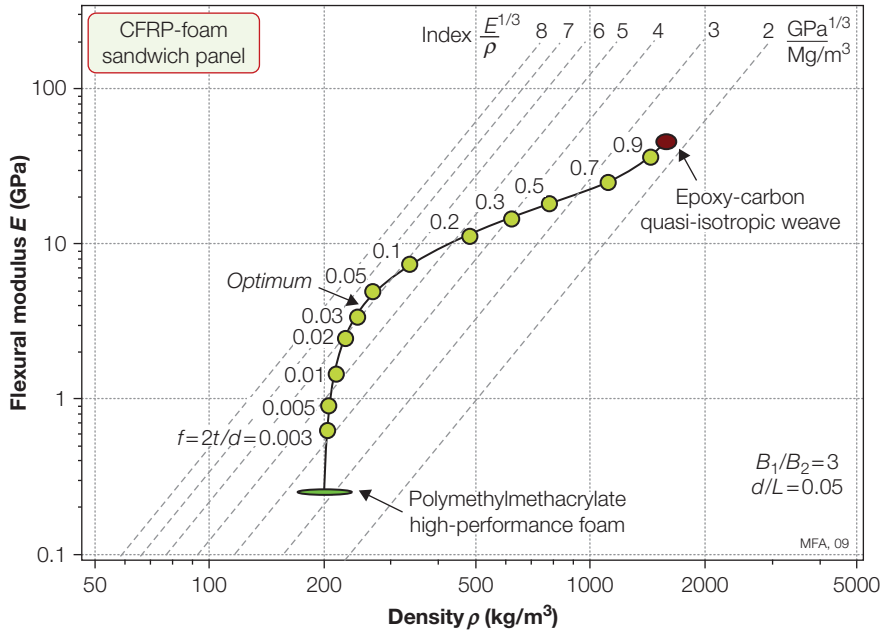


FIGURE 11.18

The equivalent modulus and density of a CFRP-foam sandwich are compared with those of monolithic materials. The contours of the index $E^{1/3}/\rho$ allow optimization of the proportions of the sandwich.

Strength (Figure 11.19) is handled in a similar way, but here there is the problem of competing mechanisms. We take the equivalent failure strength to be the least of $\tilde{\sigma}_{flex1}$, $\tilde{\sigma}_{flex2}$, and $\tilde{\sigma}_{flex3}$ (Equations (11.19), (11.21), and (11.22)), thereby properly accounting for the competition between them. For the conditions chosen here, face buckling dominates for $f < 0.025$; face yield dominates from $f = 0.025$ to $f = 0.1$, when a switch to core shear takes place. The envelope shows the achievable strength of CFRP–foam sandwich structures, and allows direct comparison with monolithic materials. Contours show the index for light strong structures:

$$M_6 = \frac{\sigma_f^{1/2}}{\rho}$$

which measures material efficiency when flexural strength is the main requirement. The optimum lies just below $f = 0.1$, at which the panel is 2.0 times lighter than a solid CFRP panel of the same strength (or $2.0^2 = 4.0$ times stronger for the same mass).

Table 11.4 Data for Face and Core of Sandwich

Face and Core Material	Density ρ (kg/m^3)	Modulus E (GPa)	Strength σ_f (MPa)
Carbon-epoxy quasi-isotropic weave	1570	46	550
Polymethacrylimide high-performance foam	200	0.255	6.8

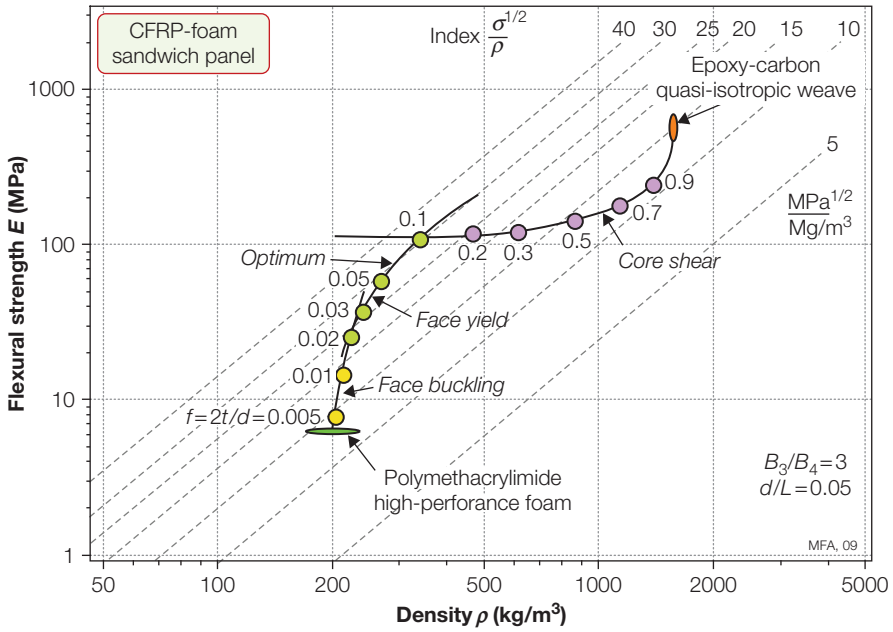


FIGURE 11.19

The equivalent strength and density of a CFRP-foam sandwich are compared with those of monolithic materials. The envelope shows the least strong of the competing failure modes. The contours of the index $\sigma^{1/2}/\rho$ allow optimization of the proportions of the sandwich. Indentation is included by imposing a minimum on the thickness ratio $2t/d$.

Indentation was not included in this competition because it is a local mechanism—it depends on the area of contact (or of impact) with the indenter, often an event for which the panel was not primarily designed. Protection is possible by estimating a "worst case" for the indentation load and area and calculating the value t/d required to withstand it. This is done using Equation (11.23) to calculate a lower safe limit for t/d, which is then applied as a constraint to the trajectory.

Thermal properties Thermal properties are treated in a similar way. The specific heat C_p follows a rule of mixtures (Equation (11.6)). The in-plane thermal conductivity $\lambda_{//}$ also follows such a rule (Equation (11.9)). The through-thickness conductivity, $\lambda_\perp$, is given by the harmonic mean

$$\tilde{\lambda}_\perp = \left(\frac{f}{\lambda_f} + \frac{(1-f)}{\lambda_c}\right)^{-1} \tag{11.24}$$

Thermal expansion in-plane is complicated by the fact that the faces and core have different expansion coefficients, but being bonded together they are forced to suffer the same strain. This constraint leads to an in-plane expansion coefficient of

$$\tilde{\alpha}_{//} = \frac{f E_f \alpha_f + (1-f) E_c \alpha_c}{f E_f + (1-f) E_c} \tag{11.25}$$

The through-thickness coefficient is simpler; it is given by the weighted mean

$$\tilde{\alpha}_\perp = f \alpha_f + (1-f)\alpha_c \tag{11.26}$$

Through-thickness thermal diffusivity is not a single-valued quantity, but depends on time. At short times heat does not penetrate the core and the diffusivity is that of the face, but at longer times the diffusivity tends to the value given by the ratio $\tilde{\lambda}/\tilde{\rho}\tilde{C}_p$.

Electrical properties The dielectric constant of a sandwich, as with composites, is given by a rule of mixtures—that is, Equation (11.12)—with $f = 2t/d$. Polymer foams have very low dielectric constants, so sandwiches with GFRP faces and polymer foam cores allow the construction of stiff, strong shells with exceptionally low dielectric loss. In-plane electrical conductivity, too, follows a rule of mixtures. Through-thickness electrical conductivity, like that of heat, is described by the harmonic mean—that is, the equivalent of Equation (11.24).

Filling property space with sandwich structures We end this section, as we did the last, with two figures illustrating how sandwich structures can expand the occupancy of material-property space. The first, Figure 11.20, is a flexural modulus-density ($E - \rho$) section. The areas filled by metals, polymers, ceramics, composites and foams appear as pale envelopes; members are labeled in gray. The stiffness and density of the CFRP–foam sandwiches of

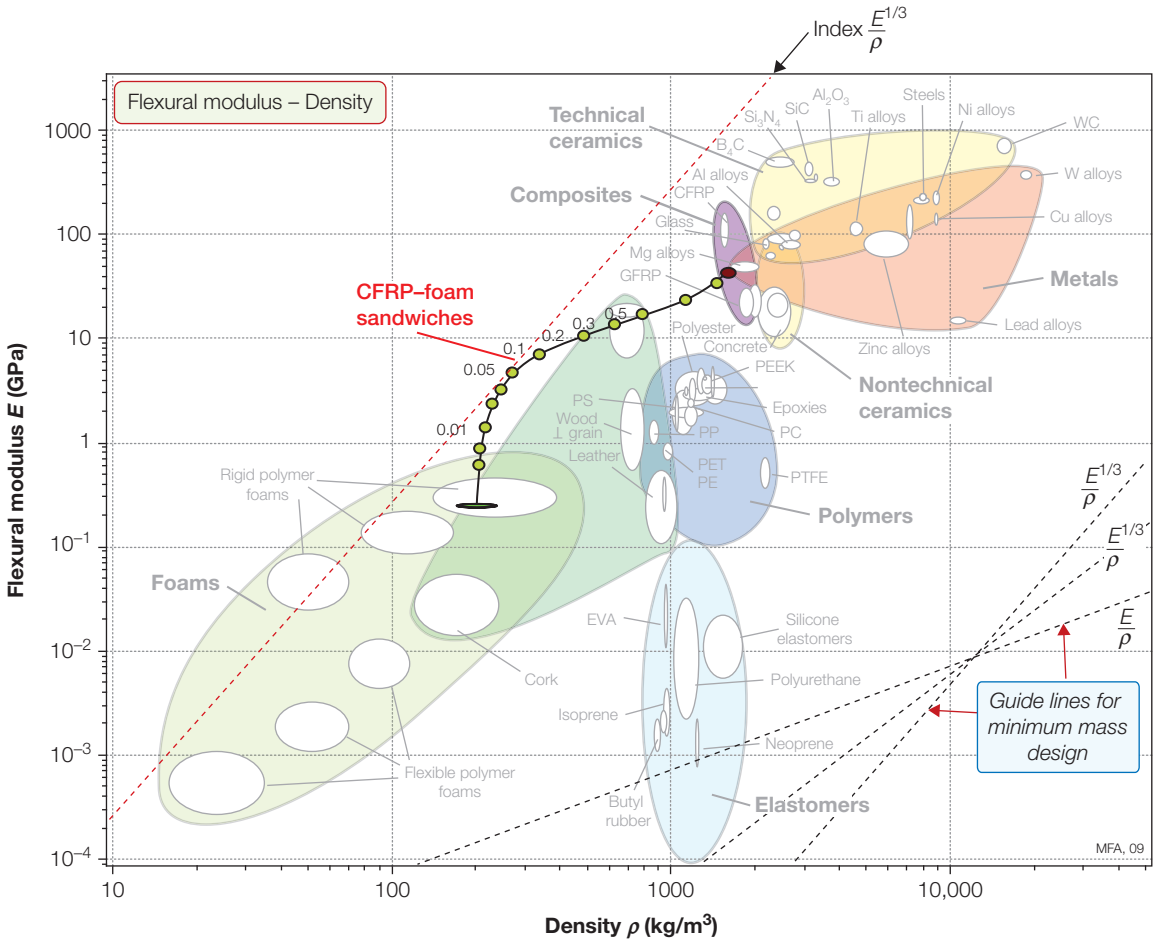


FIGURE 11.20
The sandwich data of Figure 11.18 superimposed on a modulus-density chart, showing the exceptional value of the flexural stiffness index $E^{1/3}/\rho$.

Figure 11.18 are superimposed. Those with $0.01 < f < 0.2$ extend into an area that was previously empty. Using the index $E^{1/3}/\rho$ for a light, stiff panel as a criterion of excellence, we find that sandwiches offer performance not attainable before.

Figure 11.21 tells a similar story for the strength-density ($\sigma_f - \rho$) section. The color coding is the same as that in the previous figure. The strength-density trajectory from Figure 11.19 is overlaid. Sandwiches again expand the populated area in a direction that, using the index for a light, strong panel ($\sigma_f^{1/2}/\rho$) as a criterion, offers enhanced performance.

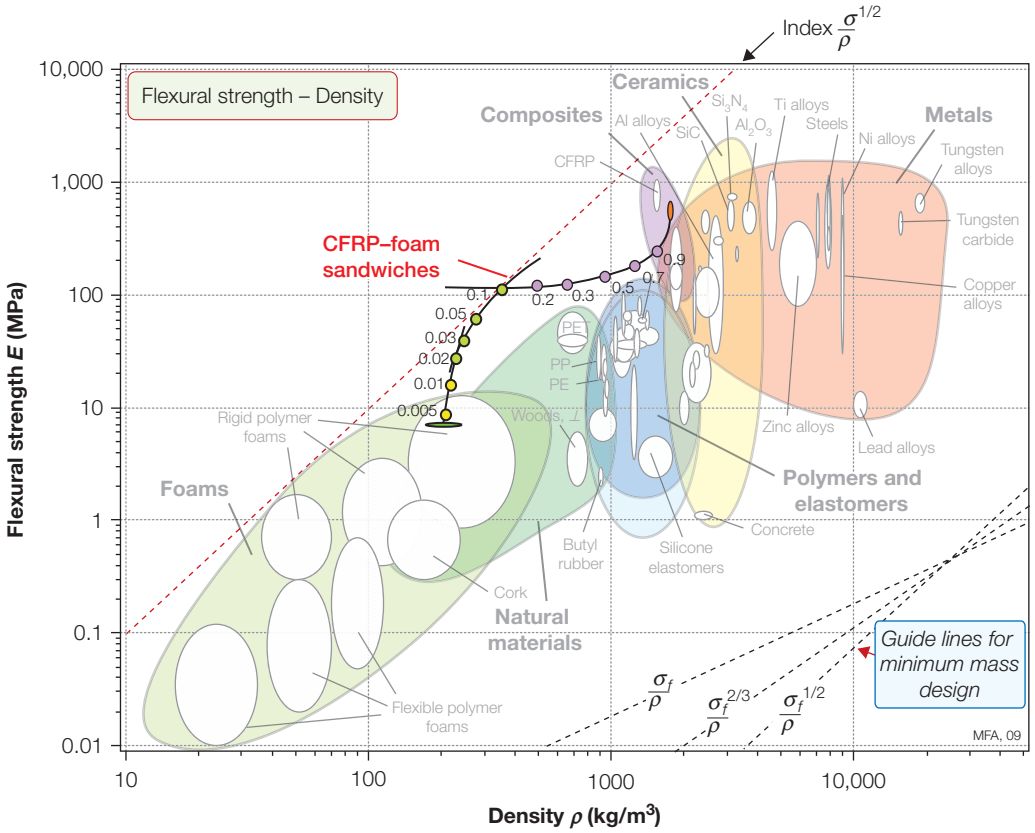


FIGURE 11.21
The sandwich data of Figure 11.19 superimposed on a strength-density chart, showing the exceptional value of the flexural stiffness index $\sigma^{1/2}/\rho$.

11.6 CELLULAR STRUCTURES: FOAMS AND LATTICES

Cellular structures—foams and lattices—are hybrids of a solid and a gas. The properties of the gas might at first seem irrelevant, but this is not so. The thermal conductivity of low-density foams of the sort used for insulation is determined by the conductivity of the gas contained in its pores; and the dielectric constant and breakdown potential, and even the compressibility, can depend on the gas properties.

There are two distinct species of cellular solid. The distinction is most obvious in their mechanical properties. The first, typified by foams, are *bending-dominated structures*; the second, typified by triangulated lattice structures, are *stretch dominated*—a distinction explained more fully next. To give an idea of

the difference: A foam with a relative density of 0.1 (meaning that the solid cell walls occupy 10% of the volume) is less stiff by a factor of 10 than a triangulated lattice of the same relative density. The word "configuration" has special relevance here.

Foams: Bending-dominated structures Foams are cellular solids made by expanding polymers, metals, ceramics, or glasses with a foaming agent—a generic term for one of many ways of introducing gas, much as yeast does in breadmaking. Figure 11.22 shows an idealized cell of a low-density foam. It consists of solid cell walls or edges surrounding a void space containing a gas or fluid. Cellular solids are characterized by their *relative density*, which for the structure shown here (with $t << L$) is

$$\frac{\tilde{\rho}}{\rho_s} \approx \left(\frac{t}{L}\right)^2 \qquad (11.27)$$

where $\tilde{\rho}$ is the density of the foam, ρ_s is the density of the solid of which it is made, L is the cell size, and t is the thickness of the cell edges.

Foams have the characteristic that, when they are loaded, their cell walls *bend*, with consequences that we now analyze.

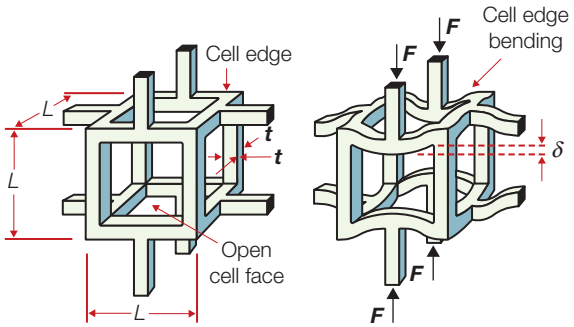


FIGURE 11.22
A cell in a low-density foam. When the foam is loaded, the cell edges bend, giving a low-modulus structure.

Mechanical properties The compressive stress-strain curve of bending-dominated foams looks like those in Figure 11.23. The material is linear elastic, with modulus $\tilde{E}$ up to its elastic limit, at which point the cell edges yield, buckle, or fracture. The foam continues to collapse at a nearly constant stress (the "plateau stress" $\tilde{\sigma}_{pl}$) until opposite sides of the cells impinge (the "densification strain" $\tilde{\varepsilon}_d$), when the stress rises rapidly. The mechanical properties are calculated in the ways developed next (details are in the texts listed in "Further reading").

A remote compressive stress σ exerts a force $F \propto \sigma L^2$ on the cell edges, causing them to bend and leading to a bending deflection δ, as shown in Figure 11.22. For the open-celled structure shown in the figure, the bending deflection (Appendix B.3) scales as

$$\delta \propto \frac{FL^3}{E_s I} \qquad (11.28)$$

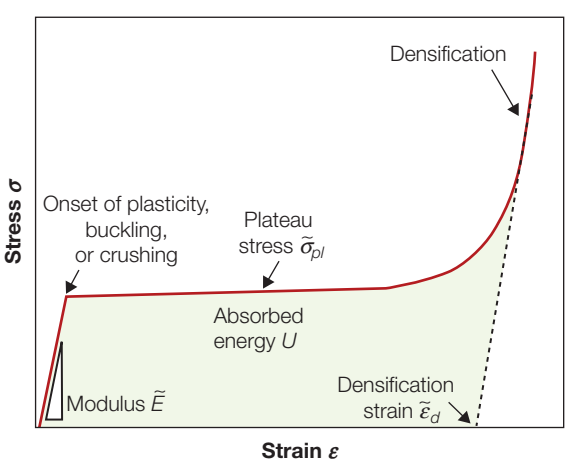


FIGURE 11.23
The plateau stress is determined by buckling, plastic bending, or fracturing of the cell walls.

where E_s is the modulus of the solid of which the foam is made and $I = \frac{t^4}{12}$ is the second moment of area of the cell edge of square cross-section, $t \times t$. The compressive strain suffered by the cell as a whole is then $\varepsilon = 2\delta/L$. Assembling these results gives the modulus $\widetilde{E} = \sigma/\varepsilon$ of the foam as

$$\frac{\widetilde{E}}{E_s} \propto \left(\frac{\widetilde{\rho}}{\rho_s}\right)^2 \quad \text{(bending – dominated behavior)} \tag{11.29}$$

Since $\widetilde{E} = E_s$ when $\widetilde{\rho} = \rho_s$, we expect the constant of proportionality to be close to unity—a speculation confirmed both by experiment and by numerical simulation. The quadratic dependence means that a small decrease in relative density causes a large drop in modulus (Figure 11.24(a)).

A similar approach can be used to model the plateau stress of the foam. The cell walls yield, as shown in Figure 11.25(a), when the force exerted on them exceeds their fully plastic moment (refer to Appendix A, Equation (A.4)),

$$M_f = \frac{\sigma_s t^3}{4} \tag{11.30}$$

where σ_s is the yield strength of the solid of which the foam is made. This moment is related to the remote stress by $M \propto FL \propto \sigma L^3$. Assembling these results gives the failure strength $\widetilde{\sigma}_{pl}$:

$$\frac{\widetilde{\sigma}_{pl}}{\sigma_{f,s}} = C\left(\frac{\widetilde{\rho}}{\rho_s}\right)^{3/2} \quad \text{(bending-dominated behavior)} \tag{11.31}$$

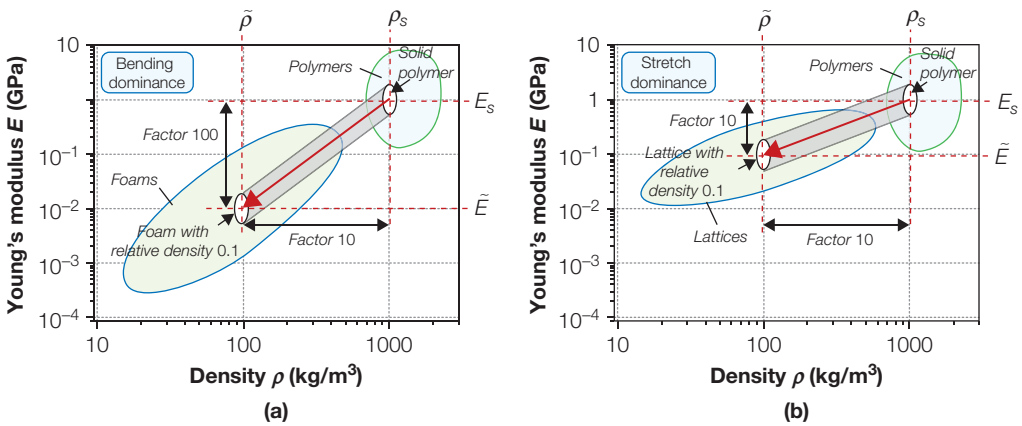


FIGURE 11.24

Foaming creates bending-dominated structures with lower modulus and density (a). Lattices that are stretch-dominated have moduli that are much greater than those of foams of the same density (b).

where the constant of proportionality, $C \approx 0.3$, has been established both by experiment and by numerical computation.

Elastomeric foams collapse not by yielding but by elastic bucking; brittle foams, by cell-wall fracture (Figures 11.25(b) and (c)). As with plastic collapse, simple scaling laws describe this behavior well. Collapse by buckling (refer to Appendix B, Table B.5) occurs when the stress exceeds $\tilde{\sigma}_{el}$, given by

$$\frac{\tilde{\sigma}_{el}}{E_s} \approx 0.05 \left(\frac{\tilde{\rho}}{\rho_s}\right)^2 \qquad (11.32)$$

and by cell-wall fracture (Appendix B.4 again) when it exceeds $\tilde{\sigma}_{cr}$:

$$\frac{\tilde{\sigma}_{cr}}{\sigma_{cr,s}} \approx 0.3 \left(\frac{\tilde{\rho}}{\rho_s}\right)^{3/2} \qquad (11.33)$$

where $\sigma_{cr,s}$ is the flexural strength of the cell-wall material. Densification, when the stress rises rapidly, is a purely geometric effect: The opposite sides of the cells are forced into contact and further bending or buckling is not possible. It is found to occur at a strain $\tilde{\varepsilon}_d$ (the densification strain) of

$$\tilde{\varepsilon}_d \approx 1 - 1.4 \left(\frac{\tilde{\rho}}{\rho_s}\right) \qquad (11.34)$$

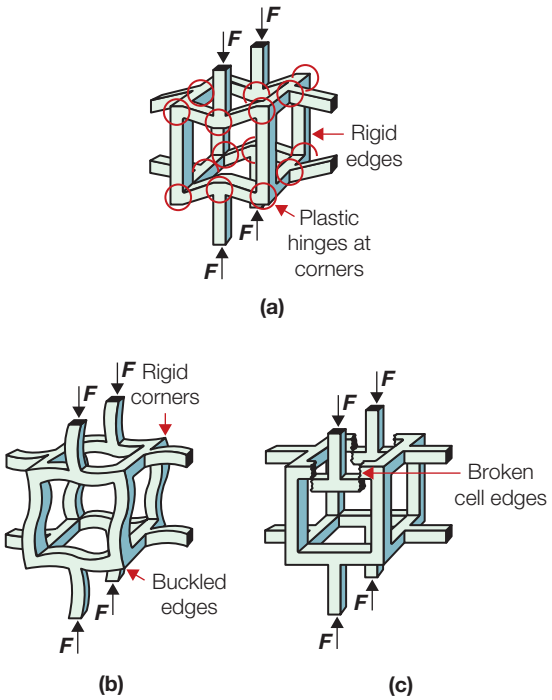


FIGURE 11.25
Collapse of foams. (a) When a foam made of a plastic materials is loaded beyond its elastic limit, the cell edges bend plastically. (b) An elastomeric foam, by contrast, collapses by the elastic buckling of its cell edges. (c) A brittle foam collapses by the successive fracturing of cell edges.

Foams are often used for cushioning and packaging and to protect against impact. The useful energy that a foam can absorb per unit volume is approximated by

$$\tilde{U} \approx \tilde{\sigma}_{pl}\, \tilde{\varepsilon}_d \qquad (11.35)$$

where $\tilde{\sigma}_{pl}$ is the plateau stress—the yield, buckling, or fracturing strength of the foam, whichever is least.

This behavior is not confined to open-cell foams with the structure idealized, as shown earlier in Figure 11.22. Most closed-cell foams also follow these scaling laws, at first sight an unexpected result because the cell faces must carry membrane stresses when the foam is loaded, and these should lead to a

linear dependence of both stiffness and strength on relative density. The explanation lies in the fact that the cell faces are very thin; they buckle or rupture at stresses so low that their contribution to stiffness and strength is small, leaving the cell edges to carry most of the load.

Thermal properties The specific heat of foams, when expressed in units of $J/m^3.K$, is given by a rule of mixtures, summing the contributions from the solid and the gas. The thermal expansion coefficient of an open cell foam is the same as that of the solid from which it is made. The same is true of rigid closed-cell foams but not necessarily of low-density elastomeric foams because the expansion of the gas within the cells can expand the foam itself, giving an apparently higher coefficient.

The cells in most foams are sufficiently small that convection of the gas within them is completely suppressed. The thermal conductivity of the foam is thus the sum of that conducted through the cell walls and that through the still air (or other gas) they contain. To an adequate approximation,

$$\widetilde{\lambda} = \frac{1}{3}\left(\left(\frac{\widetilde{\rho}}{\rho_s}\right) + 2\left(\frac{\widetilde{\rho}}{\rho_s}\right)^{3/2}\right)\lambda_s + \left(1 - \left(\frac{\widetilde{\rho}}{\rho_s}\right)\right)\lambda_g \qquad (11.36)$$

where λ_s is the conductivity of the solid and λ_g that of the gas (for dry air it is 0.025 W/m.K). The term associated with the gas is important: Blowing agents for foams intended for thermal insulation are chosen to have a low value of λ_g.

Electrical properties Insulating foams are attractive for their low dielectric constant, $\widetilde{\varepsilon}_r$, falling toward 1 (the value for air or vacuum) as the relative density decreases:

$$\widetilde{\varepsilon}_r = 1 + (\varepsilon_{r,s} - 1)\left(\frac{\widetilde{\rho}}{\rho_s}\right) \qquad (11.37)$$

where $\varepsilon_{r,s}$ is the dielectric constant of the solid of which the foam is made. The electrical conductivity follows the same scaling law as the thermal conductivity.

Lattice: Stretch-dominated structures

If conventional foams have low stiffness because the configuration of their cell edges allows them to bend, might it not be possible to devise other configurations in which the cell edges are made to stretch instead? This thinking leads to the idea of *micro-truss lattice structures*. To understand these we need one of those simple yet profound fundamental laws: the Maxwell stability criterion.

Equation (11.38) is the condition that a pin-jointed frame (meaning one that is hinged at its corners) made up of b struts and j frictionless joints, like those shown in Figure 11.26, is to be both *statically* and *kinematically*

determinate (meaning that it is rigid and does not fold up when loaded) in two dimensions:

$$M = b - 2j + 3 = 0 \qquad (11.38)$$

In three dimensions the equivalent equation is

$$M = b - 3j + 6 = 0 \qquad (11.39)$$

If $M < 0$, the frame is a *mechanism*. It has no stiffness or strength; it collapses if loaded. If its joints are locked, preventing rotation (as they are in a lattice), the bars of the frame *bend* when the structure is loaded, just as those in Figure 11.22 do. If, instead, $M \geq 0$ the frame ceases to be a mechanism. If it is loaded, its members carry tension or compression (even when pin-jointed), and it becomes a *stretch-dominated* structure. Locking the hinges now makes little difference because slender structures are much stiffer when stretched than when bent. There is an underlying

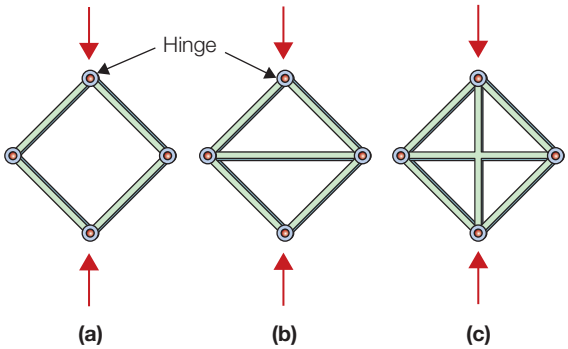


FIGURE 11.26
The pin-jointed frame at (a) is a mechanism. If its joints are welded together, the cell edges bend. The pin-jointed, triangulated frame at (b) is stiff when loaded because the transverse bar carries tension, preventing collapse. When the frame's joints are welded, its stiffness and strength hardly change. The frame at (c) is over-constrained. If the horizontal bar is tightened, the vertical bar is put in tension even when there are no external loads.

Estimating foam properties

A polyethylene foam has a density ρ of 150 kg/m^3. The density, modulus, strength, and thermal conductivity of the polyethylene are listed in the table. What would you expect the same properties of the foam to be?

	Density ρ_s (kg/m^3)	Young's Modulus E_s (GPa)	Flexural Strength $\sigma_{f,s}$ (MPa)	Thermal Conductivity λ_s (W/m.K)
High-molecular-weight polyethylene	950	0.94	33	0.195

Answer

The relative density of the foam is $\tilde{\rho}/\rho_s = 0.16$. Using Equations (11.29), (11.31), and (11.36), we find

	Young's Modulus $\tilde{E}$ (GPa)	Flexural Strength $\tilde{\sigma}_{pl}$ (MPa)	Thermal Conductivity $\tilde{\lambda}$ (W/m.K)
HMW polyethylene foam	0.024	0.63	0.04

Estimating lattice properties

A stretch-dominated polyethylene lattice has a density ρ of 150 kg/m^3. The density, modulus, strength, and thermal conductivity of the polyethylene are the same as those listed for the last example. What would you expect the properties of the lattice to be?

Answer

The relative density of the lattice is $\tilde{\rho}/\rho_s = 0.16$. Using Equations (11.40), (11.41), and (11.36), we find

	Young's Modulus $\tilde{E}$ (GPa)	Flexural Strength $\tilde{\sigma}$ (MPa)	Thermal Conductivity $\tilde{\lambda}$ (W/m.K)
HMW polyethylene lattice	0.05	1.8	0.04

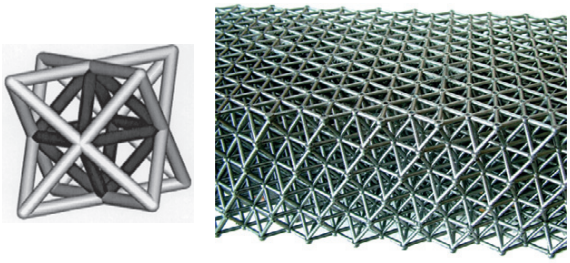


FIGURE 11.27
A micro-truss structure and its unit cell. This is a stretch-dominated structure, and it is over-constrained, meaning that it is possible for it to be in a state of self-stress.

principle here: *Stretch-dominated structures have high structural efficiency; bending-dominated structures have low.*

Mechanical properties These criteria provide a basis for the design of efficient microlattice structures. For the cellular structure of Figure 11.22, Maxwell's $M < 0$ and bending dominates. For the structure shown in Figure 11.27, however, $M > 0$ and it behaves as an almost isotropic, stretch-dominated structure. On average one-third of its bars carry tension when the structure is loaded in simple tension. Thus

$$\frac{\tilde{E}}{E_s} \approx \frac{1}{3}\left(\frac{\tilde{\rho}}{\rho_s}\right) \text{ (isotropic stretch-dominated behavior)} \qquad (11.40)$$

The modulus is linear, not quadratic, in density (Figure 11.24b), giving a much stiffer structure for the same density. Collapse occurs when the cell edges yield, giving the collapse stress

$$\frac{\tilde{\sigma}}{\sigma_{f,s}} \approx \frac{1}{3}\left(\frac{\tilde{\rho}}{\rho_s}\right) \text{ (isotropic stretch-dominated behavior)} \qquad (11.41)$$

This is an upper bound since it assumes that the struts yield in tension or compression when the structure is loaded. If the struts are slender, they may buckle before they yield. Then the "strength," like that of a buckling foam (Equation (11.32)), is

$$\frac{\tilde{\sigma}_{el}}{E_s} \approx 0.2\left(\frac{\tilde{\rho}}{\rho_s}\right)^2 \qquad (11.42)$$

Thermal and electrical properties The bending/stretching distinction influences mechanical properties profoundly, but has no effect on thermal or electrical properties, which are adequately described by the equations listed above for foams.

Filling property space with cellular structures All of the charts in Chapter 4 have an envelope labeled "Foams," indicating where the properties' commercial polymer foams lie. It is highlighted in Figure 11.28 which again

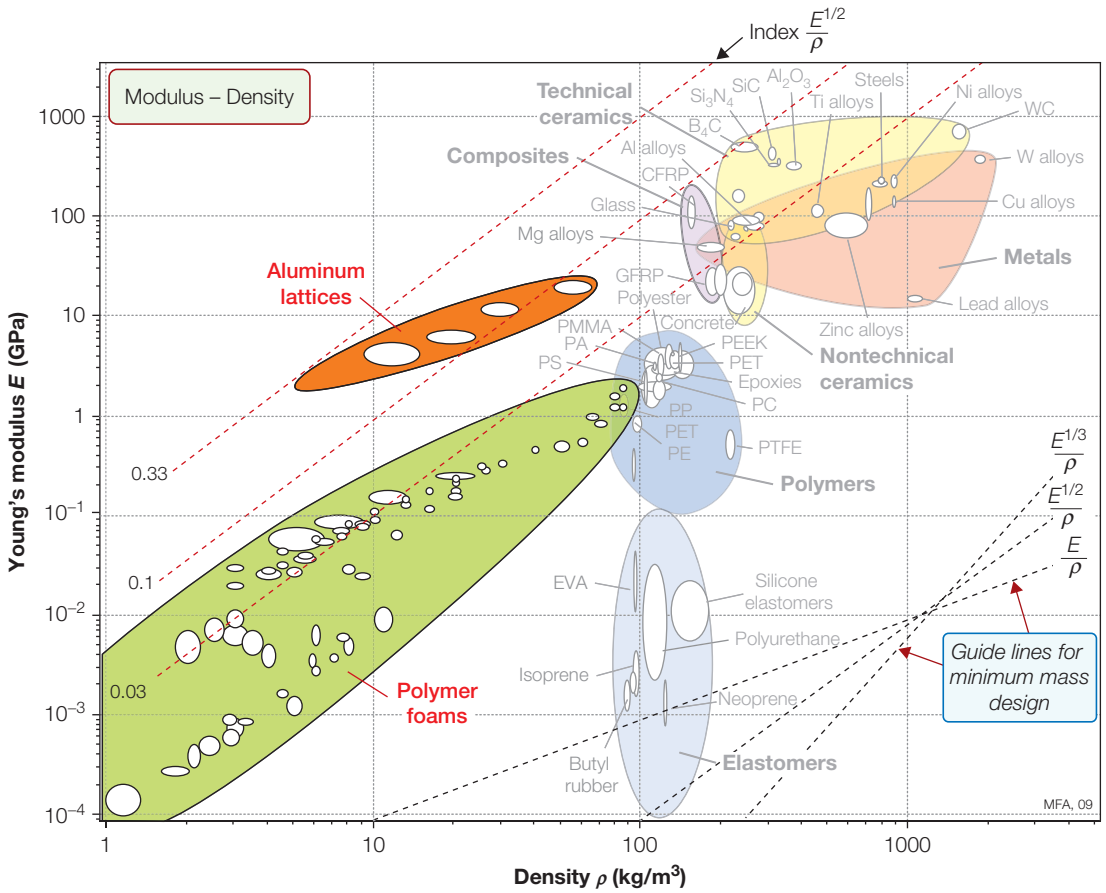


FIGURE 11.28

Foams and micro-truss structures are hybrids of material and space. Their mechanical response depends on their structure. Foams are usually bending-dominated and lie along a line of slope 2 on this chart. Micro-truss structures are stretch-dominated and lie on a line of slope 1. Both extend the occupied area of this chart by many decades.

shows the modulus-density ($E - \rho$) section through material-property space. The polymer foam envelope is extended along a line of slope 2, as predicted by Equation (11.29). Lattice structures, by contrast, extend along a line of slope 1, as Equation (11.40) predicts. Both fill areas of $E - \rho$ space that are not filled by solid materials. Lattice structures push the filled area to higher values of the indices

$$E^{1/2}/\rho \quad \text{and} \quad E^{1/3}/\rho$$

11.7 SEGMENTED STRUCTURES

Subdivision as a design variable

Shape can be used to reduce flexural stiffness and strength as well as to increase them. Springs, suspensions, flexible cables, and other structures that must flex yet have high tensile strength use shape to give a low bending stiffness. This is achieved by shaping the material into strands or leaves, as explained in Section 10.7. The slender strands or leaves bend easily but do not stretch when the section is bent: an n-strand cable is less stiff by a factor of $3/\pi n$ than the solid reference section; an n-leaf panel by a factor $1/n^2$.

Subdivision allows mixing. If one or more of the segments is replaced by a second material, combinations can be created that have property combinations not found in monolithic materials. The method is best illustrated by the example that follows.

Subdivision can be used in another way: to impart *damage tolerance*. A glass window, hit by a projectile, will shatter. One made of small glass bricks, laid as bricks usually are, will lose a brick or two but not shatter totally; it is damage-tolerant. By subdividing and separating the material, a crack in one segment does not penetrate into its neighbors, allowing local but not global failure. This is the principle of "topological toughening." Builders in stone and brick have exploited the idea for thousands of years: Stone and brick are brittle, but buildings made of them—even those made without cement ("dry-stone building")—survive ground movement, even small earthquakes, through their ability to deform with some local failure but without total collapse.

Materials for long-span power cables

In the design of suspended power cables, the objectives are to minimize electrical resistance and at the same time maximize the strength since this allows the greatest span. This is an example of multiobjective optimization, discussed in Chapter 7. As explained there, each objective is expressed, by convention, such that a minimum for it is sought. We thus seek materials with the lowest values of resistivity ρ_e and the reciprocal of yield strength, $1/\sigma_y$.

Figure 11.29 shows the result: Materials that best meet the design requirements lie toward the bottom left. But here there is a hole: All 1700 metals and alloys plotted here have properties that lie above the broken red line, none below. Those with the lowest resistance—copper, aluminum, and certain of their alloys—are not very strong; those that are strongest—drawn carbon and low-alloy steel—do not conduct very well.

Now consider a cable made by interleaving strands of copper and steel such that each occupies half the cross-section. If the steel carries no current and the copper no load (the most pessimistic scenario), the performance of the cable will lie at the point shown on the figure—it has twice the resistivity of the copper and half the strength of the steel. It lies in a part of property space that was empty, offering performance not previously possible. Other ratios of copper to steel fill other parts of the space; by varying the ratio, the shaded envelope in the figure is covered. Similar hybrids of aluminum and steel fill a different area, as is easily seen by repeating the construction using "1000 series Al alloys" in place of "OFHC copper, hard" in the combination. Their combinations of ρ_e and σ_y are less good, but they are lighter and cheaper and for this reason are widely used.

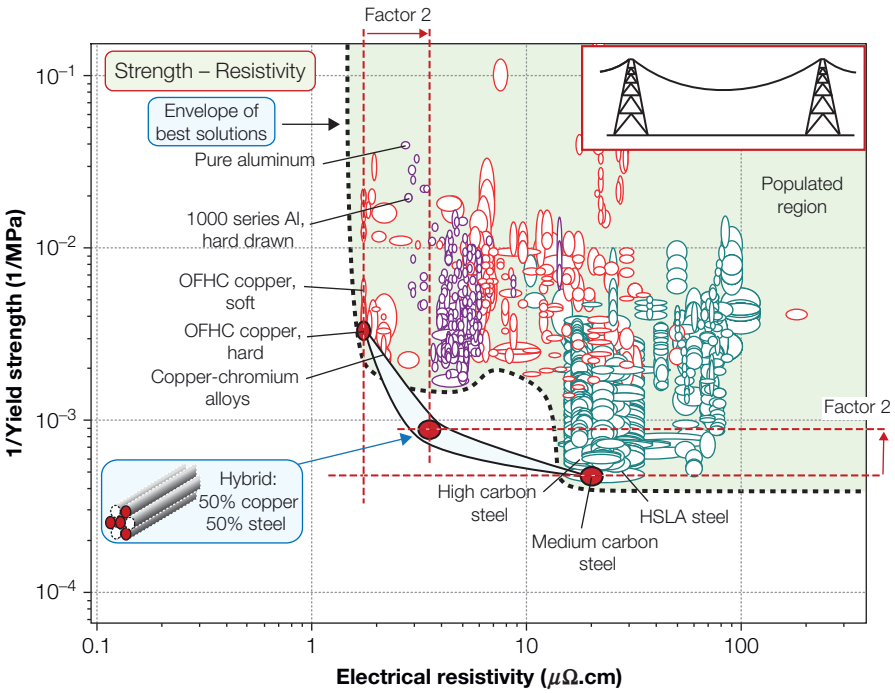


FIGURE 11.29

Designing a hybrid–here one with high strength and high electrical conductivity. The figure shows the resistivity and reciprocal of tensile strength for 1,700 metals and alloys. The construction is for a hybrid of hard-drawn OFHC copper and drawn medium carbon steel, but the figure itself allows many hybrids to be investigated.

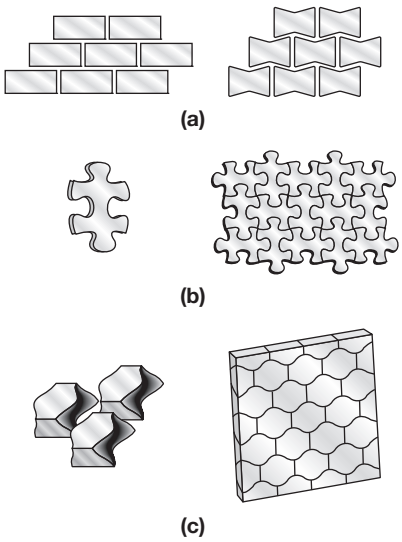


FIGURE 11.30
Examples of topological interlocking: discrete, unbonded structures that carry load. (a) Brick-like assemblies of rectangular blocks carry axial compression, but not tension or shear. (b) The 2-dimensional interlocking of a jigsaw puzzle carries in-plane loads. (c) Units that, when assembled into a continuous layer and clamped within a rigid boundary around the edges of the layer, can carry out-of-plane loads and bending moments.

Taking the simplest view, two things are necessary in order to have topological damage tolerance: discreteness of the structural units and an interlocking of the units in such a way that the array as a whole can carry a load. Brick-like arrangements (Figure 11.30(a) carry large loads and are damage-tolerant in compression and shear, but disintegrate under tension. Strand- and layer-like structures are damage-tolerant in tension because if one strand fails, the crack does not penetrate its neighbors—the principle of multistrand ropes and cables. The jigsaw puzzle configuration (Figure 11.30(b)) carries in-plane tension, compression and shear, but at the cost of introducing a stress concentration factor of about $\sqrt{R/r}$, where R is the approximate radius of a unit and r that of the interlock.

The sources listed in Section 11.9 explore a particular set of topologies that rely on compressive or rigid boundary conditions to create continuous layers that tolerate out-of-plane forces and bending moments, illustrated in Figure 11.30(c). This is done by creating interlocking units with nonplanar surfaces that have curvature both in the plane of the array and normal to it. Provided the array is constrained at its periphery, the nesting shapes limit the relative motion of the units, locking them together. Topological interlocking of this sort allows the formation of continuous layers that can be used for ceramic claddings or linings to give surface protection. And of course the units need not be made of one material alone. Since the only requirement is that of interlocking shape, the segments can be made of different materials. Just as a builder constructing a brick wall can use porous bricks for ventilation and transparent bricks to admit light, the designer of a segmented hybrid can add functionality through the choice of material for the units.

11.8 SUMMARY AND CONCLUSIONS

The properties of engineering materials can be thought of as defining the axes of a multidimensional space, with each property as a dimension. Chapter 4 showed how this space can be mapped. The maps reveal that some areas of

property space are occupied and others are empty—there are holes. The holes can sometimes be filled by making hybrids: combinations of two (or more) materials in a chosen configuration and scale. Design requirements isolate a small box in a multidimensional material-property space. If it is occupied by materials, the requirements can be met. However, if the box hits a hole in any one of the dimensions, we need a hybrid. A number of families of configuration exist, each offering different combinations of functionality. Individual configurations are characterized by a set of bounds, bracketing their effective properties. The methods developed in this chapter provide tools for exploring alternative material configuration combinations. Chapter 12 gives examples of their use.

11.9 FURTHER READING

Hybrid materials—general

Bendsoe, M.P., & Sigmund, O. (2003). *Topology optimization, theory, methods, and applications*. Springer-Verlag, ISBN 3-540-42992-1.
The first comprehensive treatment of emerging methods for optimizing configuration and scale.

Kromm, F.X., Quenisset, J.M, Harry, R., & Lorriot, T. (2001). An example of multimaterial design. *Proc Euromat '01*. Rimini, Italy.
One of the first papers to address hybrid design.

McDowell, D.L., Allen, J., Mistree, F., Panchal, J., & Choi, H-J (2009). *Integrated design of multiscale materials and products*. Elsevier, ISBN 978-1-85617-662-0.
McDowell et al. seek to integrate the "materials by design approach" of Olson/Ques Tek LLC with the "materials selection" approach developed in this book and implemented in the CES software.

Composites

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CHAPTER 12

Case Studies: Hybrids



Racing yachts used hybrids throughout—mast and boom: epoxy-carbon fiber; hull: GRFP and CFRP skinned polymethyl-methyl-acrylamid foamed core sandwich; sails: Kevlar-nylon mix weave with thermally bonded PET skin.

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12.1 INTRODUCTION AND SYNOPSIS

Chapter 11 explored hybrids of four types: composites, sandwiches, lattices, and segmented structures. Each is associated with a set of models that allow its properties to be estimated. In this chapter we illustrate the models' use to design hybrids to fill specified needs—needs that cannot be filled by single material choices. The cover picture is a reminder of the widespread use of hybrids to maximize performance: The yacht is almost entirely made of hybrid materials.

12.2 DESIGNING METAL MATRIX COMPOSITES

The most common state of loading in structures is that of bending. One measure of excellence in designing materials to carry bending moments at minimum weight is the index $E^{1/2}/\rho$, where E is Young's modulus and is the ρ the density. Alloys of aluminum and of magnesium rank highly by this criterion; titanium alloys and steels are not as good. How could the performance of magnesium (the best of the lot) be enhanced further? Table 12.1 summarizes the challenge.

Table 12.1 Design Requirements for the Material of the Panel

Function	Light, stiff beam
Constraint	Magnesium matrix
Objective	Maximize stiffness to weight in bending (index $E^{1/2}/\rho$)
Free variable	Choice of reinforcement and volume fraction

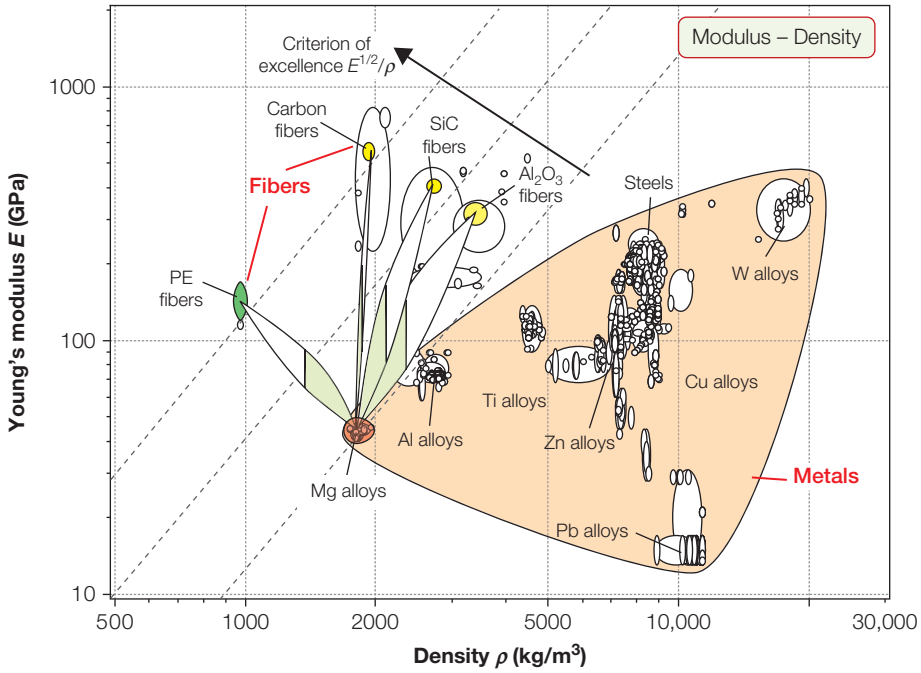


FIGURE 12.1

Possible magnesium-matrix composites. The lozenges show the areas bracketed by the upper and lower bounds of Table 12.2. The green shaded areas within them extend up to a volume fraction of 0.5.

Table 12.2 Superposition Rules for Composite Density and Modulus*

Property	Lower Bound	Upper Bound
Density	$\tilde{\rho} = f\rho_r + (1-f)\rho_m$ (exact)	
Modulus	$\tilde{E}_L = \dfrac{E_m E_r}{f E_m + (1-f)E_r}$	$\tilde{E}_u = f E_r + (1-f)E_m$

* *Subscripts m and r mean "matrix" and "reinforcement"; f = volume fraction.*

The method Figure 12.1 shows a chart of E and ρ for metals and fibers. Magnesium alloys appear at the extreme left of the red "Metals" envelope. The criterion of excellence is shown as a set of diagonal contours, increasing toward the top left. Magnesium ranks slightly higher than aluminum and much higher than titanium and steel.

Table 12.2 summarizes the superposition rules for density and modulus. They are plotted as envelopes of attainable performance for four magnesium-based composites; the upper edge of each envelope is the upper bound; the lower edge, the lower bound. There is an upper limit for the volume fraction, which

we will set at 0.5. It is shown as a vertical bar within each envelope. Only the shaded part of the envelope below the bar is accessible. The diagonal lines plot the criterion of excellence, $E^{1/2}/\rho$. The combinations with the highest values of this quantity offer the greatest gain in stiffness per unit weight.

The results It is immediately apparent that the greatest promise is shown by the composites of magnesium with drawn polyethylene (PE) or carbon fibers; magnesium-SiC is less good. Magnesium-Al_2O_3 offers almost no gain at all. The method allows the potential of rapidly exploring alternative choices.

Postscript Magnesium-PE composites look good and are promising, but there remains the challenge of actually making them. Polyethylene fibers are already used in ropes and cables because of their high stiffness, strength, and low weight. But they are destroyed by temperatures much above 120°C, so casting or sintering of the magnesium around the fibers is not an option. One possibility is to use drawn PE sheets rather than fibers, and fabricate a multilayer laminate by bonding the PE sheets between sheets of magnesium. A second possibility is to explore ternary composites: dispersing magnesium powder in an epoxy and using this mix as the matrix to contain the PE fibers, for example. Otherwise we must fall back on magnesium-carbon, an attractive option.

Related case study

6.12 "Stiff, high damping materials for shaker tables"

12.3 FLEXIBLE CONDUCTORS AND PERCOLATION

A material is required for demountable seals in specialized equipment. The material must conform to the curved surfaces between which it is clamped, it must be electrically conducting to prevent charge build-up, and it should be moldable. Table 12.3 summarizes the requirements.

The method and results Metals, carbon, and some carbides and intermetallics are good conductors, but they are stiff and cannot be molded (Figure 12.2). Thermoplastic and thermosetting elastomers can be molded and are flexible, but they do not conduct. How are they to be combined? Metal coating of polymers is workable if the product will be used in a protected environment, but coatings are easily damaged. If a robust, flexible

Table 12.3 Materials for Flexible Conductors

Function	Flexible conducting solid
Constraints	Low Young's modulus to allow conformation Low resistivity to permit conduction (ρ_e < 1000 $\mu\Omega$.cm)
Objective	Able to be molded
Free variable	Choice of matrix, reinforcement, configuration, and volume fraction

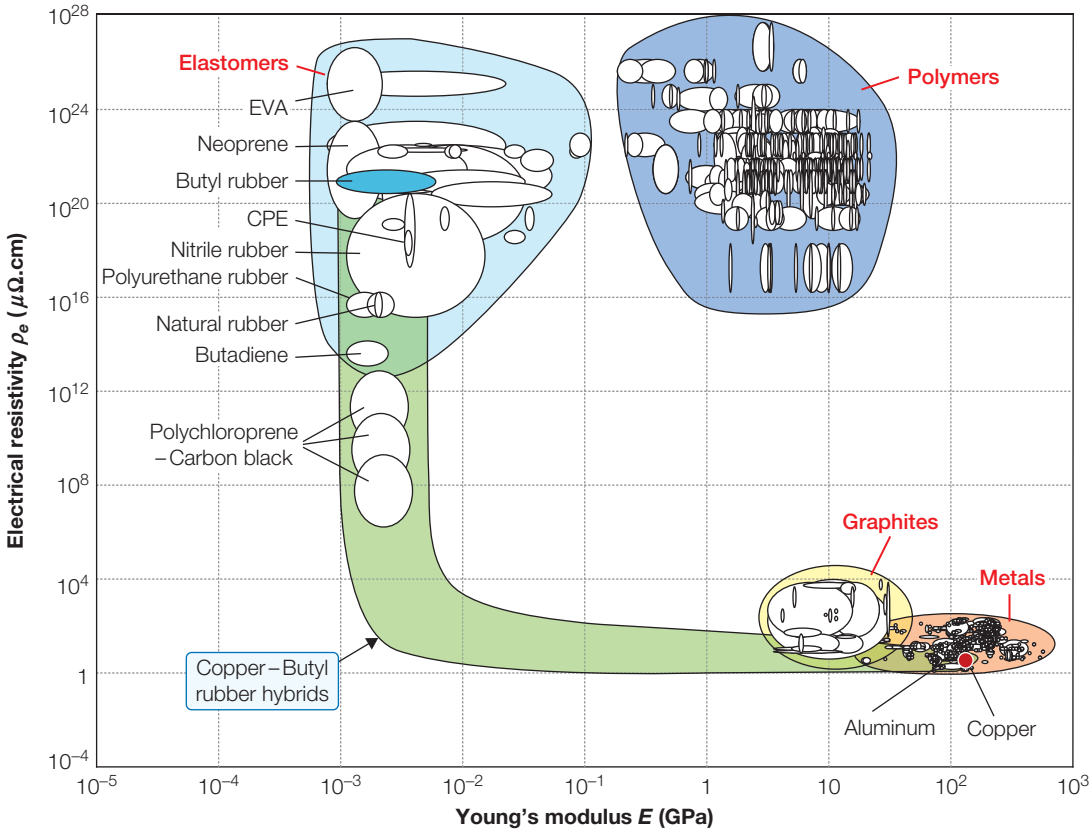


FIGURE 12.2

When conducting particles or fibers are mixed into an insulating elastomer, a hole in the material-property space is filled. Carbon-filled butyl rubbers lie in this part of the space.

product is needed, bulk rather than surface conduction is essential. This can be achieved by dispersing conducting material in a polymer. But how much? And in what shape?

Think of mixing conducting and insulating spheres of the same size by shaking them up in a jar. If there are only a few conducting spheres, they do not touch and the array as a whole is an insulator. If each conducting sphere contacts just one other, there is still no connecting path. If, on average, each touches two, there is still no path. Adding more conducting spheres gives larger clusters, but they can be large yet still discrete. For bulk conduction we need *connectivity*: The array first becomes a conductor when a single trail of contacts links one surface to the other, that is, when the volume fraction f of the conducting spheres reaches the *percolation threshold* f_c. Percolation problems are easy to describe but difficult to solve. Research since 1960 has provided

approximate solutions to most of the percolation problems associated with the design of hybrids (see "Further reading" for a review). For simple cubic packing $f_c = 0.248$; for close packing $f_c = 0.180$. For a random array (the sphere shaken in a jar) it is somewhere in between—approximately 0.2.[1]

Making the spheres smaller smears the transition out. The percolation threshold is still 0.2, but the first connecting path is now thin and extremely devious—it is the only one out of the vast number of almost complete paths that actually connects. Increase the volume fraction and the number of conducting paths increases initially as $(f - f_c)^2$, then linearly, reverting to a rule of mixtures. If the particles are very small, as much as 40% may be needed to give good conduction. But a loading of 40% seriously degrades the moldability and flexibility of the polymer.

Shape gives a way out. If the spheres are replaced by slender fibers, they touch more easily and the percolation threshold falls. If their aspect ratio is $\beta = L/d$ (where L is the fiber length, d the diameter), then, empirically, the percolation threshold falls from f_c to roughly $f_c/\beta^{1/2}$, so an aspect ratio of 9 reduces p_c, the volume fraction of the conducting phase, by a factor of 3.

The concept of percolation is a necessary tool in designing hybrids. Electrical conductivity works that way; so too does the passage of liquids through foams or porous media—no connected path and no fluid flows; just one (out of a million possibilities) and there is a leak. Add a few more connections and there is a flood. Percolation ideas are particularly important in understanding the transport properties of hybrids: Properties that determine the flow of electricity or heat, of fluid, or of flow by diffusion, especially when the differences in properties of the components are extreme, because it is then that connected paths matter.

Figure 12.2 is a chart of electrical resistivity and modulus with metals, polymers, and composites plotted. There is a hole at the lower left where materials with the property combination we want would lie. The pale-green L-shaped envelope encloses the properties of hybrids of rubber with chopped copper wires. The resistivity drops sharply at the percolation threshold, which, for wires, falls to a few percent. The material retains all the flexibility of the elastomer but behaves as a bulk conductor.

Postscript Conducting elastomers exploiting these ideas are widely available. Such materials find application in antistatic clothing and mats, as pressure sensing elements, and even as solderless connections.

Related case study

12.4 "Extreme combinations of thermal and electrical conduction"

[1] These results are for infinite, or at least very large, arrays. Experiments generally give values in the range 0.19 to 0.22, with some variability because of the finite size of the samples.

12.4 EXTREME COMBINATIONS OF THERMAL AND ELECTRICAL CONDUCTION

Materials that are good electrical conductors are always good thermal conductors too. Copper, for example, excels at both. Most polymers, by contrast, are electrical insulators (meaning that their conductivity is so low that for practical purposes they do not conduct at all), and as solids go they are also poor thermal conductors—polyethylene is an example. Thus the "high-high" and "low-low" combinations of conduction can be met by monolithic materials, and there are plenty of them. The "high-low" and "low-high" combinations are a different matter: Nature gives us very few of either. The challenge is summarized in Table 12.4: Using only copper and polyethylene, devise hybrid materials that achieve both of these combinations (data for both are in Table 12.5).

The method and results Figure 12.3 shows two possible hybrid configurations. Both are of type "composite" but with very different configurations and volume fractions. The first is a tangle of fine copper wires embedded in a PE matrix. To describe its performance we draw on the bounds of Section 11.4. Electrical conductivity requires percolation; as explained in Section 12.3, the percolation threshold is minimized by using conducting wires of high aspect ratio. Above the percolation threshold the conductivity tends toward a rule of mixtures (Equation (11.9), with thermal conductivity replaced by electrical conductivity):

$$\widetilde{\kappa}_1 = f\,\kappa_{cu} + (1-f)\,\kappa_{PE} \tag{12.1}$$

Table 12.4 Requirements for the Conducting Hybrids of Copper and Polyethylene

Function	Extreme conduction combinations
Constraint	Materials: copper and polyethylene
Objective	Maximize difference between electrical and thermal conductivities
Configuration	Free choice
Free variable	Configuration and relative volume fractions of the two materials

Table 12.5 Data for Copper and HDPE

Material	Electrical Conductivity ($1/\mu\Omega$.cm)	Thermal Conductivity (W/m.K)
High conductivity copper	0.6	395
High density polyethylene	1×10^{-25}	0.16

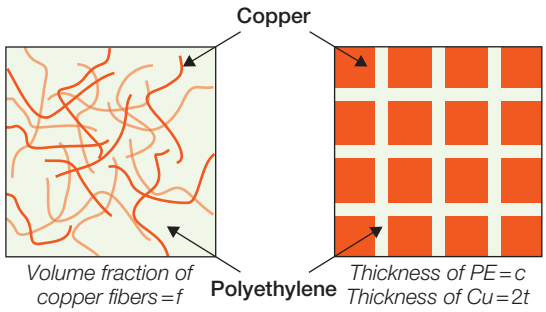


FIGURE 12.3
Two alternative configurations of copper and polyethylene, shown here in two dimensions but easily generalized to three. The one on the left has high electrical conductivity but low thermal conductivity; the one on the right has the opposite.

The thermal conductivity for a random array like this will lie near the lower bound (Equation (11.10)):

$$\widetilde{\lambda}_1 = \lambda_{PE}\left(\frac{\lambda_{Cu} + 2\lambda_{PE} - 2f(\lambda_{PE} - \lambda_{Cu})}{\lambda_{Cu} + 2\lambda_{PE} + f(\lambda_{PE} - \lambda_{Cu})}\right) \tag{12.2}$$

The composite properties $\widetilde{\kappa}_1$ and $\widetilde{\lambda}_1$ are plotted in Figure 12.4, stepping through values of f, giving the upper curve.

The second hybrid is a multilayered composite with three orthogonal families of PE sheet separating blocks of copper. We draw again on the bounds of Section 11.4. When the PE layers are thin, the through-thickness thermal resistances add; the same is true of the electrical resistances. This means that both the electrical and thermal conductivities are given by the harmonic means (Equation (11.24)):

$$\widetilde{\kappa}_2 = \left(\frac{f}{\kappa_{Cu}} + \frac{(1-f)}{\kappa_{PE}}\right)^{-1} \tag{12.3}$$

$$\widetilde{\lambda}_2 = \left(\frac{f}{\lambda_{Cu}} + \frac{(1-f)}{\lambda_{PE}}\right)^{-1} \tag{12.4}$$

The composite properties $\widetilde{\kappa}_2$ and $\widetilde{\lambda}_2$ are also plotted in Figure 12.4, giving the lower curve. The shape of the curve is a consequence of the very different ranges of values of the two properties: a factor of 1000 for λ, a factor of 10^{25} for κ. The two hybrids differ dramatically in their behavior. The hybridization has allowed the creation of "materials" with extreme combinations of conductivities.

Postscript Hybrids of the first type are widely used to give electrical screening to computer and TV cabinets. Those of the second type are less common, but

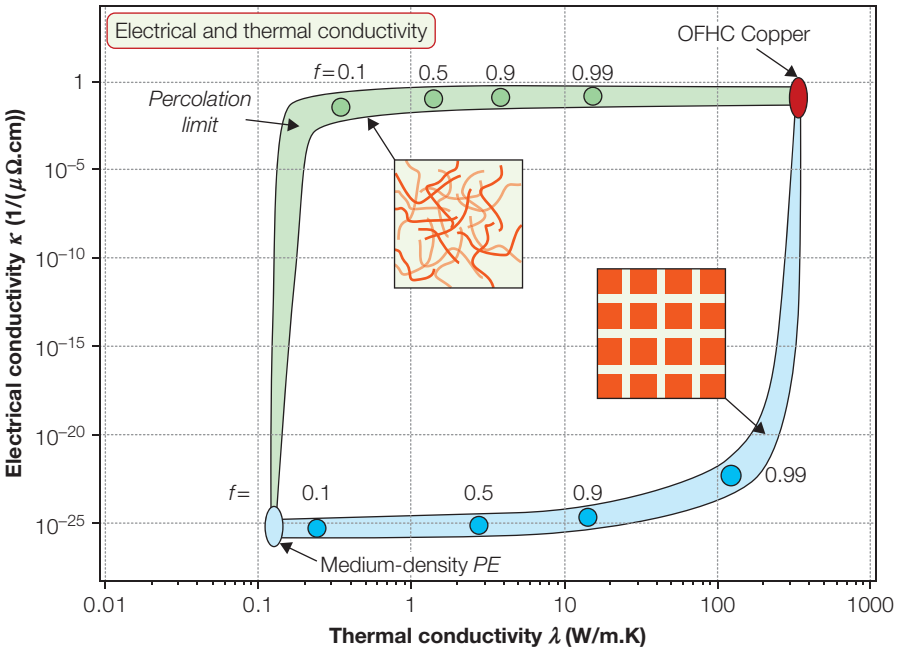


FIGURE 12.4

Two alternative hybrid configurations of copper and polyethylene give very different combinations of thermal and electrical conductivity, and create new "materials" with properties that are not found in homogeneous materials.

could find application as heat sinks for power electronics in which large-scale electrical conduction would lead to coupling and eddy current losses. How would you make the second type of hybrid? Perhaps by thermally bonding a stack of copper sheets interleaved with polyethylene film, dicing the stack normal to the layers and restacking the slices with PE film interlayers, and finally dicing and restacking a third time to give the last set of layers.

Related case studies

6.17 "Materials for heat exchangers"
12.3 "Flexible conductors and percolation"
12.7 "Connectors that don't relax their grip"

12.5 REFRIGERATOR WALLS

The panels of a refrigerator or freezer like that in Figure 12.5 perform two primary functions. The first is to insulate, and for this the through-thickness thermal conductivity must be minimized. The second is mechanical: The walls provide stiffness and strength, and support the shelves on which the contents rest. For a given thickness of panel, the first is achieved by



FIGURE 12.5
A refrigerator. The panels of the container unit must insulate, protect against the external environment, and be stiff and strong in bending.

Table 12.6 Design Requirements for the Insulating Panel

Function	Insulating panel
Constraints	Sufficient stiffness to suppress vibration and support internal loads Low cost Protect against environment Not too thick
Objective	Minimize heat transfer through thickness
Free variable	Material for faces and core; their relative thicknesses

Table 12.7 Superposition Rules for Sandwich Stiffness and Conductivity

Property	Lower Bound	Upper Bound
Flexural modulus	$\widetilde{E}_{flex} = \left(1 - \left(1 - \frac{2t}{d}\right)^3\right) E_f K_s$	$\widetilde{E}_{flex} = \left(1 - \left(1 - \frac{2t}{d}\right)^3\right) E_f$
Through-thickness conductivity	$\widetilde{\lambda}_\perp = \left(\frac{2t/d}{\lambda_f} + \frac{(1-2t/d)}{\lambda_c}\right)^{-1}$ (exact)	

minimizing $\widetilde{\lambda}$, where $\widetilde{\lambda}$ is the appropriate thermal conductivity. The second is achieved by seeking materials or hybrids that maximize $\widetilde{E}_{flex}$, where $\widetilde{E}_{flex}$ is the flexural modulus. (See Table 12.6.)

The method and results We select a hybrid of type "sandwich" and explore how the performance of various combinations of face and core compare with each other and with monolithic materials. The quantity $\widetilde{E}_{flex}$ for the sandwich is given by Equation (11.18) and its through-thickness thermal conductivity is given by Equation (11.24). Simplified equations describing them are assembled in Table 12.7, in which t is the face sheet thickness, d the panel thickness, E_f the modulus of the face material, λ_f and λ_c the conductivities of face and core, and K_s the knock-down factor to allow for core shear, ideally equal to 1 (no shear) but potentially as low as 0.5.

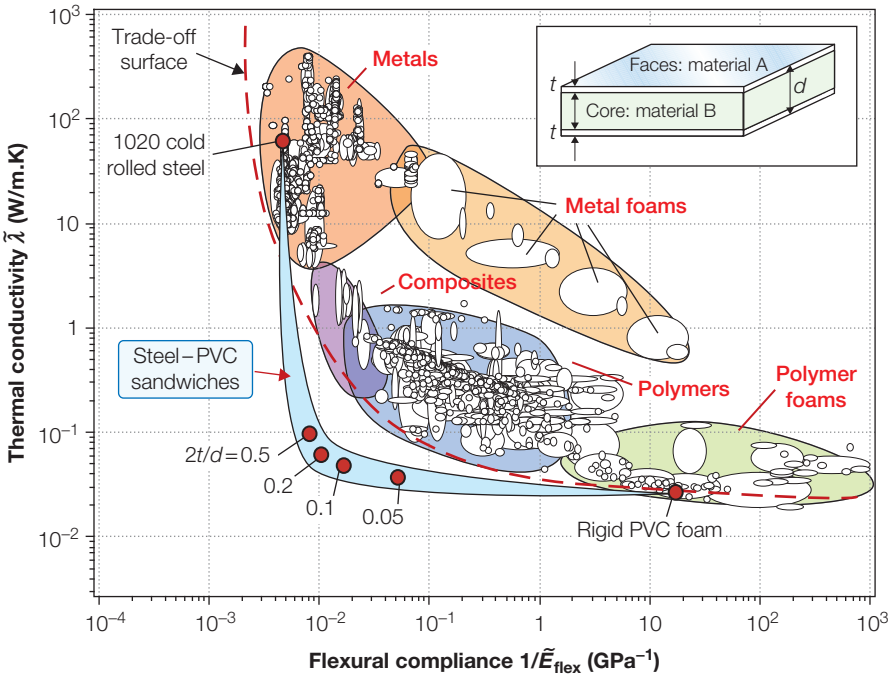


FIGURE 12.6

The blue L-shaped trajectory plots the performance of steel-faced, PVC foam cored sandwiches. The thermal performance is plotted on the vertical axis, the mechanical performance on the horizontal one. Both are to be minimized.

Figure 12.6 shows the appropriate plot, using the thermal conductivity $\widetilde{\lambda}_{\perp}$ and the flexural compliance $1/\widetilde{E}_{flex}$ instead of its inverse so that a minimum is sought for both quantities. The approximate performance of a sandwich with mild steel faces and a core of rigid foamed PVC is plotted using the equations in the table for four values of skin-to-core thickness; all other values are contained in the blue band. The panel offers combinations of stiffness and insulation that cannot be matched by monolithic metals, composites, polymers, or foams. Other combinations of face and core (aluminum or SMC faces with foamed polystyrene core, for example) can be evaluated efficiently using the same chart. Neither combination performs as well as the steel-polystyrene combination, though both come close.

Postscript Adhesive technology has advanced rapidly over the last two decades. Adhesives are now available to bond almost any two materials, and to do so with high bond strength (though some adhesives are expensive). Fabricating the sandwich should not be a problem.

Related case studies

6.14 "Energy-efficient kiln walls"

12.6 "Materials for microwave-transparent enclosures"

12.6 MATERIALS FOR MICROWAVE-TRANSPARENT ENCLOSURES

Microwave-transparent radomes were introduced in Section 6.19. The radome is a thin panel or shell, requiring flexural stiffness and strength but also requiring as low a dielectric constant, ε_r, as possible. Could hybrids offer better performance than monolithic materials?

The method and results Sandwich structures offer flexural stiffness and strength, and allow some control of electrical properties. We therefore explore these, seeking to meet the requirements of Table 12.8.

Figure 12.7 shows the flexural strength, σ_{flex}, of foams, polymers, and ceramics, plotted against the dielectric constant, ε_r. Many polymers have dielectric constants between 2 and 5. Dielectric response is an extensive property—if these polymers are foamed, the dielectric constant of the foam falls linearly with the relative density, approaching 1 at low densities (Equation (11.37)):

$$\widetilde{\varepsilon}_r = 1 + (\varepsilon_{r,s} - 1)\left(\frac{\widetilde{\rho}}{\rho_s}\right) \tag{12.5}$$

where $\varepsilon_{r,s}$ is the dielectric constant of the solid of which the foam is made and $\widetilde{\rho}/\rho_s$ is its relative density. Foams, however, are not very strong. GFRP, with a dielectric constant of 5, is much stronger. Based on a survey of possible faces and cores from among those plotted in Figure 12.7, we choose to explore a sandwich with faces of GFRP and a core of a low-/medium-density expanded polymer foam. The flexural strength of the sandwich, provided it is properly manufactured and the core material has sufficient strength, is (Equation (11.19))

$$(\widetilde{\sigma}_{flex})_{UI} = (1 - (1-f)^2)\,\sigma_f + (1-f)^2\sigma_c \tag{12.6}$$

The dielectric constant for the panel (Equation (11.12) with $f = 2t/d$) is

$$\widetilde{\varepsilon}_r = f\,\varepsilon_f + (1-f)\varepsilon_c \tag{12.7}$$

Table 12.8 Requirements for Low Dielectric Constant Radome Skin

Function	Material for protection of microwave detector
Constraint	Must meet constraints on flexural strength σ_{flex}
Objective	Minimize dielectric constant ε_r
Free variables	Choice of material for face and core Relative thickness of the two materials

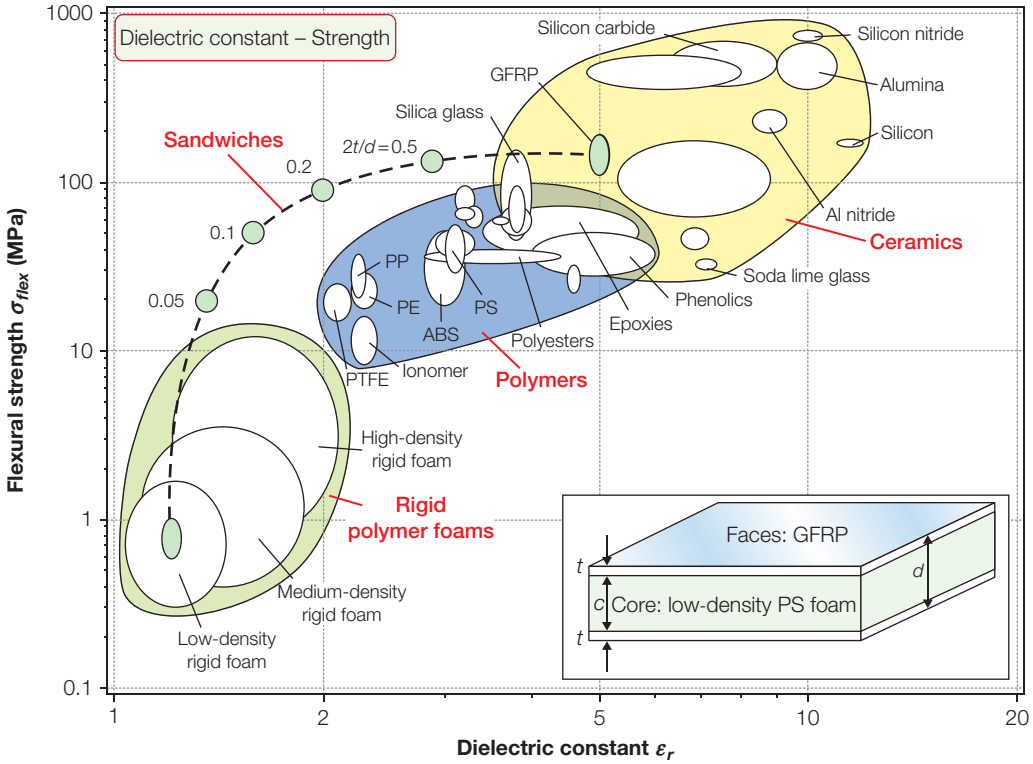


FIGURE 12.7

A plot of flexural modulus and dielectric constant for low dielectric constant materials. The trajectory shows the possibilities offered by hybrids of GFRP and polymer foam.

The simplest way to explore dielectric constants is to plot the two using $f = 2t/d$ as a parameter to link them. Figure 12.7 shows the results. The sandwich allows the creation of a set of materials with combinations of flexural strength and dielectric constant that outperform all the homogeneous materials that appear in the figure—indeed it outperforms even the best composites (not shown here). The figure works by identifying the desired flexural strength, σ_{flex}, and reading off the values of $2t/d$ and dielectric constant.

Related reading

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Related case study

6.19 "Materials for radomes"

12.7 CONNECTORS THAT DON'T RELAX THEIR GRIP

There are kilometers of wiring in a car. The transition to drive-by-wire control systems will increase this further. Wires have ends; they don't do much unless the ends are connected to something. The connectors are the problem: They loosen with time until, eventually, the connection is lost.

Car makers, responding to market forces, now design cars to run for at least 300,000 kilometers and last, on average, 10 years. The electrical system is expected to operate *without servicing* for the lifetime of the car. Its integrity is vital: You would not be happy in a drive-by-wire car with loose connectors. With increasing instrumentation on engine and exhaust systems, many of the connectors get hot; some have to maintain good electrical contact at temperatures up to 200°C (see Table 12.9).

The primary material choice for connectors today is a copper-beryllium alloy, Cu 2% Be: It has excellent conductivity and the high strength needed to act as a spring to give the required clamping force at the connection. But the maximum long-term service temperature of copper-beryllium alloys is only about 130°C; at higher temperatures, creep relaxation causes the connector to loosen its grip. The challenge: to suggest a way to solve this problem.

The method and results The answer is to separate the functions, select the best material for each, check for compatibility, and combine the materials to make a hybrid. So here goes. Function 1: Conduct electricity. Copper excels at this; no other affordable material is as good. Its alloys (among them copper-beryllium) are stronger, but at the cost of some loss of conductivity and a great increase in price. We choose copper to provide the conduction. Function 2: Provide clamping force for the lifetime of the vehicle. The material chosen to fill function 2 will have to be bonded to the copper, and if the combination is not to distort when heated, it must have the same expansion coefficient.

Figures 12.8 and 12.9 guide the choice. The first shows the maximum service temperature and expansion coefficient for copper, Cu-2% Be, and a range of steels. The box encloses materials with the same expansion coefficient as copper. Type 302 and 304 austenitic stainless steels match copper in expansion coefficient and can be used at much higher temperatures. But do they

Table 12.9 Design Requirements for the Connector

Function	Hybrid conductor for electrical connector
Constraints	Provide good electrical connection Maintain clamping force at 200°C for life of vehicle
Objective	Minimize cost
Free variable	Material 1 and 2; their relative thicknesses

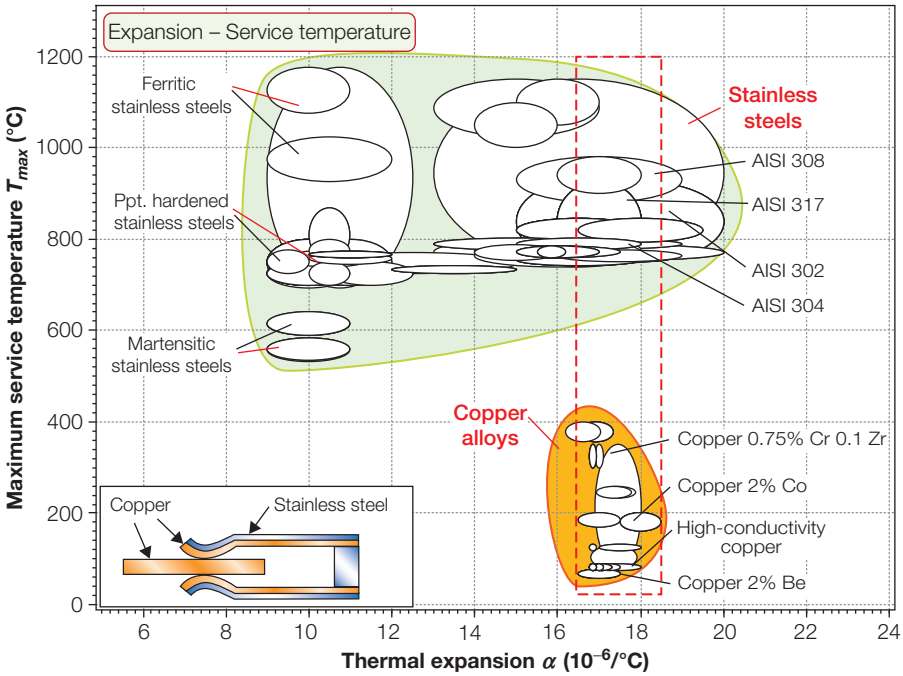


FIGURE 12.8

A hybrid connector. We seek materials with matching thermal expansion, one of which retains strength and stiffness well above 200°C. Copper is chosen for Material 1 because of its excellent electrical conductivity. Type 302 or 304 stainless steel is a good choice for Material 2.

make good springs? And are they affordable? Figure 12.9 answers these questions. Good materials for springs (Section 6.7) are those with high values of σ_f^2/E—this appears as one axis of the chart. The other axis is approximate price/kg. The chart shows that both 302 and 304 stainless steels, in the wrought condition, are almost as good as Cu 2% Be as a spring and considerably cheaper.

Postscript The proposed solution, then, is a hybrid of copper and type 302 stainless steel, roll-bonded to form a bilayer like that shown in the inset on both figures. Further detailed design will, of course, be needed to establish the thicknesses of each layer, the best degree of cold work, the formability, and the resistance to the environment in which it will be used. But the method has guided us to a sensible concept, quickly and efficiently.

Related case studies

6.7 "Materials for springs"

10.8 "Ultra-efficient springs"

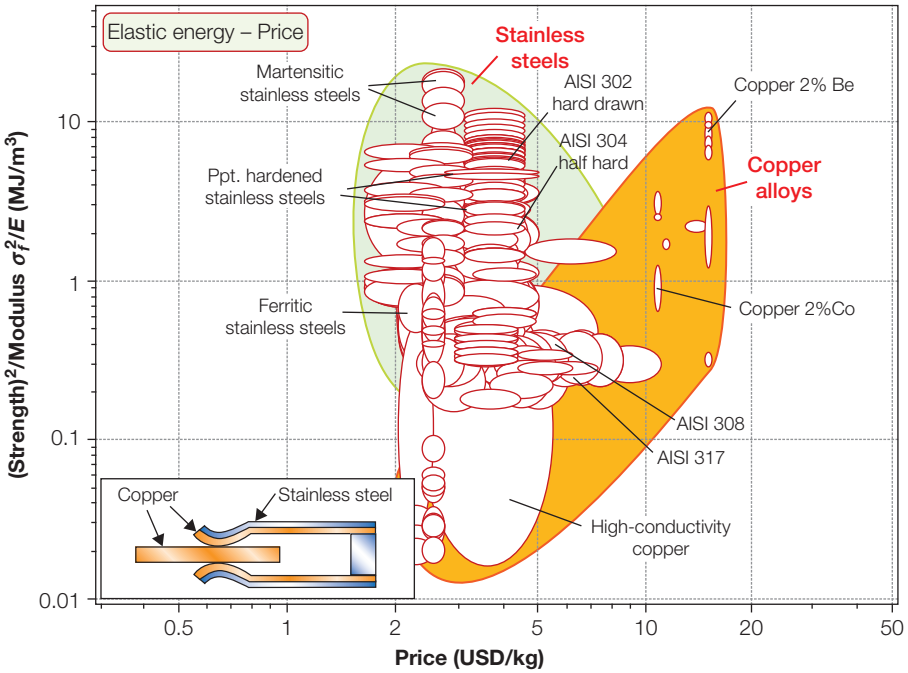


FIGURE 12.9

The connector has two jobs—to conduct and to exert a clamping force that does not relax. High-conductivity copper and 304 stainless steel are both much cheaper that Cu 2% Be. Roll-bonding them will of course add cost, but in large volume production it could be competitive—and it solves the relaxation problem.

12.8 EXPLOITING ANISOTROPY: HEAT-SPREADING SURFACES

A saucepan made from a single material, when heated on an open flame, develops hot spots that can burn its contents at those spots. That is because the saucepan is thin; heat is transmitted through the thickness more quickly than it can be spread transversely to bring the entire pan surface to a uniform temperature. The metals of which saucepans are usually made—cast iron, aluminum, stainless steel, or copper—have thermal conductivities that are isotropic, the same in all directions. What we clearly want is a thermal conductivity that is higher in the transverse direction than in the through-thickness direction. A bilayer (or multilayer) hybrid can achieve this. Table 12.10 summarizes the situation.

The method and results Heat transmitted in the plane of a bilayer (Figure 12.10, *inset*) has two parallel paths; the total heat flux is a sum of

Table 12.10 Design Requirements for the Panel	
Function	Heat-spreading surface
Constraint	Temperature up to 200°C without distortion
Objective	Maximize thermal anisotropy while maintaining good conduction
Free variable	Choice of materials and their relative thicknesses

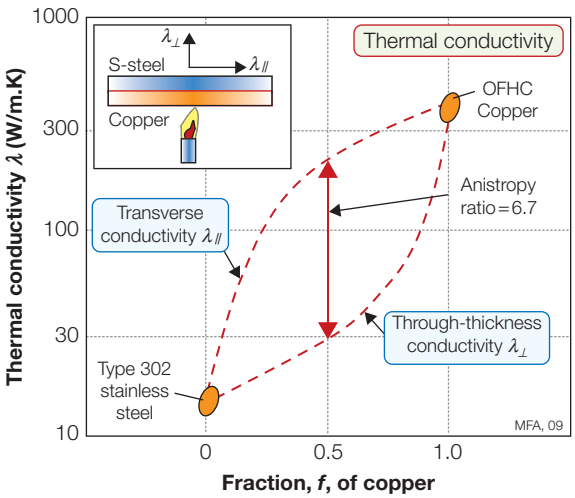


FIGURE 12.10
Creating anisotropy. A bilayer of copper and stainless steel creates a "material" with good conductivity and an anisotropy ratio greater than 6.

the heat flux in each of the paths. If it is made of a layer of material 1 with thickness t_1 and conductivity λ_1, bonded to a layer of material 2 with thickness t_2 and conductivity λ_2, the conductivity parallel to the layers is

$$\tilde{\lambda}_{//} = f\lambda_1 + (1-f)\lambda_2 \tag{12.8}$$

(Equation (11.9) with $f = t_1/(t_1 + t_2)$. Perpendicular to the layers the conductivity is

$$\tilde{\lambda}_{\perp} = \left(\frac{f}{\lambda_1} + \frac{(1-f)}{\lambda_2}\right)^{-1} \tag{12.9}$$

(the harmonic mean of Equation (11.24)).

If the bilayer is made of materials that differ greatly in thermal expansion coefficient, the pan will distort when heated. We therefore seek a pair of

materials with almost the same expansion coefficient but that differ as much as possible in conductivity in order to maximize the difference between Equations (12.10) and (12.11).

The results Copper is an excellent thermal conductor. We saw, in Case Study 12.7, that the thermal expansion of copper matches that of 302 or 304 stainless steel. The thermal conductivities of the two, however, differ. Figure 12.10 shows the conductivities $\widetilde{\lambda}_{\perp}$ and $\widetilde{\lambda}_{//}$ of the bilayer as a function of the relative thickness f for a bilayer of copper (λ_1 = 390 W/m.K) and austenitic stainless steel (λ_2 = 16 W/m.K). The maximum separation between $\widetilde{\lambda}_{\perp}$ and $\widetilde{\lambda}_{//}$ occurs broadly at f = 0.5 (each material occupies about half the thickness) when the ratio of the two conductivities (the anisotropy ratio) is 6.7. The hybrid has extended the occupied area of property space along an unusual dimension—thermal anisotropy.

Related case studies

6.15 "Materials for passive solar heating"

12.7 "Connectors that don't relax their grip"

12.9 THE MECHANICAL EFFICIENCY OF NATURAL MATERIALS

> "As a general principle natural selection is continually trying to economize every part of the organization."
>
> —Charles Darwin, writing 150 years ago

Natural materials are remarkably efficient. By efficient we mean that they fulfill the complex requirements posed by the way plants and animals function and that they do so using as little material as possible. Many of these requirements are mechanical in nature: the need to support static and dynamic loads created by the mass of the organism or by wind loading, the need to store and release elastic energy, the need to flex through large angles, and the need to resist buckling and fracture.

Virtually all natural materials are hybrids. They consist of a relatively small number of polymeric and ceramic components or building blocks that are often composites themselves. Plant cell walls, for instance, combine cellulose, hemicellulose, and pectin, and can be lignified; animal tissue consists largely of collagen, elastin, keratin, chitin, and minerals such as salts of calcium or silica. From these limited "ingredients" nature fabricates a remarkable range of structured hybrids. Wood, bamboo, and palm consist of cellulose fibers in a lignin/hemicellulose matrix, shaped to hollow prismatic cells of varying wall thickness. Hair, nail, horn, wool, reptilian scales, and hooves are made

of keratin while insect cuticle contains chitin in a matrix of protein. The dominant ingredient of mollusk shell is calcium carbonate, bonded with a few percent of protein. Dentine, bone, and antler are formed of "bricks" of hydroxyapatite cemented together with collagen. Collagen is the basic structural element for soft and hard tissues in animals, such as tendon, ligament, skin, blood vessels, muscle, and cartilage, often used in ways that exploit shape.

From a mechanical point of view, there is nothing very special about the individual building blocks. Cellulose fibers have Young's moduli that are about the same as those of nylon fishing line, but a lot less than those of steel; and the lignin-hemicellulose matrix in which they are embedded has properties very similar to those of epoxy. Hydroxyapatite has a fracture toughness comparable with that of human-made ceramics; that is to say it is brittle. It is thus the *configuration* of the components that give rise to the striking efficiency of natural materials.

The complex hierarchies of configurations that make up wood and bone are illustrated in Figures 12.11 and 12.12. The stiffness, strength, and toughness of wood derive largely from the stiffness, strength, and toughness of the cellulose molecule, shown on the left of Figure 12.11. Crystalline microfibrils are built up of aligned molecules some 30 to 60 nm in length. These form the reinforcing fibers of the lamellae, the matrix of which is amorphous lignin and hemicellulose. Stacks of lamellae in the four-layer pattern and fiber orientation (*right*) become the structural material of the cell wall. The fibrils of the primary cell wall interweave randomly, as in cotton wool. In

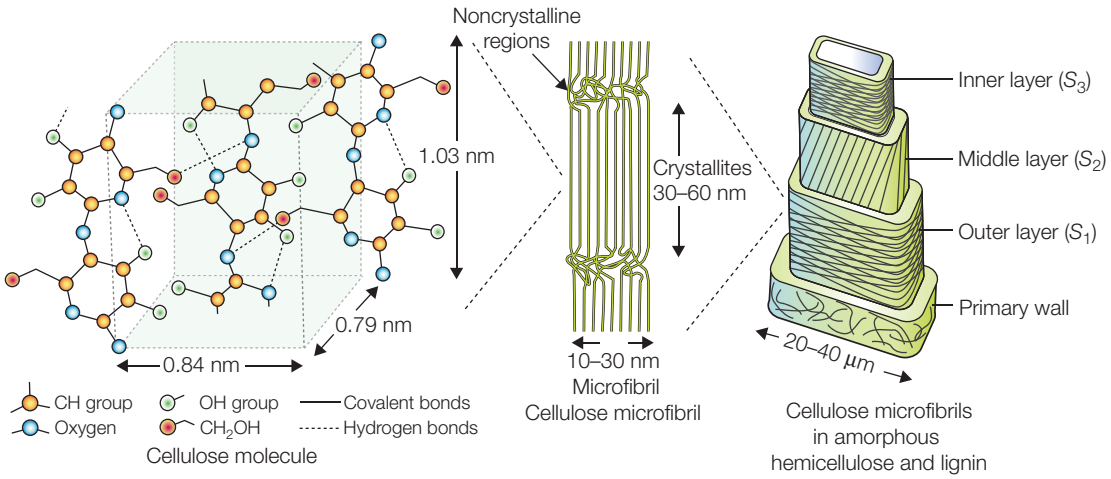


FIGURE 12.11
The hierarchical structure of wood.

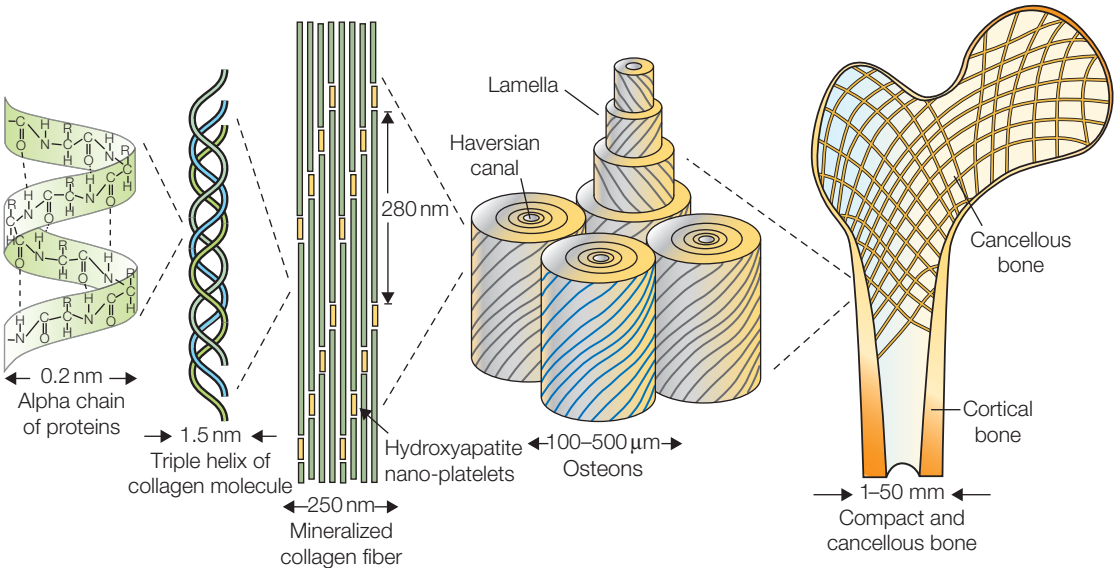


FIGURE 12.12
The hierarchical structure of bone.

subsequent layers the fibrils are parallel and closely packed. The outer S_1 layer has lamellae with alternate right- and left-handed spiral winding of fibrils. Beneath it, the thicker S_2 layer has fibrils that are more nearly oriented along the axis of the cell. The innermost S_3 layer has a lay-up like that of S_1. The cell as a whole is bonded to its neighbors by the middle lamella (not shown), a lignin-pectin complex devoid of cellulose.

There is an interesting parallel between the hierarchical structure of bone and that of wood, despite the great differences in their molecular chemistry (Figure 12.12). The starting point here is the triple-helical structure of the collagen molecule, shown on the left. But—unlike wood—this becomes the matrix, not the reinforcement of the mineralized tissue of bone. Hydroxyapatite nanoplatelets deposit in the nascent tissue and increase in volume fraction over time to produce the mature osteons, with an ordered lay-up of highly mineralized fibers with the strength and stiffness to support structural loads that bone must carry in a mature organism. At the most macro scale, nearly fully dense compact bone provides the outer structure of whole bone while highly porous trabecular bone fills the vertebrae, shell-like bones such as the skull, and the ends of the long bones such as the femur.

The same as with engineering materials, the building blocks of natural materials can be grouped into classes: *bio-ceramics* (calcite, aragonite, hydroxyapatite), *biopolymers* (the organic building blocks: polysaccharides; cellulose;

and the proteins chitin, collagen, silk, and keratin), and *natural elastomers* (elastin, resilin, abductin, skin, artery, and cartilage). These combine to give a range of hybrids, among them composites and sandwiches (bone, antler, enamel, dentine, shell, cuttle, and coral), cellular structures (natural cellular materials such as wood, cork, palm, bamboo, trabecular bone), and segmented structures (scales, hair). Their properties, like those of engineering materials, can be explored and compared using material property charts. We conclude this chapter by examining charts for natural hybrids and the building blocks from which they are made.

The Young's modulus–density chart Figure 12.13 shows data for Young's modulus, E, and density, ρ. Those for the classes of natural materials are circumscribed by large envelopes; class members are shown as smaller bubbles within them. Data for woods, palms, corks, trabecular (foam-like)

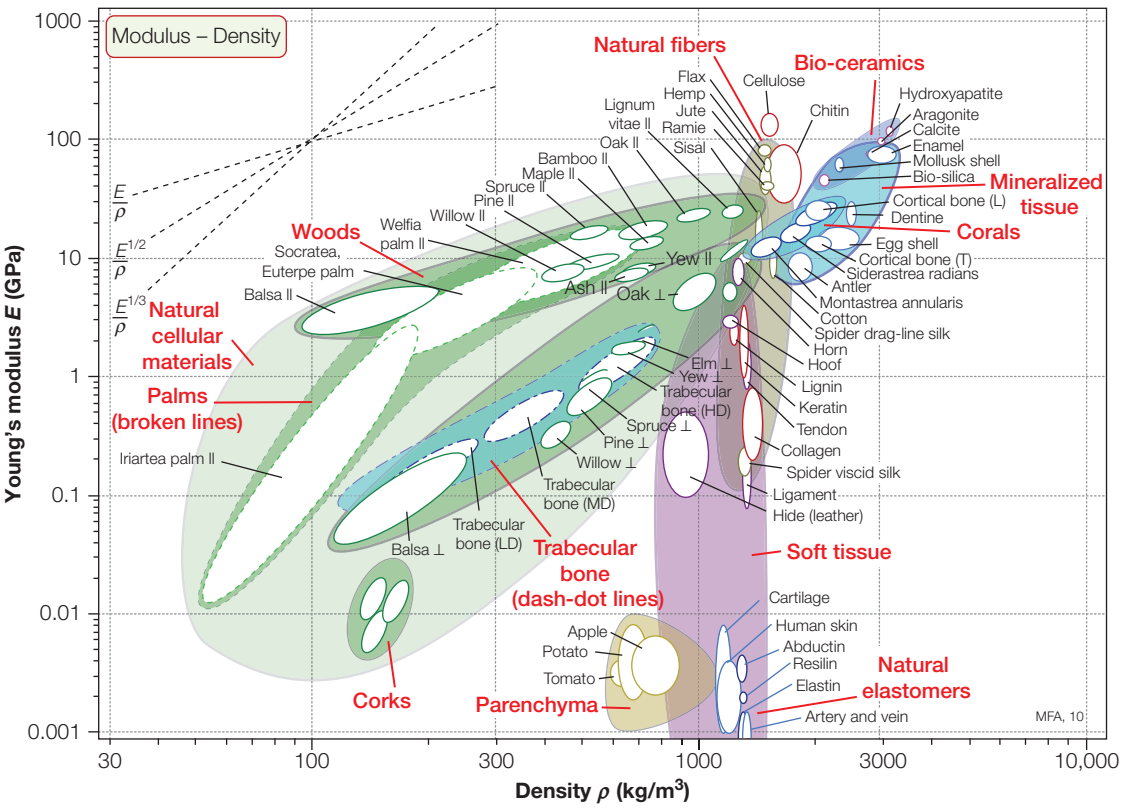


FIGURE 12.13
A material-property chart for natural materials, plotting Young's modulus against density. The guide lines identify structurally efficient materials that are light and stiff.

bone, and corals are enclosed in subsidiary envelopes. Natural fibers (silk, flax, jute, and the like) have their own envelope, as does mineralized tissue (bone, shell, etc.) and soft tissue (ligament, cartilage, and so forth). Where they differ, the moduli parallel (symbol ||) and perpendicular (symbol ⊥) to the fiber orientation or grain have been plotted separately. Trabecular bone exhibits a particularly wide range of densities; for these, three bubbles for high density (HD), medium density (MD), and low density (LD) are plotted. The familiar stiffness guidelines E/ρ, $E/p^{1/2}/\rho$, and $E^{1/3}/\rho$ are shown, each representing the material index for a particular mode of loading.

The natural polymer with the highest efficiency in tension, measured by the index E/ρ, is cellulose; it exceeds that of steel by a factor of about 2.6. The high values for the fibers flax, hemp, and cotton derive from this. Wood, palm, and bamboo are particularly efficient in bending and resistant to buckling, as indicated by the high values of the flexure index $E^{1/2}/\rho$ when loaded parallel to the grain. That for balsa wood, for example, can be five times greater than that for steel.

The tensile strength–density chart Figure 12.14 shows data for the strength, σ_f, and density, ρ, of natural materials. The color coding and envelope scheme parallels that of Figure 12.13. For natural ceramics the tensile strength is identified with the flexural strength (modulus of rupture) in the beam-bending symbol (T). For natural polymers and elastomers, the strengths are tensile strengths. And for natural cellular materials, the tensile stress is either the plateau stress, or flexural strength, symbol (T), depending on the nature of the material. Where they differ, the strengths parallel (symbol ||) and perpendicular (symbol ⊥) to the fiber orientation or grain have been plotted separately.

Evolution to provide tensile strength would—we anticipate—lead to materials with high values of σ_f/ρ, where strength in bending or buckling is required we expect to find materials with high $\sigma_f^{2/3}/\rho$. Silk and cellulose have the highest values of σ_f/ρ; that of silk is even higher than that of carbon fibers. The fibers of flax, hemp, and cotton, too, have high values of this index. Bamboo, palm, and wood have high values of $\sigma_f^{2/3}/\rho$, giving resistance to flexural failure.

The Young's modulus–strength chart Data for the strength, σ_f, and modulus, E, of natural materials are shown in Figure 12.15. Two of the combinations are significant. Materials with large values of σ_f^2/E store elastic energy and make good springs; those with large values of σ_f/E have exceptional resilience. Both are plotted in the figure. Silks (including the silks of spider webs) stand out as exceptionally efficient, having values of σ_f^2/E that exceed those of spring steel or rubber. High values of the other index, σ_f/E,

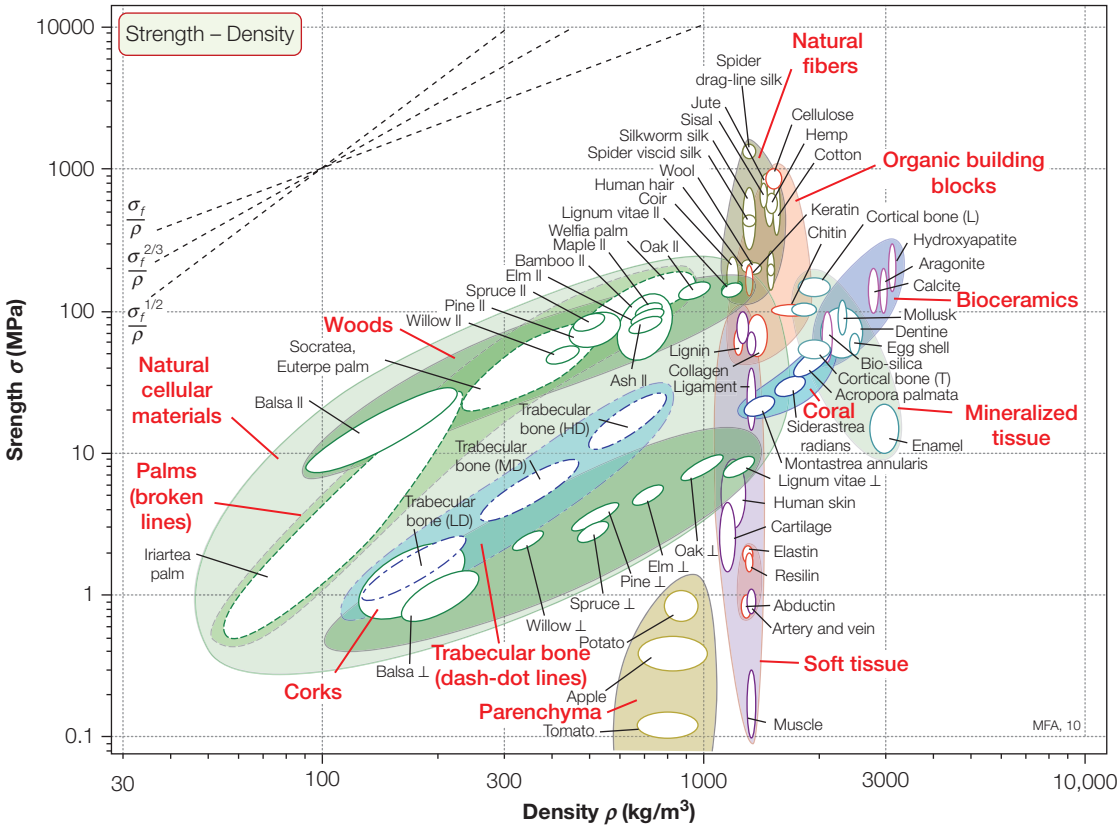


FIGURE 12.14

A material-property chart for natural materials, plotting strength against density. Guide lines identify structurally efficient materials that are light and strong.

mean that a material allows large, recoverable deflections and, for this reason, makes good elastic hinges. Palm (coconut timber) has a higher value of this index than wood, allowing palms to flex in a high wind. Nature makes much use of these: Skin, leather, and cartilage are all required to act as flexural and torsional hinges.

The toughness–Young's modulus chart The toughness of a material measures its resistance to the propagation of a crack. The limited data for the toughness, J_c, and Young's modulus, E, of natural materials are shown in Figure 12.16. When the component is required to absorb a given *impact energy* without failing, the best material will have the largest value of J_c. These materials lie at the top of Figure 12.16: Antler, hoof, horn, bamboo, and woods stand out. When instead a component containing a crack must

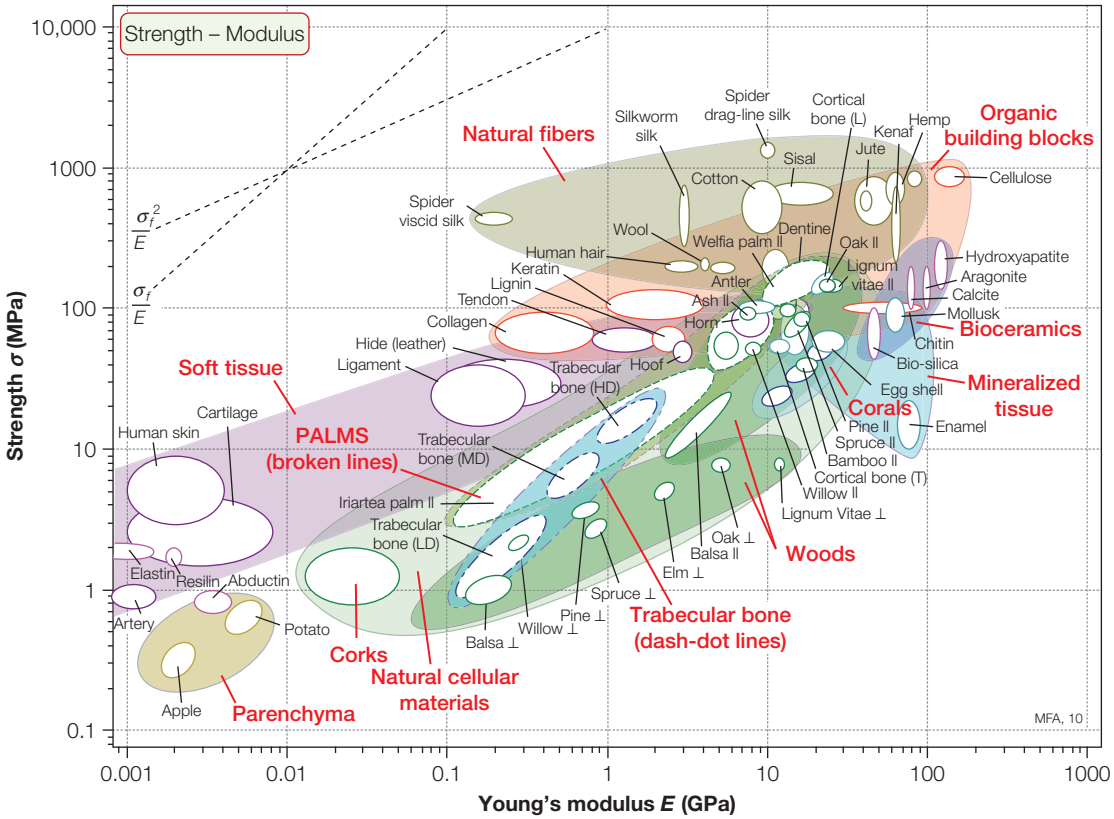


FIGURE 12.15

A material-property chart for natural materials, plotting Young's modulus against strength. The guide lines identify materials that store the most elastic energy per unit volume and that make good elastic hinges.

carry a given *load* without failing, the safest choice of material is that with the largest values of the fracture toughness $K_{1c} \approx (EJ_c)^{1/2}$. Diagonal contours sloping from the upper left to the lower right in Figure 12.16 show values of this index. Mollusk shell and tooth enamel stand out.

Many engineering materials (e.g., steels, aluminum, alloys) have values of J_c and K_{1c} that are much higher than those of the best natural materials. However, the toughness of natural ceramics like nacre, dentine, cortical (dense) bone, and enamel is an order of magnitude higher than that of conventional engineering ceramics like alumina. Their toughness derives from their segmented structure: platelets of ceramics such as calcite, hydroxyapatite or aragonite, bonded by a small volume fraction of polymer, usually collagen; it increases with decreasing mineral content and increasing collagen content.

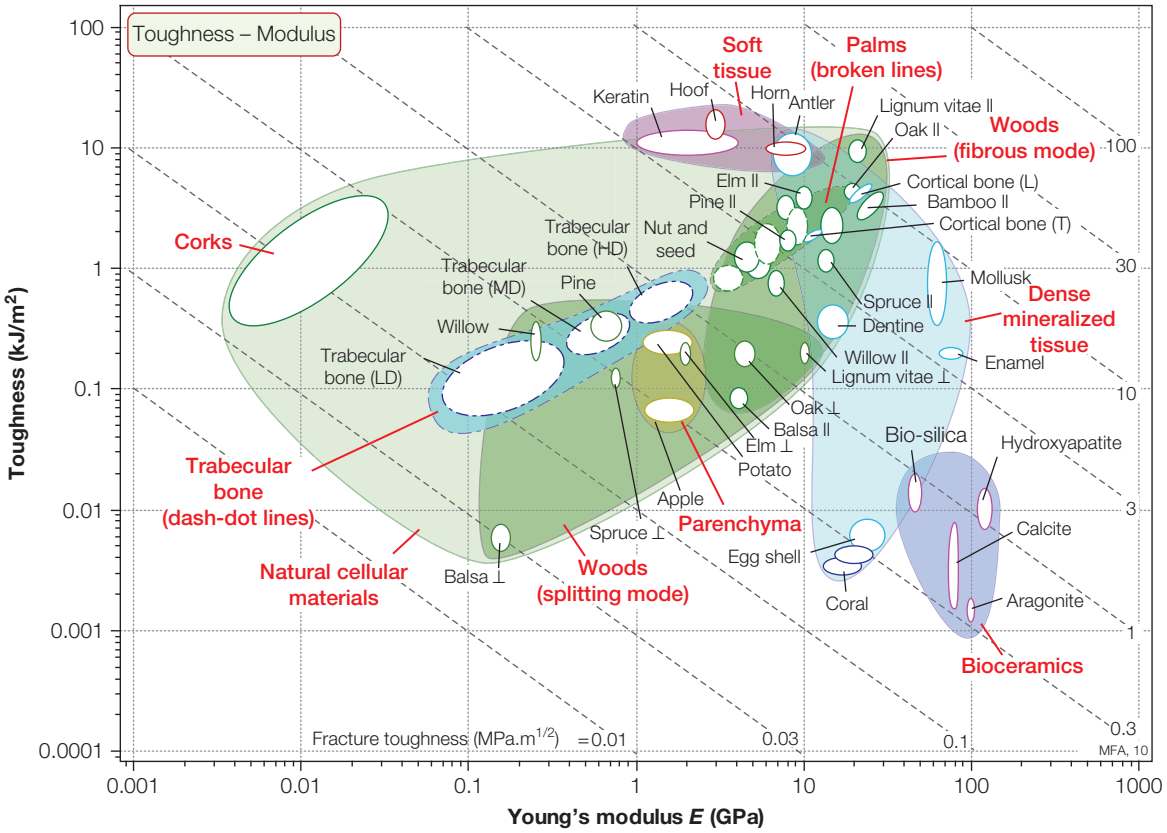


FIGURE 12.16

A material-property chart for natural materials, plotting toughness against Young's modulus. The guide lines show fracture toughness $(EJ_c)^{1/2}$ $(\text{MPa})^{1/2}$.

Postscript The charts bear out Darwin's idea that natural materials evolve to make the most of what is available to them—they are efficient, in the sense used earlier. But there is more to it than that. For deeper insight, see the books listed in "Further reading."

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CHAPTER 13

Processes and Process Selection



Die casting. Image courtesy of Thomas Publishing, New York.

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13.1 INTRODUCTION AND SYNOPSIS

A *process* is a method of shaping, joining, or finishing a material. *Sand casting*, *injection molding*, *fusion welding*, and *electro-polishing* are all processes; there are hundreds of them. The choice, for a given component, depends on the material of which it is to be made; on its size, shape, and required precision; and on how many are to be made—in short, on the *design requirements*.

To select processes we must first classify them. Section 13.2 develops the classification. It is used to structure Section 13.3, in which process *families* and their *attributes* are described: the materials they can handle, the shapes they can make, and the precision with which they can do it.

Processing has dual functions. The obvious one is that of shaping, joining, and finishing. The less obvious one is that of property control. Metals are strengthened by rolling and forging; steels are heat-treated to enhance hardness and toughness; polymers are drawn to increase modulus and strength; and ceramics are hot-pressed, again to increase strength. Process-property relationships are explored more closely in Section 13.4.

Process selection—finding the best match between process attributes and design requirements—is the subject of Sections 13.5 and 13.6. In using the methods developed there, one should not forget that material, shape and processing interact (Figure 13.1). Material properties and shape limit the choice of

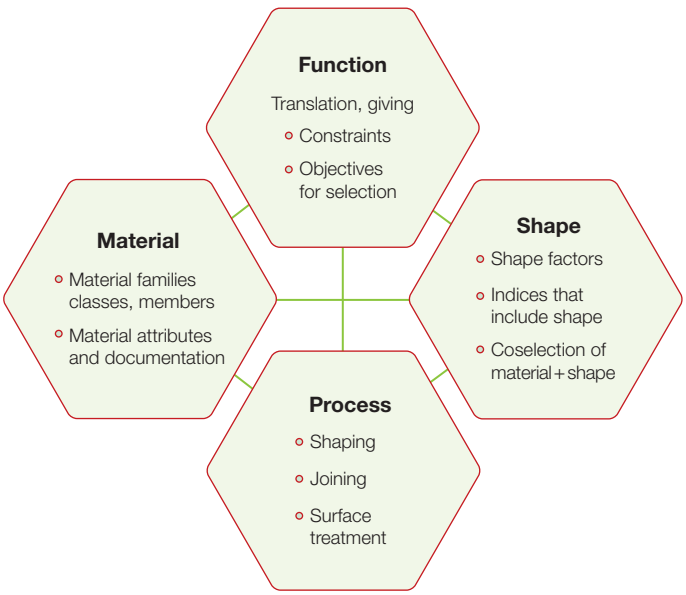


FIGURE 13.1
Processing selection depends on material and shape. The "process attributes" are used as criteria for selection.

process: Ductile materials can be forged, rolled, and drawn; those that are brittle may have to be shaped using powder methods. Materials that melt at modest temperatures to low-viscosity liquids can be cast; those that do not have to be processed by other routes. Shape, too, can influence the choice of process. Slender shapes can be made easily by rolling or drawing but not by casting. Hollow shapes cannot be made by forging, but they can by casting or molding. Conversely, processing affects properties. Rolling and forging change the hardness and texture of metals and align the inclusions they contain, enhancing strength and ductility. Heat treatment allows manipulation of strength, ductility, and toughness. Composites do not exist until they are processed; before processing, they are just a soup of polymer and a sheaf of fibers.

Like the other aspects of design, process selection is an iterative procedure. The first iteration gives one or more possible process routes. The design must then be rethought to adapt it, as far as possible, to ease manufacture by the most promising route. The final choice is based on a comparison of *process cost*, requiring the use of cost models developed in Section 13.6, and on *documentation*: guidelines, case histories, and examples of process routes used for related products. Documentation also helps in dealing with the coupling between process and material properties.

The chapter ends, as always, with a summary and annotated recommendations for further reading.

13.2 CLASSIFYING PROCESSES

Manufacturing processes can be classified under the headings shown in Figure 13.2. *Primary shaping* create shapes. The first row lists six classes of primary forming processes: casting, molding, deformation, powder methods, methods for forming composites, and special methods such as rapid prototyping. *Secondary processes* modify shapes or properties; here they are shown as "machining," which adds features to an already shaped body, and "heat treatment," which enhances surface or bulk properties. Below these come *joining* an, *surface treatment* or *finishing*.

Figure 13.2 has merit as a flow chart; it shows a progression through the manufacturing route. It should not be treated too literally, however; the order of the steps can be varied to suit the needs of the design. The point it makes is that there are three broad process families: shaping, joining, and finishing. The attributes of one family differ so greatly from those of another that, in assembling and structuring data for them, they must be treated separately.

To organize processes in more detail, we need a hierarchical classification like that used for materials in Chapter 5. Figure 13.3 shows part of it. The process universe has three families: shaping, joining, and finishing. In this figure, the shaping family is expanded to show classes: casting, deformation,

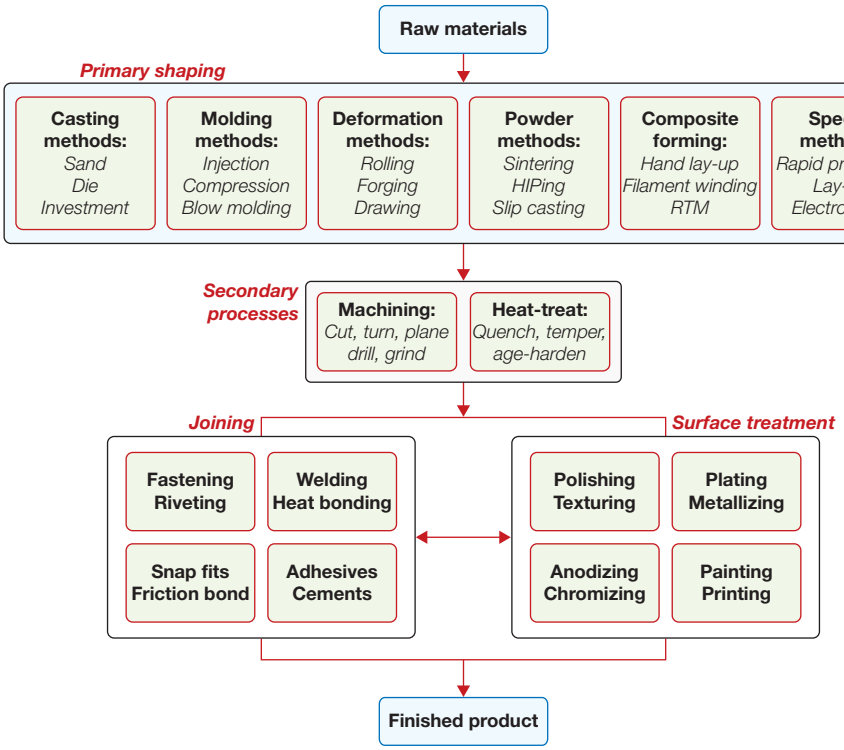


FIGURE 13.2

The classes of process. The first row contains the family of shaping processes; below lie the secondary processes of machining and heat treatment, followed by the families of joining and finishing (surface treatment) processes.

molding, and so on. One of these—molding—is again expanded to show its members: rotation molding, blow molding, injection molding, and so forth. Each of these has certain attributes: the materials it can handle, the shapes it can makes, their size, precision, and an optimum batch size (the number of units that molding can make economically).

Expanding members of the shaping family

Expanding the casting family at about the same level of detail as that used for molding in the figure gives the result:

1. Sand casting
2. Shaping–casting
3. Die casting
4. Investment casting

The other two families are partly expanded in Figure 13.4. The joining family contains three broad classes: adhesives, welding, and fasteners. In this

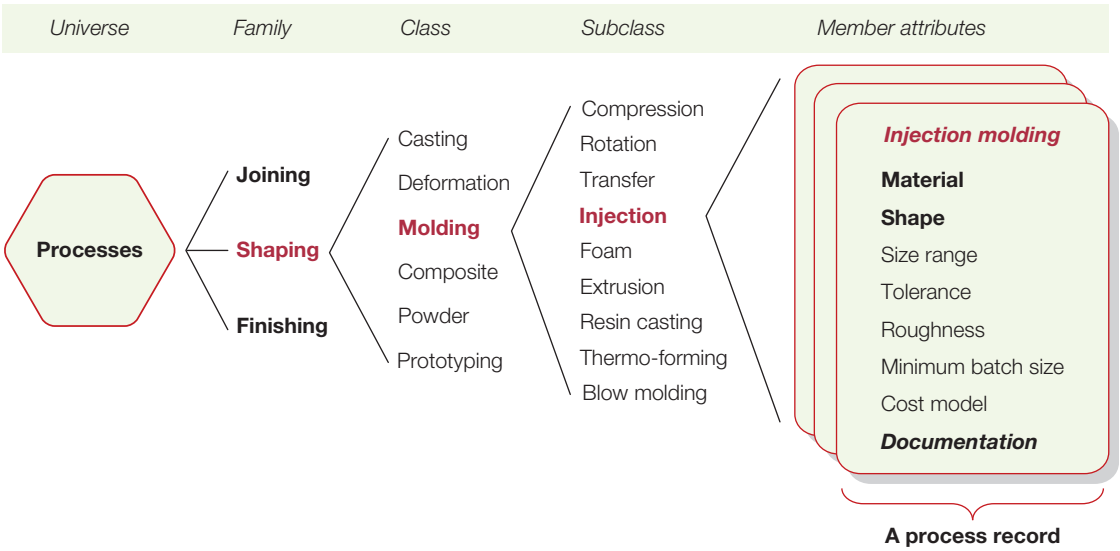


FIGURE 13.3
The taxonomy of the kingdom of process with part of the *shaping* family expanded. Each member is characterized by a set of attributes. Process selection involves matching these to the requirements of the design.

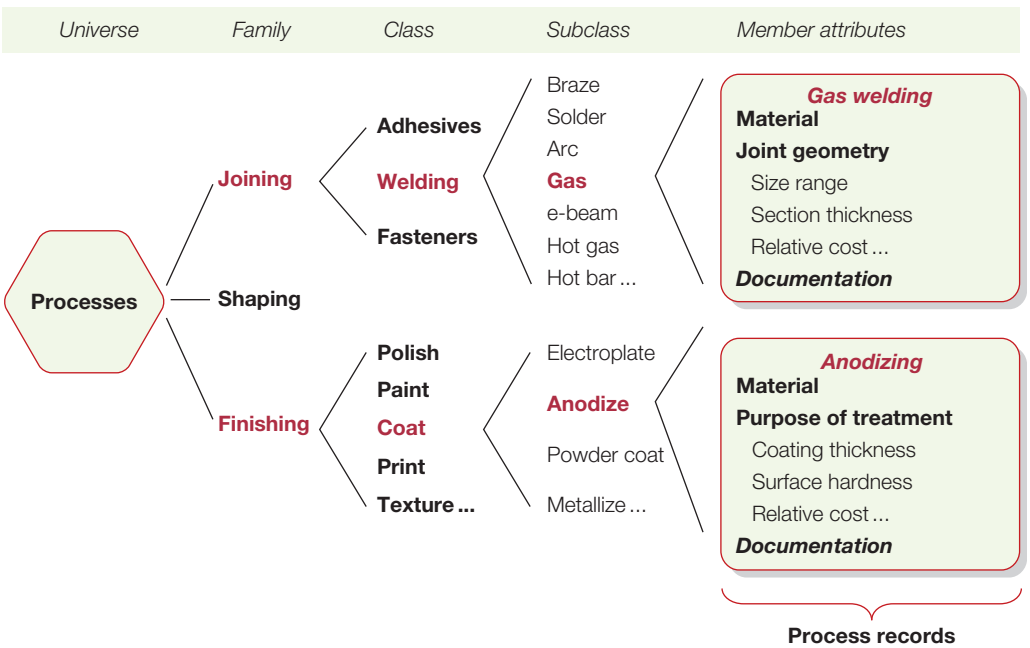


FIGURE 13.4
The taxonomy of the process kingdom again, with the families of *joining* and *finishing* partly expanded.

figure one of them—welding—is expanded to show its members. As before, each member has attributes. The first is the materials it can join. After that the attribute list differs from that for shaping. Here the geometry of the joint and the way it will be loaded are important, as are requirements that the joint can or cannot be disassembled, that it be watertight, that it be electrically conducting, and the like.

The lower part of the figure expands the family of finishing. Some of the classes it contains are shown; one—coating—is expanded to show some of its members. As with shaping and joining, the material to be coated is an important attribute but the others differ. Most important is the purpose of the treatment (protection, surface hardening, decoration, etc.), followed by properties of the coating itself.

Expanding the joining family

Expanding the fasteners family at about the same level of detail as that used for molding in the figure, gives the result:

1. Rivets and staples
2. Joining—fasteners
3. Threaded fasteners
4. Sewing
5. Snap fits

With this background we can embark on a lightning tour of processes. It will be kept as concise as possible; details can be found in the numerous books listed in "Further reading" (Section 13.9).

13.3 THE PROCESSES: SHAPING, JOINING, FINISHING

Shaping processes

In ***casting*** (Figure 13.5), a liquid is poured or forced into a mold where it solidifies by cooling. Casting is distinguished from molding, which comes next, by the low viscosity of the liquid: It fills the mold by flow under its own weight (as in gravity sand and investment casting) or under a modest pressure (as in die casting and pressure sand casting). Sand molds for one-off castings are cheap; metal dies for die casting large batches can be expensive. Between these extremes lie a number of other casting methods: shell, investment, plaster mold, and so forth.

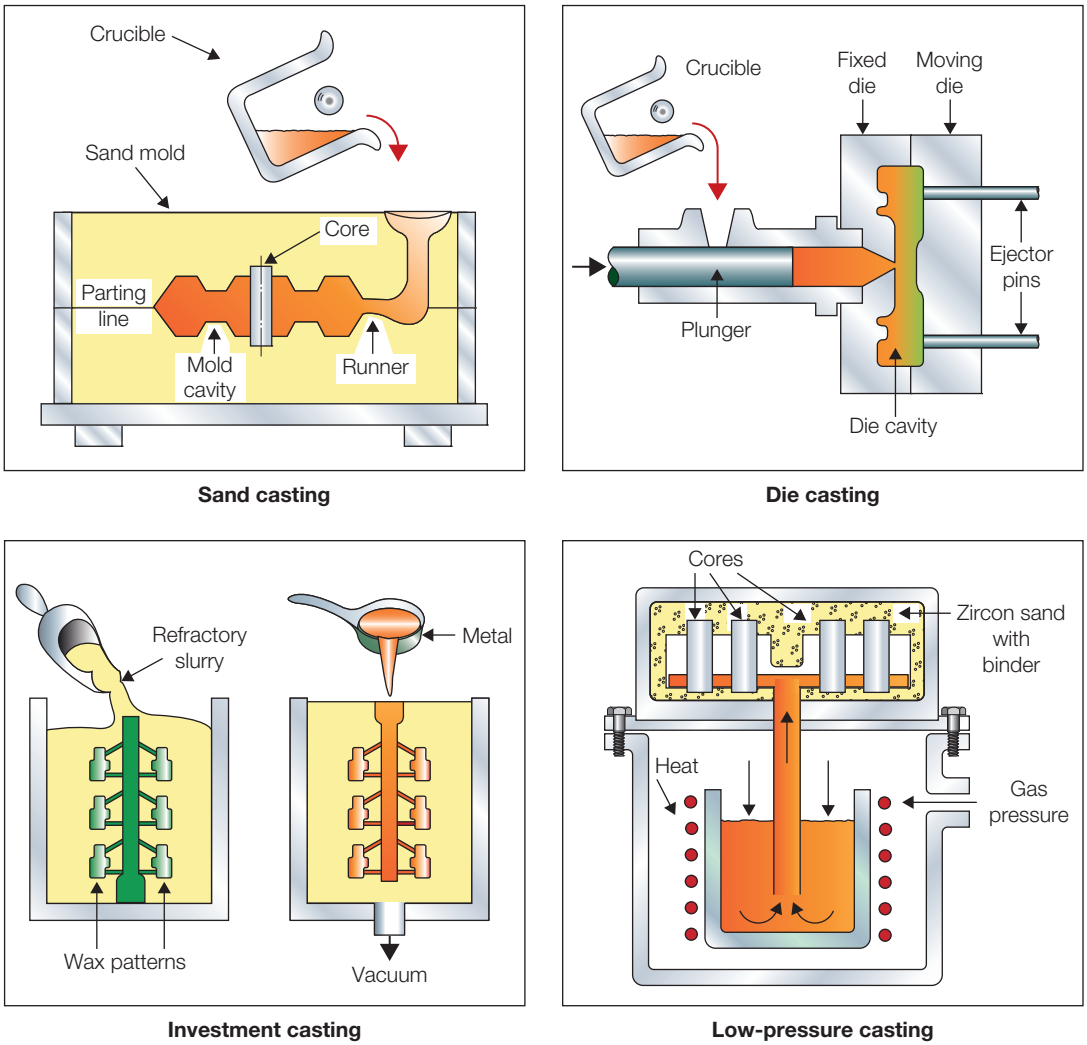


FIGURE 13.5

Casting processes. In *sand casting*, liquid metal is poured into a split sand mold. In *die casting*, liquid is forced under pressure into a metal mold. In *investment casting*, a wax pattern is embedded in a refractory, melted out, and the cavity filled with metal. In *pressure casting*, a die is filled from below, giving control of atmosphere and of the flow of metal into the die.

Cast shapes must be designed for easy flow of liquid to all parts of the mold, and for progressive solidification that does not trap pockets of liquid in a solid shell, giving shrinkage cavities. Whenever possible, section thicknesses are made uniform (the thickness of adjoining sections should not

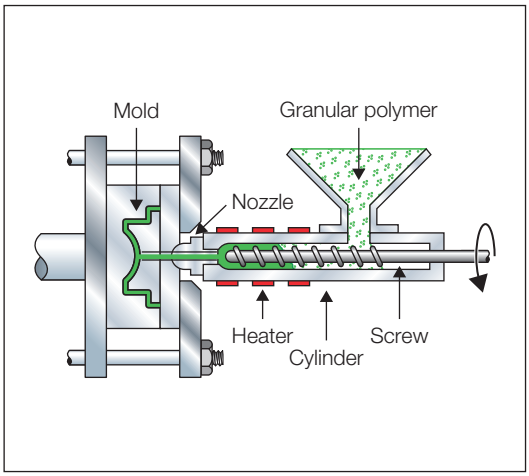
differ by more than a factor of 2). The shape is designed so that the pattern and the finished casting can be removed from the mold. Keyed-in shapes are avoided because they lead to "hot tearing" (a tensile creep fracture) as the solid cools and shrinks. The tolerance and surface finish of a casting vary from poor for sand casting to excellent for precision die castings; they are quantified in Section 13.6.

When metal is poured into a mold, the flow is turbulent, trapping surface oxide and debris within the casting, giving casting defects. These are avoided by filling the mold from below in such a way that flow is laminar, driven by a vacuum or gas pressure, as shown in Figure 13.5.

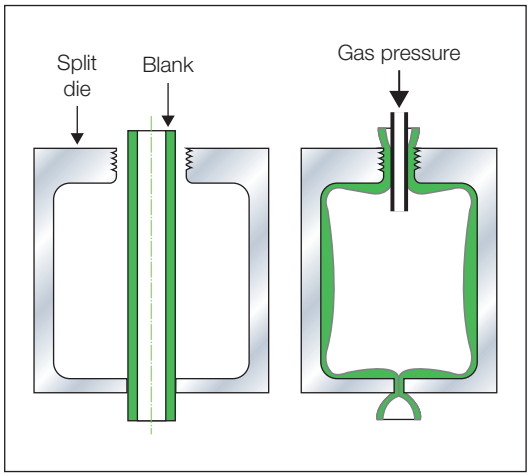
Molding Figure 13.6 shows casting, adapted to materials that are very viscous when molten, particularly thermoplastics and glasses. The hot, viscous fluid is pressed or injected into a die under considerable pressure, where it cools and solidifies. The die must withstand repeated application of pressure, temperature, and the wear involved in separating and removing the part, and therefore it is expensive. Elaborate shapes can be molded, but at the penalty of complexity in die shape and in the way it separates to allow removal. The molds for thermo-forming, by contrast, are cheap. Variants of the process use gas pressure or vacuum to press a heated polymer sheet onto a single-part mold. Blow molding, too, uses a gas pressure to expand a polymer or glass blank into a split outer die. It is a rapid, low-cost process well suited for mass production of cheap parts like milk bottles. Polymers, such as metals, can be extruded; virtually all rods, tubes, and other prismatic sections are made in this way.

Deformation processing Figure 13.7 shows that the deformation process can be hot, warm, or cold—cold, that is, relative to the melting point T_m of the material being processed. Extrusion, hot forging, and hot rolling ($T > 0.55\ T_m$) have much in common with molding, though the material is a true solid, not a viscous liquid. The high temperature lowers the yield strength and allows simultaneous recrystallization, both of which lower the forming pressures. Warm working ($0.35\ T_m < T < 0.55\ T_m$) allows recovery but not recrystallization. Cold forging, rolling, and drawing ($T < 0.35\ T_m$) exploit work hardening to increase the strength of the final product, but at the penalty of higher forming pressures.

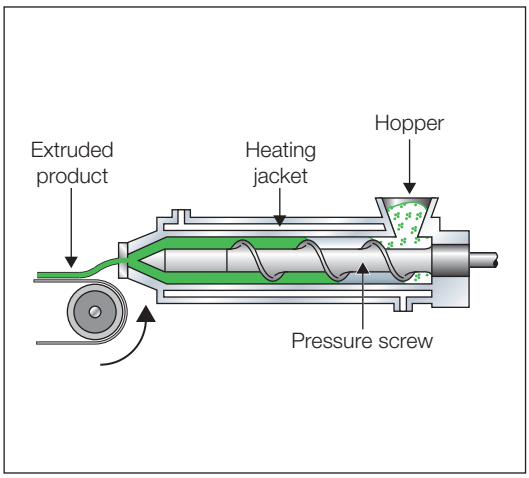
Forged parts are designed to avoid sudden changes in thickness and sharp radii of curvature since both require large local strains that can cause the material to tear or to fold back on itself ("lapping"). Hot forging of metals allows larger changes of shape but generally gives a poor surface and tolerance because of oxidation and warpage. Cold forging gives greater precision and finish, but forging pressures are higher and the deformations are limited by work hardening.



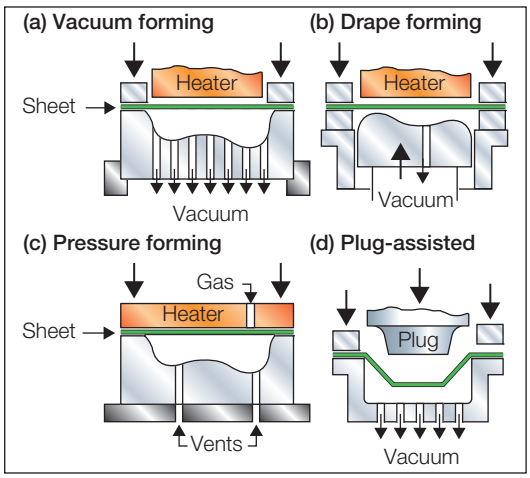
Injection molding



Blow molding



Polymer extrusion

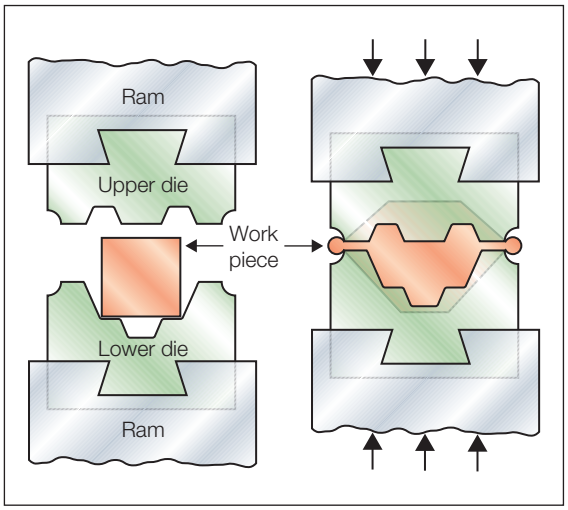


Thermo-forming

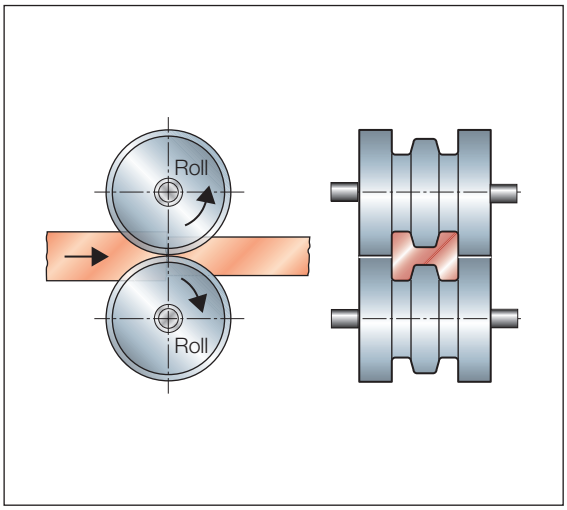
FIGURE 13.6

Molding processes. In *injection molding,* a granular polymer (or filled polymer) is heated, compressed, and sheared by a screw feeder, forcing it into the mold cavity. In *blow molding*, a tubular blank of hot polymer or glass is expanded by gas pressure against the inner wall of a split die. In *polymer extrusion*, shaped sections are formed by extrusion through a shaped die. In *thermo-forming*, a sheet of thermoplastic is heated and deformed into a female die by vacuum or gas pressure.

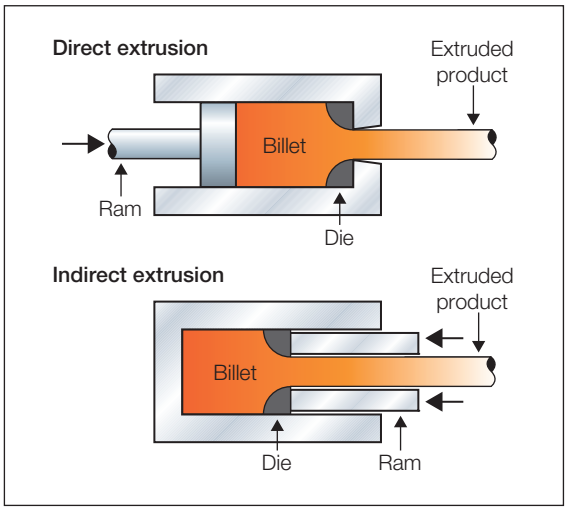
Powder methods Figure 13.8 shows that powder methods create the shape by pressing and then sintering fine particles of the material. The powder can be cold-pressed and then sintered (heated at up to 0.8 T_m to give diffusion bonding); it can be pressed in a heated die ("die pressing"); or, contained in a thin preform, it can be heated under a hydrostatic pressure ("hot isostatic pressing"



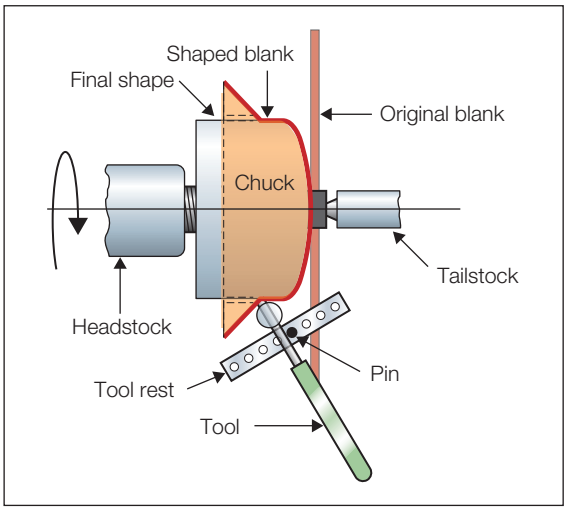
Forging



Rolling



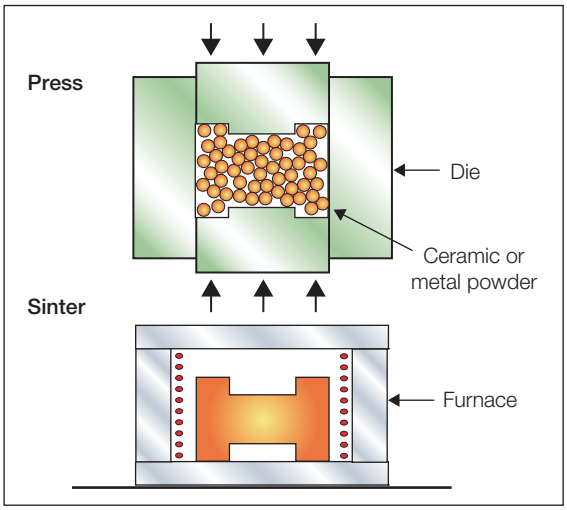
Extrusion



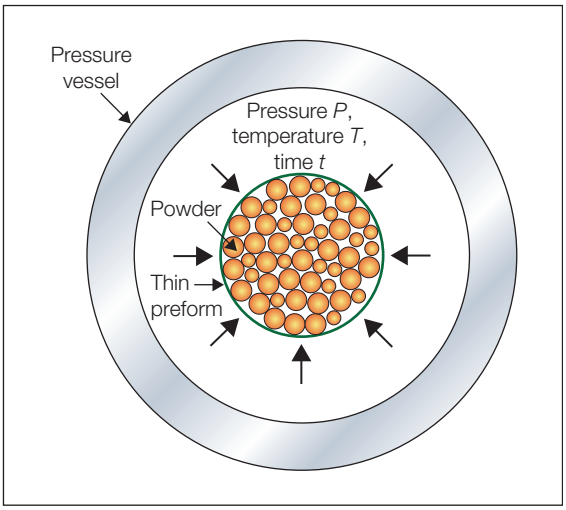
Spinning

FIGURE 13.7

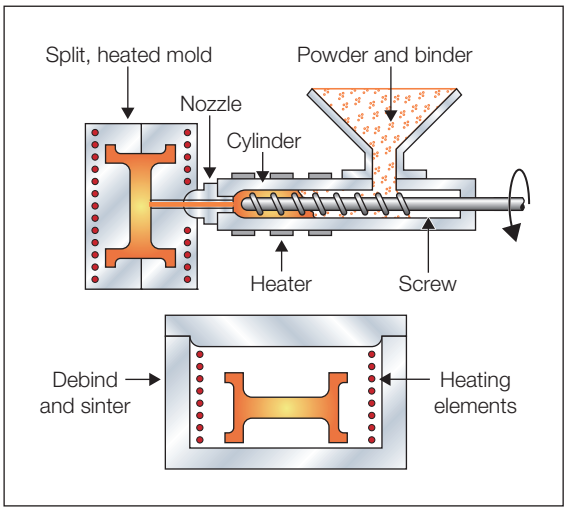
Deformation processes. In *forging*, a slug of metal is shaped between two dies held in the jaws of a press. In *rolling*, a billet or bar is reduced in section by compressive deformation between the rolls. In *extrusion*, metal is forced to flow through a die aperture to give a continuous prismatic shape. All three processes can be hot ($T > 0.85\ T_m$), warm ($0.55\ T_m < T < 0.85\ T_m$), or cold ($T < 0.35\ T_m$). In *spinning*, a spinning disc of ductile metal is shaped over a wooden pattern by repeated sweeps of the smooth, rounded tool.



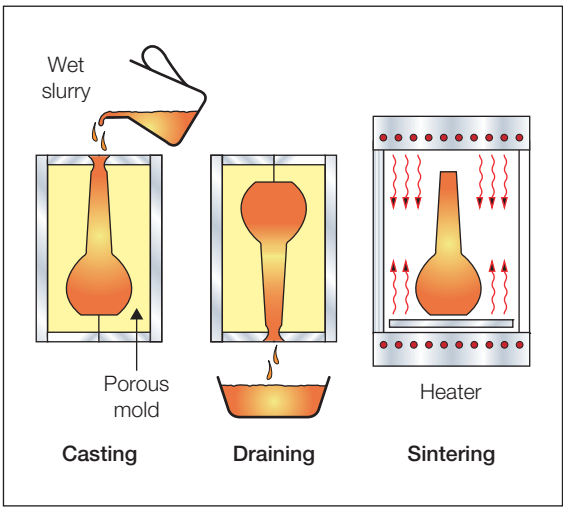
Die-pressing and sintering



Hot isostatic pressing



Powder injection molding



Slip casting

FIGURE 13.8

Powder processing. In *die-pressing* and *sintering* the powder is compacted in a die, often with a binder, and the green compact is then fired to give a more or less dense product. In *hot isostatic pressing*, powder in a thin, shaped shell or preform is heated and compressed by an external gas pressure. In *powder injection molding*, powder and binder are forced into a die to give a green blank that is then fired. In *slip casting,* a water-based powder slurry is poured into a porous plaster mold that absorbs the water, leaving a powder shell that is subsequently fired.

or "HIPing"). Metals that are too high-melting to cast and too strong to deform can be made (by chemical methods) into powders and then shaped in this way. But the processes are not limited to "difficult" materials; almost any material can be shaped by subjecting it, as a powder, to pressure and heat.

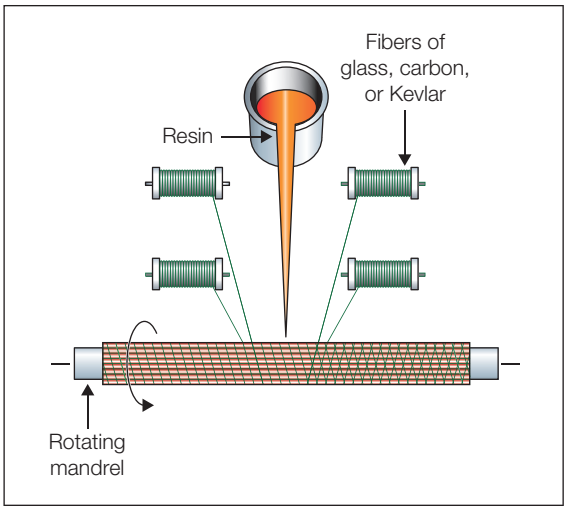
Powder processing is most widely used for small metallic parts like gears and bearings for cars and appliances. It is economic in its use of material; it allows parts to be fabricated from materials that cannot be cast, deformed, or machined; and it can give a product that requires little or no finishing. Since pressure is not transmitted uniformly through a bed of powder, the length of a die-pressed powder part should not exceed 2.5 times its diameter. Sections must be near uniform because the powder will not flow easily around corners. And the shape must be simple and easily extracted from the die.

Ceramics, difficult to cast and impossible to deform, are routinely shaped by powder methods. In slip casting, a water-based powder slurry is poured into a plaster mold. The mold wall absorbs water, leaving a semi-dry skin of slurry over its inner wall. The remaining liquid is drained out, and the dired slurry shell is fired to give a ceramic body. In powder injection molding (the way spark-plug insulators are made) a ceramic powder in a polymer binder is molded in the conventional way; the molded part is fired, burning the binder and sintering the powder.

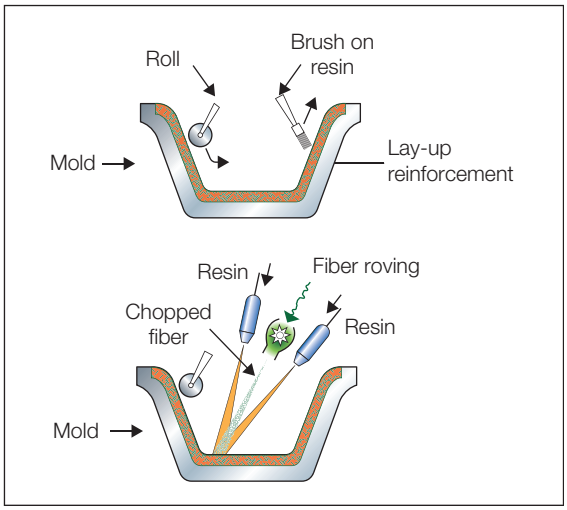
Composite forming methods, as shown in Figure 13.9, make polymer-matrix composites reinforced with continuous or chopped fibers. Large components are fabricated by filament winding or by laying up preimpregnated mats of carbon, glass, or Kevlar fiber ("pre-preg") to the required thickness, pressing, and curing. Parts of the process can be automated, but it remains a slow manufacturing route; and, if the component is a critical one, extensive ultrasonic testing may be necessary to confirm its integrity. Higher integrity is given by vacuum-bag or pressure-bag molding, which squeezes bubbles out of the matrix before it polymerizes. Lay-up methods are best suited to a small number of high-performance, tailor-made components.

More routine components (car bumpers, tennis racquets) are made from chopped-fiber composites by pressing and heating a "dough" of resin containing the fibers, known as bulk molding compound (BMC) or sheet molding compound (SMC), in a mold, or by injection molding a rather more fluid mixture into a die. The flow pattern is critical in aligning the fibers, so the designer must work closely with the manufacturer to exploit the composite properties fully.

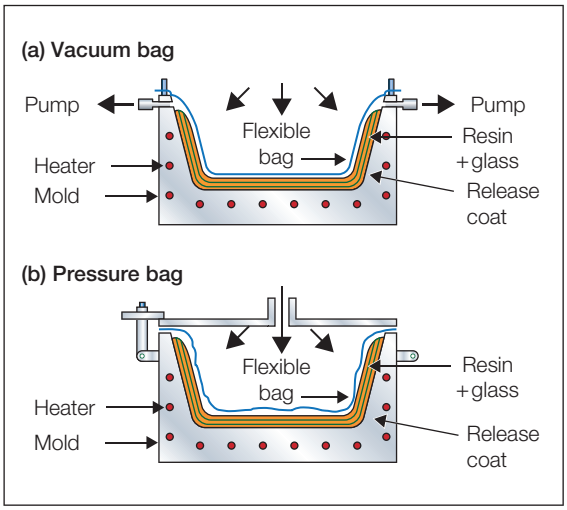
Rapid prototyping systems (RPS) allow single examples of complex shapes to be made from numerical data generated by CAD solid-modeling software (see Figure 13.10). The motive may be that of visualization: The aesthetics of an object may be evident only when viewed as a prototype. It may be



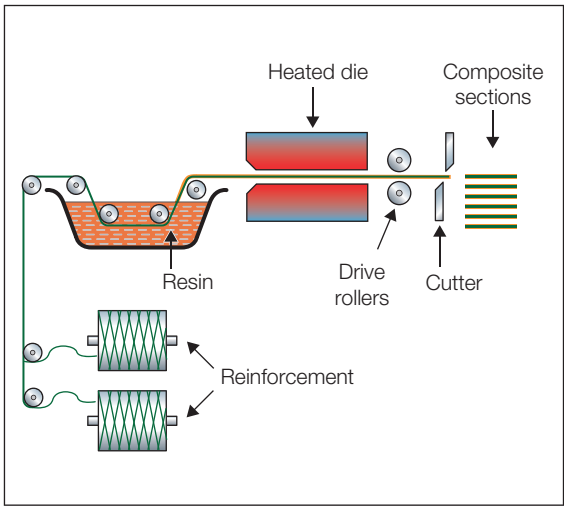
Filament winding



Hand and spray lay-up



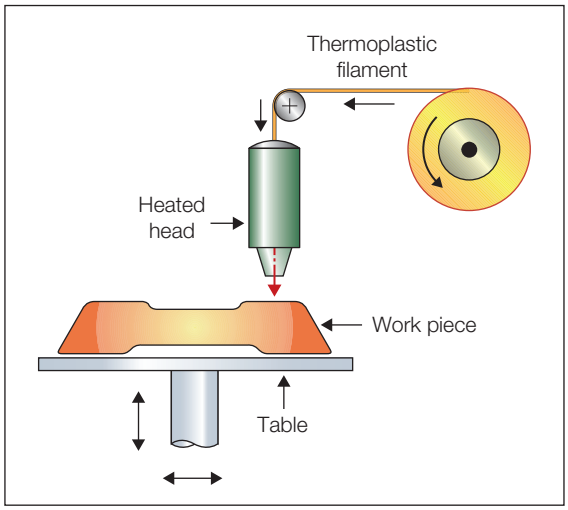
Vacuum and pressure-bag molding



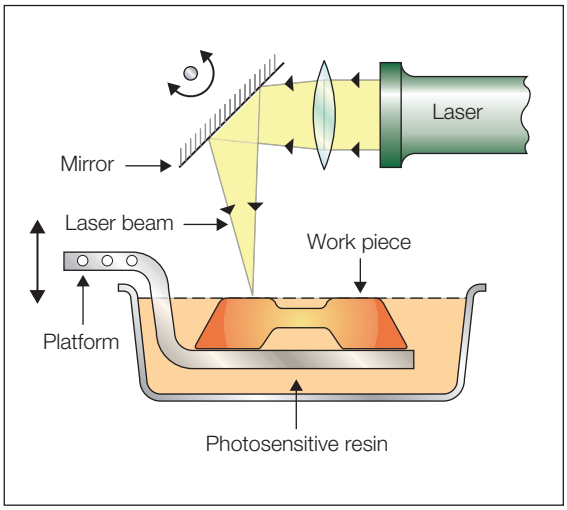
Pultrusion

FIGURE 13.9

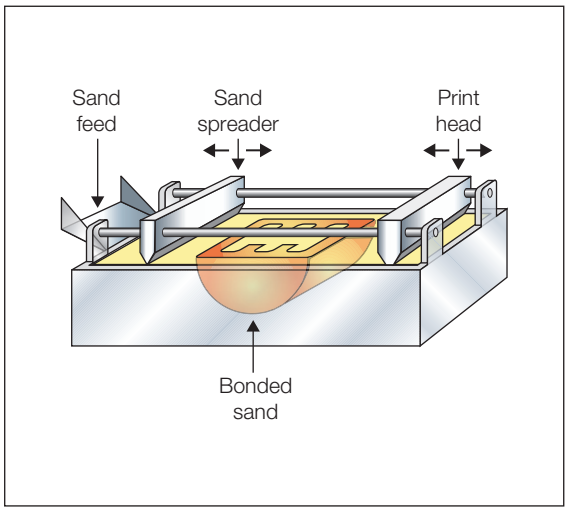
Composite forming methods. In *filament winding*, fibers of glass, Kevlar, or carbon are wound onto a former and impregnated with a resin-hardener mix. In *roll* and *spray lay-up*, fiber reinforcement is laid up in a mold onto which the resin-hardener mix is rolled or sprayed. In *vacuum-bag* and *pressure-bag molding*, laid-up fiber reinforcement, impregnated with resin-hardener mix, is compressed and heated to cause polymerization. In *pultrusion*, fibers are fed through a resin bath into a heated die to form continuous prismatic sections.



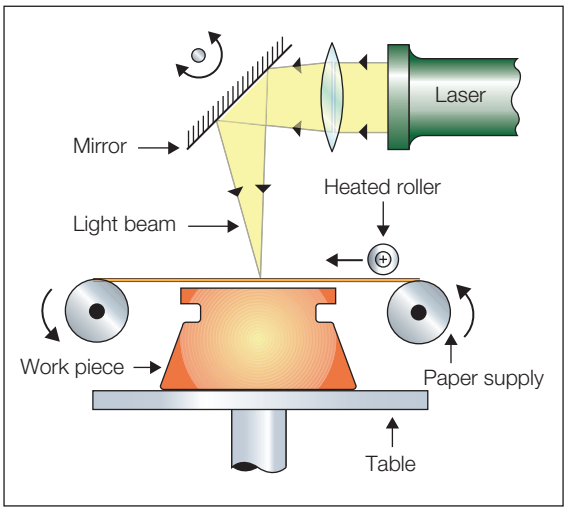
Deposition modeling



Stero-lithography (SLA)



Direct mold modeling



Laminated object manufacture (LOM)

FIGURE 13.10

Rapid prototyping. In *deposition modeling* and *ballistic particle manufacture* (BPM), a solid body is created by the layer-by-layer deposition of polymer droplets. In *stereo-lithography* (SLA), a solid shape is created layer by layer by laser-induced polymerization of a resin. In *direct mold modeling*, a sand mold is built up layer by layer by selective spraying of a binder from a scanning print-head. In *laminated object manufacture* (LOM), a solid body is created from layers of paper, cut by a scanning laser beam and bonded with a heat-sensitive polymer.

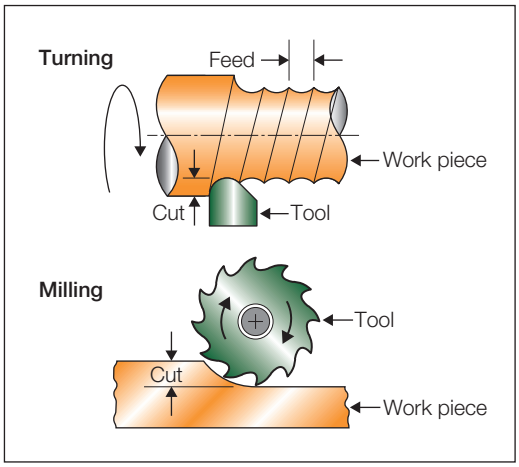
that of pattern-making: The prototype becomes the master from which molds for conventional processing, such as casting, can be made. Or—in complex assemblies—it may be that of validating intricate geometry, ensuring that parts fit, can be assembled, and are accessible.

All RPSs can create shapes of great complexity with internal cavities, overhangs, and transverse features, though the precision, at present, is limited to ± 0.3 mm at best. All RP methods build shapes layer by layer, rather like three-dimensional printing, and are slow (typically 4–40 hours per unit). There are at least six broad classes.

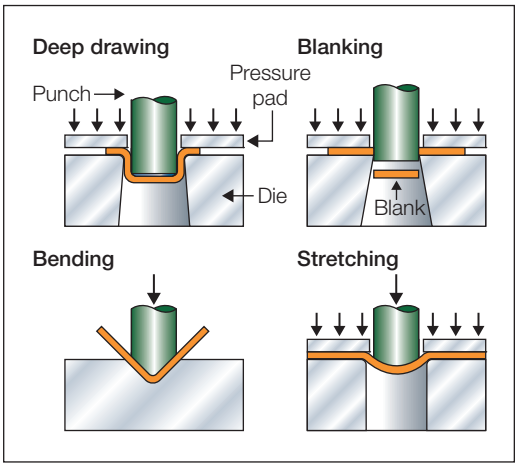
1. The shape is built up from a thermoplastic fed to a single scanning head that extrudes it like a thin layer of toothpaste (*fused deposition modeling* or FDM), exudes it as tiny droplets (*ballistic particle manufacture*, BPM), or ejects it in a patterned array like a bubble-jet printer ("3D printing").
2. Scanned-laser–induced polymerization of a photo-sensitive monomer (*stereo-lithography* or SLA). After each scan, the work piece is incrementally lowered, allowing fresh monomer to cover the surface.
3. Scanned laser cutting of bondable paper elements. Each paper-thin layer is cut by a laser beam and heat-bonded to the next one.
4. Screen-based technology like that used to produce microcircuits (*solid ground curing* or SGC). A succession of screens admits UV light to polymerize a photo-sensitive monomer, building shapes layer by layer.
5. *Selected laser sintering* (SLS) allows components to be fabricated directly in thermoplastic, metal, or ceramic. A laser, as in SLA, scans a bed of particles, sintering a thin surface layer where the beam strikes. A new layer of particles is swept across the surface and the laser-sintering step is repeated, building up a three-dimensional body.
6. *Bonded sand molding* offers the ability to make large complex metal parts easily. Here a multi-jet print-head squirts a binder onto a bed of loose casting sand, building up the mold shape much as selected laser sintering does, but more quickly. When complete, the mold is lifted from the remaining loose sand and used in a conventional casting process.

To be useful, prototypes made by RPS are used as silicone molding masters, allowing replicas to be cast using high-temperature resins or metals.

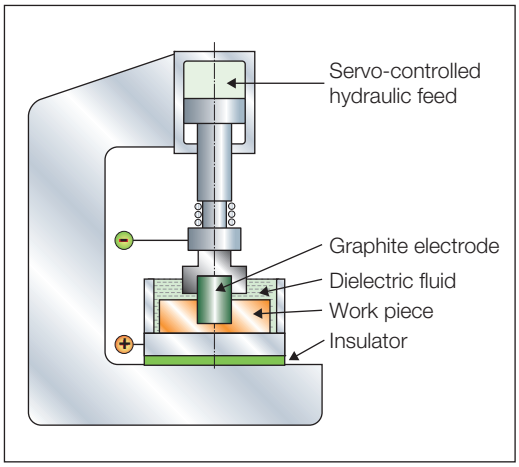
Almost all engineering components, whether made of metal, polymer, or ceramic, are subjected to some kind of ***machining*** (Figure 13.11) during manufacture. To make this possible they should be designed to make gripping and jigging easy and to keep the symmetry high: Symmetric shapes need fewer operations. Metals differ greatly in their *machinability*, a measure of the ease of chip formation, the ability to give a smooth surface, and the ability to give economical tool life (evaluated in a standard test). Poor machinability means higher cost.



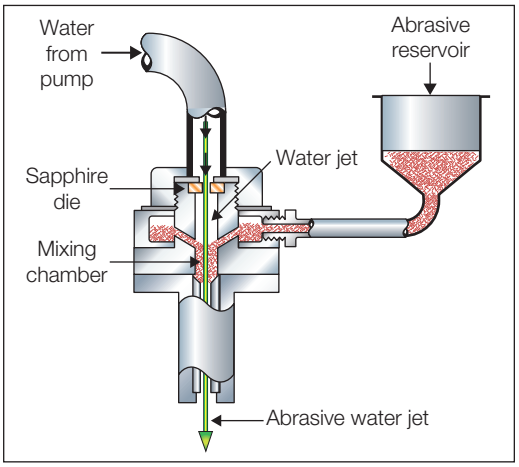
Turning and milling



Drawing, blanking, bending, and stretching



Electro-discharge machining



Water-jet cutting

FIGURE 13.11

Machining operations. In *turning* and *milling*, the sharp, hardened tip of a tool cuts a chip from the work piece surface. In *drawing, blanking*, and *stretching*, sheet is shaped and cut to give flat and dished shapes. In *electro-discharge machining*, electric discharge between a graphite electrode and the work piece, submerged in a dielectric such as paraffin, erodes the work piece to the desired shape. In *water-jet cutting*, an abrasive entrained in a high-speed water jet erodes the material in its path.

Most polymers are molded into a final shape. When necessary they can be machined but their low moduli mean that they deflect elastically during the machining operation, limiting tolerance. Ceramics and glasses can be ground and lapped to high tolerance and finish (think of the mirrors of telescopes). There are many "special" machining techniques with particular applications

including: electro-discharge machining (EDM), ultrasonic cutting, chemical milling, and cutting by water and sand jets and electron and laser beams.

Sheet metal forming involves punching, bending, and stretching. Holes cannot be punched to a diameter less than the thickness of the sheet. The minimum radius to which a sheet can be bent, its *formability*, is sometimes expressed in multiples of the sheet thickness t: A value of 1 is good; one of 4 is average. Radii are best made as large as possible, and never less than t. The formability also determines the amount the sheet can be stretched or drawn without necking and failing. The *forming limit diagram* gives more precise information: It shows the combination of principal strains in the plane of the sheet that will cause failure. The part is designed so that the strains do not exceed this limit.

Machining is often a secondary operation applied to castings, moldings, or powder products to increase finish and tolerance. Higher finish and tolerance mean higher cost; over-specifying either is a mistake.

Joining processes

Joining Figure 13.12 shows that joining is made possible by a number of techniques. Almost any material can be joined with adhesives, though ensuring a sound, durable bond can be difficult. Bolting, riveting, stapling, and snap fitting are commonly used to join polymers and metals, and have the feature that they can be disassembled if need be. Welding, the largest class of joining processes, is widely used to bond metals and polymers; specialized techniques have evolved to deal with each class. Friction welding and friction-stir welding rely on the heat and deformation generated by friction to create a bond between different metals. Ceramics can be diffusion-bonded to themselves, to glasses, and to metals.

If components are to be welded, the material of which they are made must be characterized by a high *weldability*. Like *machinability*, it measures a combination of basic properties. A low thermal conductivity allows welding with a low rate of heat input, but can lead to greater distortion on cooling. Low thermal expansion gives small thermal strains with less risk of distortion. A solid solution is better than an age-hardened alloy because, in the heat-affected zone on either side of the weld, over-aging and softening can occur.

Welding always leaves internal stresses that are roughly equal to the yield strength of the parent material. They can be relaxed by heat treatment but this is expensive, so it is better to minimize their effect by good design. To achieve this, parts to be welded are made of equal thickness whenever possible, the welds are located where stress or deflection is least critical, and the total number of welds is minimized.